Diversification and Convexity in Trend Following*

Dennis Jung
Quintet Private Bank
Technical University of Darmstadt
dennis_rene_jung@yahoo.com

Abstract

How does the cross-sectional filter of a trend-following portfolio determine its Sharpe ratio and its convexity, and can a filter raise the first without giving up the second? In a Gaussian benchmark, the return of any filter is a weighted sum of independent single-market trend followers, weighted by the eigenvalues of the filter times the correlation matrix. For the classic equal-risk portfolio, convexity breadth never exceeds Sharpe breadth: 73 futures markets offer about fifteen independent Sharpe bets but only four convexity bets, and skewness stops falling at about 16 markets. Targeting the volatility of current positions removes most of the remaining convexity. The Sharpe-optimal filter weights directions by trend premium and earns lead–lag premia. Estimated by a risk-premium PCA, it raises the walk-forward Sharpe ratio from 0.60 to 1.33 with no detectable loss of convexity; net of costs, it nearly matches a volatility-targeted reference system but keeps the skewness that the target removes.

Keywords: Trend Following, Managed Futures, CTA, Skewness, Convexity, Diversification, Crisis Alpha, Co-Trending, Cross-Sectional Filter, Risk-Premium PCA, Volatility Targeting.

JEL codes: G11, G12, G23

*This version: October 11, 2026. Version with the Appendix. Disclaimer: The views and opinions expressed herein are those of the author and do not necessarily reflect the views of Quintet Private Bank, its affiliates, or its employees. The results in the paper do not constitute an offer or solicitation of an offer, or any advice or recommendation, by Quintet Private Bank, to purchase any securities or other financial instruments and may not be construed as such.

1 Introduction

Investors hold trend-following programs, commodity trading advisors (CTAs) and managed-futures funds, for their convexity: trend followers lose small amounts often and gain large amounts in sustained market moves, so their returns are positively skewed and tend to be high when equity markets fall (Fung and Hsieh, 2001; Potters and Bouchaud, 2006; Moskowitz, Ooi, and Pedersen, 2012; Hurst, Ooi, and Pedersen, 2017). For a single trend-following position this convexity is structural: the skewness of the aggregated return of an exponentially weighted trend filter is positive even when the market has no drift and no autocorrelation (Sepp and Lucic, 2026). Diversified programs are far less skewed. Sepp and Lucic (2026) report a quarterly skewness of about 0.5 for their diversified reference portfolio after 2000, against about 2.3 for a single contract, and read the gap as a matter of filter design and sample period, since in their view aggregation across markets preserves skewness through the common trend component. Practitioners hold that diversification buys Sharpe ratio at the expense of convexity (Capital Fund Management, 2018; Dao, Nguyen, Deremble, Lempérière, Bouchaud, and Potters, 2017), but the size of that trade-off, and how a program could do better than accept it, are open.

This article asks how the cross-sectional filter of a trend-following portfolio determines its Sharpe ratio and its convexity, and whether a filter can raise the Sharpe ratio without giving up convexity. A cross-sectional filter is the symmetric matrix C that maps the vector of trend signals S_{t-1} into positions $x_{t-1} = CS_{t-1}$; it fixes which markets the portfolio trades and how each position draws on the signals of all markets. The classic CTA, which trades every market on its own signal with equal risk, is the diagonal case $C = W$. We measure convexity, as Sepp and Lucic (2026) do, by the skewness of horizon returns, and report the convex response to equity-market returns as a secondary measure. The answer has three parts, one for each result section: the spectral law and what it implies for the classic CTA (Section 4); the features of markets and of the portfolio risk rule that decide whether the

convexity of a filter reaches the investor (Section 5); and the choice of the filter (Section 6).

The spectral law. In the Gaussian benchmark of no predictability, the T-day return of any cross-sectional filter has the distribution of a weighted sum of independent single-market trend followers, the weights being the eigenvalues of CP, with P the correlation matrix of returns; every cumulant is the single-market cumulant times $\mathrm{tr}((CP)^n)$ (Proposition 3). For the classic CTA this gives an exact eigen-trend representation (Lemma 1) and three breadths, numbers of effectively independent bets on the Sharpe ratio, on skewness, and on kurtosis, ordered $K_4 \leq K_3 \leq K_2$ (Corollary 1). On 73 futures markets the Sharpe breadth is 14.5 but the convexity breadth only 3.8. Skewness therefore falls from 2.13 for a single market to 1.07 for 16 markets and then stops falling, while the Sharpe ratio keeps rising from 0.17 to 0.72 (Exhibit 2), and across portfolios and over time skewness follows convexity breadth; given breadth, the number of markets adds little (R^2 of 0.82 against 0.80). Convexity sits in the leading eigen-trends: in the benchmark, the first carries 51% of the variance but 71% of the convexity, and realized contributions follow (Exhibit A14 of the Appendix).

What breaks the benchmark. Without the benchmark, the third moment of any trend-following portfolio splits exactly into a position channel, through which positions grow after gains, a co-movement channel, through which conditional variances and correlations move after gains, and a drift part (Proposition 2). The correlation part of the co-movement channel contributes 0.98 to the skewness of the 73-market portfolio, against 0.08 in phase-randomized surrogates that keep all linear dependence (Exhibit 7): after trends have paid, correlations rise in the direction of the positions. A negative drift part, trend reversals after crises, gives most of this back: net, the realized skewness of 1.06 is not above that of the surrogates. The predictable part of the drift has an observable counterpart in commodity futures: the roll yield, known one day ahead, earns 61% of the trend premium of a commodity trend follower and contributes nothing to its skewness (Exhibit A16). Both channels run through the conditional variance of the positions, so a rule that holds the conditional variance of the portfolio constant, given current positions and current covariances, removes them; one based

on a static covariance removes only the position channel (Corollary 3). In implementable portfolios net of costs, in a comparison added after the pre-specified backtest, targeting the covariance of current positions lowers the 55-day skewness of the classic CTA from 1.35 to 0.04, whereas targeting realized P&L volatility keeps 0.79 (Exhibit 9). A confirmatory test with predictions fixed in advance gives the same ordering on the commodity panel: without a rule the skewness is 1.65, with the ex-ante rule 0.43, and with the ex-post rule 0.72, although both thresholds were narrowly missed: the ex-ante rule kept 26% of the skewness (predicted: at most 25%) and the ex-post rule 43% (predicted: at least 50%) (Exhibit A18).

Choosing the filter. The trend premium of a filter is $\mathrm{tr}(C\Gamma)$, with Γ the cross-moment of returns and lagged signals; its off-diagonal elements are lead–lag premia that the classic CTA cannot earn (Lemma 2). The Sharpe-optimal filter is, to first order, $C^* \propto P^{-1}\Gamma_s P^{-1}$, and its restriction C^+ to positive semidefinite filters, which trades every direction as a trend follower, is convex in the benchmark (Proposition 4). The classic CTA is optimal only in the knife-edge case $\Gamma_s \propto PWP$. Because the plug-in estimate of C^+ inverts a noisy correlation matrix, we estimate it with a risk-premium PCA in the spirit of Lettau and Pelger (2020b), which chooses the eigen-trends by variance and premium together (Proposition 5). On designs fixed in advance, premium-weighted RP eigen-trends raise the test-sample Sharpe ratio of the classic CTA, gross of costs and with positions fixed in the training sample, from 0.45 to 1.28 in 2013–2026 and, in a reverse split that trains on the later period, from 0.98 to 1.49 in 2001–2012; a quarter to three fifths of the premium is lead–lag premium. The gain holds in a walk-forward test with yearly re-estimation (0.60 to 1.33, higher in 21 of 23 years), lies far outside what the same estimators fit in simulated data without predictability ($p \leq 0.007$), and survives a deflation for the 29 filters and grid points evaluated in this comparison (Exhibit 12). We detect no loss of convexity: the test-sample skewness of the premium-weighted filter is 1.36 against 1.20 for the classic CTA in 2013–2026 and 0.76 against 0.90 in 2001–2012, and no 90% bootstrap interval of a skewness difference lies below zero, although with twelve years of test data only differences of about one unit of

skewness are detectable. The premium-aware filter is not less diversified than the classic CTA but differently: it holds slightly fewer Sharpe bets and twice as many convexity bets (Exhibit A20). The static spectrum predicts the skewness of risk-based filters to within 0.3 but understates that of premium-weighted ones, which load on the directions in which trends pay and so on the channels of the second part. Implemented with monthly re-estimation and net of costs, the same filter applied to the signal of the capped reference system raises its net Sharpe ratio from 0.63 to 0.84 without a portfolio volatility target (difference 0.21, 90% interval 0.01 to 0.41, Ledoit–Wolf $p = 0.07$), but only from 0.87 to 0.90 with the ex-ante target (0.03, interval -0.14 to 0.20); without a portfolio target it earns about the net Sharpe ratio that the reference system reaches only with the ex-ante target (0.84 against 0.87) but keeps a 55-day skewness of 0.98 instead of 0.07 (Exhibit 10). On a separately constructed panel of 37 commodities, where trends rarely lead each other, it does not help.

The closest results are the following. The cumulants of Gaussian quadratic forms are classical (Mathai and Provost, 1992), and Jusselin, Lezmi, Malongo, Masselin, Roncalli, and Dao (2017) derive a trace formula for the skewness of the instantaneous P&L of a multi-asset trend model; we apply them to horizon returns of every cross-sectional filter and derive the breadths and their empirical content. Kelly, Malamud, and Pedersen (2023) show that the optimal linear strategy built from a prediction matrix $\mathbb{E}[R_{t+1}S_t^\top]$ is given by its eigen-structure (principal portfolios); our optimal filter is the trend-following case of this result, and what we add is its third moment: the convexity of every filter, the convexity cost of the Sharpe-optimal one, and the channels through which convexity arises beyond the benchmark. We restrict filters to symmetric matrices, for which the spectral law holds; a symmetric filter earns only the symmetric part of the prediction matrix, and the antisymmetric part, the principal alpha portfolios of Kelly et al. (2023), is left aside. Grebenkov and Serror (2015) and Valeyre (2024) derive Sharpe-optimal allocations of trend-following strategies across correlated assets; we add the third moment of every filter and the convexity of the optimum. Benichou, Lempérière, Sérié, Kockelkoren, Seager, Bouchaud, and Potters (2017)

trade signals in the basis of principal components with equal risk (agnostic risk parity), and Baltas and Kosowski (2020) scale time-series momentum by the average pairwise correlation; the spectral law gives the convexity of such rotations, and Corollary 3 bears on such risk rules. The split of a premium into own and cross-serial parts goes back to Lo and MacKinlay (1990). Dao et al. (2017) show that a diversified trend follower is convex relative to risk parity and that some convexity is lost to diversification; we measure the loss and explain it by the spectrum. Pitkäjärvi, Suominen, and Vaittinen (2020) document cross-asset predictability of time-series momentum; we show exactly how much of it a filter earns and which filter earns it best. Lettau and Pelger (2020b) introduce risk-premium PCA for factors of returns; for a trend follower the premium of a direction is a quadratic form, and the method becomes an estimator of the optimal filter. We do not claim that trend premia exist or explain why crises produce co-trending.

Section 2 develops the theory and Section 3 the data and methodology. Sections 4, 5, and 6 report the three results, Section 7 robustness and limitations, and Section 8 concludes. All proofs are in appendix A; the online supplement documents the pre-specification of the analyses, the systems, the data, and additional results.

The article adds to the closed-form analysis of trend following (Potters and Bouchaud, 2006; Grebenkov and Serror, 2014; Martin, 2023; Firoozye and Koshiyama, 2019), to work on the convexity and crisis performance of diversified CTAs (Hoffman and Kaminski, 2016; Hamill, Rattray, and Van Hemert, 2016; Safari and Schmidhuber, 2026), to the choice of markets and signal designs (Moore, Goodall, and Sanchez Martin, 2026; Masturzo, 2025), to latent factor models with premia (Lettau and Pelger, 2020a), and to the study of carry (Erb and Harvey, 2006; Gorton and Rouwenhorst, 2006; Koijen, Moskowitz, Pedersen, and Vrugt, 2018) and volatility targeting (Moreira and Muir, 2017; Harvey, Hoyle, Korgaonkar, Rattray, Sargaison, and Van Hemert, 2018).

2 A Spectral Theory of Trend-Following Portfolios

This section develops the theory in four steps. Section 2.1 fixes the notation and two sets of assumptions, a static Gaussian benchmark and a general conditional model. Section 2.2 shows that a portfolio of trend followers is a portfolio of trend followers on eigen-portfolios and derives the spectral law of skewness and the breadths. Section 2.3 drops the benchmark and splits the third moment into channels, and Section 2.4 extends the spectral law to every cross-sectional filter, derives the Sharpe-optimal filter and its convexity, and proposes a risk-premium PCA to estimate it. All proofs are in Section A of the Appendix.

2.1 Setting

There are N markets with daily normalized returns $z_t = (z_{1,t}, \dots, z_{N,t})^\top$, returns divided by a point-in-time volatility estimate (Equation (17)), and a filtration $(\mathcal{F}_t)$ with $\sigma(z_s : s \le t) \subseteq \mathcal{F}_t$, which may contain further information known at t, such as futures curves. In every market a trend follower holds the position

$$S_{t-1} = \sum_{j\ge 1} \beta_j \, z_{t-j}, \qquad \sum_{j\ge 1} \beta_j^2 = 1, \qquad \sum_{j\ge 1} |\beta_j| < \infty, \tag{1}$$

a linear filter of past returns with the same weights β_j in every market. The leading example is the exponentially weighted moving average (EWMA) of Sepp and Lucic (2026),

$$S_{i,t} = a \sum_{j\ge 0} (1-\nu)\nu^j z_{i,t-j}, \qquad \nu = 1 - \frac{2}{\text{span}+1}, \qquad a = \sqrt{\frac{1+\nu}{1-\nu}}, \tag{2}$$

for which $\beta_j = \sqrt{1-\nu^2}\,\nu^{j-1}$; the loading a gives the signal unit variance when returns are white noise. The signal autocorrelation is $\rho_S(h) = \sum_{j\ge 1} \beta_j \beta_{j+h}$. With risk weights $w_i > 0$ and $W = \text{diag}(w_1, \dots, w_N)$, the trend follower in market i earns $f_{i,t} = S_{i,t-1} z_{i,t}$ on day t, the

portfolio earns

$$g_t = \sum_{i=1}^{N} w_i f_{i,t} = S_{t-1}^{\top} W z_t, \tag{3}$$

and its return over a horizon of T days is $F_T = \sum_{t=1}^{T} g_t$; $\mathbf{1}$ is the vector of ones. Skewness is $\varsigma(F_T) = \kappa_3(F_T)/\kappa_2(F_T)^{3/2}$, with κ_n the n-th cumulant; κ_2 is the variance and κ_3 the third central moment.

Assumption 1 (Static benchmark)**.** The vectors z_t are independent over time and Gaussian, $z_t \sim \mathcal{N}(0, P)$, with P a correlation matrix.

Assumption 2 (Conditional model)**.** For every t: (i) $\mathbb{E}[z_t \mid \mathcal{F}_{t-1}] = m_t$; (ii) the innovation $\eta_t = z_t - m_t$ has the conditional covariance matrix $\mathbb{E}[\eta_t \eta_t^{\top} \mid \mathcal{F}_{t-1}] = \Sigma_t$; (iii) its conditional third moments vanish, $\mathbb{E}[\eta_{i,t}\eta_{k,t}\eta_{l,t} \mid \mathcal{F}_{t-1}] = 0$ for all i, k, l; and (iv) $\sup_t \mathbb{E}\|z_t\|^6 < \infty$.

Assumption 1 is the benchmark of Sepp and Lucic (2026) for many markets: no drift, no autocorrelation, constant correlations, and therefore no expected trend-following return. Assumption 2 allows a predictable drift m_t, which captures autocorrelation and the trends from which trend followers earn, and conditional variances and correlations Σ_t that depend on the history, for example on the state of the trends. Condition (iii) rules out skewness in daily innovations, so that all convexity of the portfolio comes from the dynamics; Assumption 1 is the special case $m_t = 0$, $\Sigma_t = P$ with Gaussian innovations.

Exhibit 1 summarizes the notation.

2.2 Eigen-trends and the spectral law

We first take the classic filter $C = W$ and show that it trades uncorrelated portfolios. Let $M = W^{1/2} P W^{1/2} = \sum_{k=1}^{N} \mu_k q_k q_k^{\top}$ be the spectral decomposition, with eigenvalues $\mu_1 \geq \cdots \geq \mu_N \geq 0$ and orthonormal eigenvectors q_k. The eigenvalues of M are those of WP, so $\mathrm{tr}((WP)^n) = \sum_k \mu_k^n$; with equal risk, $W = I/N$, they are the eigenvalues of P divided by

Symbol	Meaning
z_t, P	normalized returns of the N markets; their correlation matrix
$S_{t-1} = \sum_j \beta_j z_{t-j}$, $\rho_S(h)$	trend signal (linear filter, $\sum_j \beta_j^2 = 1$); signal autocorrelation
$W = \mathrm{diag}(w)$	risk weights of the markets
$f_{i,t} = S_{i,t-1} z_{i,t}$, $g_t = S_{t-1}^\top W z_t$	daily P&L of market i and of the portfolio
$F_T = \sum_{t<T} g_t$, $\varsigma(F_T)$	T-day portfolio return and its skewness
$M = W^{1/2} P W^{1/2} = \sum_k \mu_k q_k q_k^\top$	eigenvalues μ_k and eigenvectors q_k
$y_{k,t}$, $\Phi_{k,t}$	eigen-portfolio return and eigen-trend P&L (Lemma 1)
K_2, K_3, K_4	Sharpe, convexity, and tail breadth (Corollary 1)
m_t, Σ_t	conditional mean and covariance of z_t (Assumption 2)
e_t, V_t	innovation of g_t and its conditional variance (Proposition 2)
$\mathcal{C}_T^{\mathrm{pos}}$, $\mathcal{C}_T^{\mathrm{var}}$, $\mathcal{C}_T^{\mathrm{cor}}$, Δ_T	position and co-movement channels; drift part
ℓ_t	portfolio scaling factor of a risk rule (Corollary 3)
C, $x_{t-1} = C S_{t-1}$, ϑ_k	cross-sectional filter, its positions, eigenvalues of CP
$\Gamma = \mathbb{E}[z_t S_{t-1}^\top]$, Π	trend cross-moment and premium matrix (Lemma 2)
$\pi_k = q_k^\top \Pi q_k$, γ	eigen-trend premium; weight on the premium in the RP basis
λ_k, C^*, C^+	generalized eigenvalues of (Γ_s, P); Sharpe-optimal filter and its PSD restriction

Exhibit 1: **Notation.** Vectors are columns; tr is the trace, κ_n the n-th cumulant.

N. The *eigen-portfolio* k has the return $y_{k,t} = q_k^\top W^{1/2} z_t$, and the *eigen-trend* k is the trend follower on it, with position $s_{k,t-1} = q_k^\top W^{1/2} S_{t-1}$ and daily P&L $\Phi_{k,t} = s_{k,t-1}\, y_{k,t}$.

Lemma 1 (Eigen-trend representation)**.** *(a) For every realization of the returns,* $s_{k,t-1} = \sum_{j\geq 1} \beta_j\, y_{k,t-j}$ *and*

$$g_t = \sum_{k=1}^{N} \Phi_{k,t}. \tag{4}$$

(b) Under Assumption 1, $y_{k,t} = \sqrt{\mu_k}\, \xi_{k,t}$ *with* $\xi_{k,t}$ *independent standard normal over* k *and* t*, and* $\Phi_{k,t} = \mu_k\, \phi_{k,t}$*, where* $\phi_{k,t} = (\sum_{j\geq 1} \beta_j \xi_{k,t-j})\, \xi_{k,t}$ *is the P&L of a single-market trend follower on the white noise* ξ_k*. The processes* $(\phi_{k,t})_t$*,* $k = 1, \ldots, N$*, are independent and identically distributed.*

Part (a) is an identity: the P&L of a portfolio of trend followers is the sum of the P&L of trend followers on the eigen-portfolios, because the filter is linear and the same in every market. Part (b) turns the identity into a statistical statement: in the benchmark, a diversified trend follower is a weighted sum of independent single-market trend followers, the weights being the eigenvalues of WP. Every distributional property of the portfolio follows.

Proposition 1 (Spectral law)**.** *Under Assumption 1, let* Ψ *denote the* T*-day P&L of a*

single-market trend follower on white noise. Then F_T has the same distribution as $\sum_k \mu_k \Psi_k$ with Ψ_k independent copies of Ψ, and for every $n \geq 2$

$$\kappa_n(F_T) = \kappa_n(\Psi) \sum_{k=1}^{N} \mu_k^n = \kappa_n(\Psi)\, \mathrm{tr}\big((WP)^n\big). \tag{5}$$

In particular, $\mathrm{Var}[F_T] = T\,\mathrm{tr}((WP)^2)$, $\kappa_3(F_T) = 6\sum_{h=1}^{T-1}(T-h)\,\beta_h \rho_S(h)\,\mathrm{tr}((WP)^3)$, *and the skewness and excess kurtosis of the portfolio's T-day return are*

$$\varsigma_P(T) = \varsigma_1(T) \cdot \frac{\mathrm{tr}\big((WP)^3\big)}{\mathrm{tr}\big((WP)^2\big)^{3/2}}, \qquad \delta_P(T) = \delta_1(T) \cdot \frac{\mathrm{tr}\big((WP)^4\big)}{\mathrm{tr}\big((WP)^2\big)^2}, \tag{6}$$

with $\varsigma_1(T)$ and $\delta_1(T)$ the skewness and excess kurtosis of a single market (Sepp and Lucic, 2026, Proposition D.2).

The proposition separates horizon and cross-section: the filter and the horizon determine the single-market profile $\varsigma_1(T)$, and the correlation structure and the risk weights scale it uniformly at every horizon. The spectral factor in Equation (6) never exceeds one and equals one only if WP has rank one: aggregation preserves skewness only if one common trend drives all markets. Section A of the Appendix gives two proofs, one through Lemma 1 and a direct one that extends the pairing argument of Sepp and Lucic (2026), in which correlation enters through the closed cycles $w_i w_k w_l P_{ik} P_{kl} P_{li}$ that sum to $\mathrm{tr}((WP)^3)$. Exhibit A1 of the Appendix confirms the formula by simulation.

The spectrum also governs the Sharpe ratio. Suppose that every market's trend P&L has the same small mean $\bar{\pi}$ per day, generated by weak autocorrelation of order ε as in Proposition 3(b). To first order the variance is that of Proposition 1, so a single market has the Sharpe ratio $\mathrm{SR}_1 = \bar{\pi}\sqrt{T}$ at horizon T, and, since $\sum_k \mu_k = \mathrm{tr}(WP) = \sum_i w_i$, the portfolio has mean $\bar{\pi} T \sum_k \mu_k$ and variance $T \sum_k \mu_k^2 (1 + O(\varepsilon))$.

Corollary 1 (Breadths)**.** *Under Assumption 1, and for the Sharpe ratio under the weak-autocorrelation perturbation of Proposition 3(b), define for a nonnegative vector $x \neq 0$ the*

Sharpe breadth $K_2(x)$*, the convexity breadth* $K_3(x)$*, and the tail breadth* $K_4(x)$*, and write* $K_n = K_n(\mu)$*,*

$$K_2(x) = \frac{(\sum_k x_k)^2}{\sum_k x_k^2}, \qquad K_3(x) = \frac{(\sum_k x_k^2)^3}{(\sum_k x_k^3)^2}, \qquad K_4(x) = \frac{(\sum_k x_k^2)^2}{\sum_k x_k^4}. \tag{7}$$

Then, to first order in $\bar{\pi}$*, the portfolio Sharpe ratio is* $\mathrm{SR}_1\sqrt{K_2}$*, its skewness* $\varsigma_1(T)/\sqrt{K_3}$*, and its excess kurtosis* $\delta_1(T)/K_4$*, and*

$$1 \;\le\; K_4 \;\le\; K_3 \;\le\; K_2 \;\le\; \mathrm{rank}(P) \;\le\; N, \tag{8}$$

where $K_4 = K_3 = K_2 = \mathrm{rank}(P)$ *if and only if all nonzero eigenvalues* μ_k *are equal, and* $K_4 = 1$ *if and only if* P *has rank one.*

All three breadths count effectively independent bets: with k equal blocks of perfectly correlated markets and equal risk, $K_2 = K_3 = K_4 = k$. For any other spectrum, the higher the moment, the fewer the effective bets, because higher powers weight the leading eigenvalues, the common trends, more heavily. Diversification therefore raises the Sharpe ratio faster than it dilutes skewness, and skewness reaches its floor, set by the leading eigenvalues, long before the Sharpe ratio stops improving. The same weighting governs the distribution of convexity across eigen-trends.

Corollary 2 (Concentration of convexity)**.** *Under Assumption 1, the share of eigen-trend* k *in the variance of* F_T *is* $v_k = \mu_k^2/\sum_l \mu_l^2$ *and its share in the third moment is* $c_k = \mu_k^3/\sum_l \mu_l^3$*. For every* $1 \le p \le N$*,*

$$\sum_{k\le p} c_k \;\ge\; \sum_{k\le p} v_k, \tag{9}$$

with equality for some $1 \le p < \mathrm{rank}(P)$ *if and only if all nonzero eigenvalues are equal.*

The convexity of a diversified trend follower thus sits in fewer dimensions than its risk. The result connects Corollary 1 to the factor structure of returns: weak factors, those with

small eigenvalues, add to the Sharpe breadth but hardly to convexity, because their convexity shares are of third order. In the instantaneous P&L of a continuous-time multi-asset trend model, Jusselin et al. (2017, Appendix A.5.6) obtain a skewness formula in traces of matrix powers, which, without trends and with equal allocations, reduces to $\sqrt{8}$ times the factor in Equation (6). Lemma 1 shows why traces appear: the trend P&L of a correlated universe is that of independent eigen-trends. The spectral law rests on the benchmark of Assumption 1; the next subsection drops it.

2.3 Two channels of convexity

In the data, trends exist and correlations move with them, so we now work under Assumption 2. The results below hold for any portfolio with $\mathcal{F}_{t-1}$-measurable positions x_{t-1}, $\sup_t \mathbb{E}\|x_{t-1}\|^6 < \infty$, that earns $g_t = x_{t-1}^\top z_t$, with $S_{t-1}^\top W$ replaced by $x_{t-1}^\top$ below; for clarity we state them for the classic CTA, $x_{t-1} = WS_{t-1}$. Under Assumption 2, the daily portfolio P&L of Equation (3) has the predictable drift $d_t = \mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = S_{t-1}^\top W m_t$, the innovation $e_t = g_t - d_t = S_{t-1}^\top W \eta_t$, and the conditional variance

$$V_t = \mathbb{E}[e_t^2 \mid \mathcal{F}_{t-1}] = S_{t-1}^\top W \Sigma_t W S_{t-1}. \tag{10}$$

The horizon return splits into a martingale part $E_T = \sum_t e_t$ and a drift part $D_T = \sum_t d_t$. For a fixed correlation matrix $\bar{P}$, such as the unconditional correlation, write $\Sigma_t = \bar{P} + (\Sigma_t - \bar{P})$ and split the deviation into its diagonal and off-diagonal parts. The conditional variance splits accordingly into $V_t = V_t^{\text{pos}} + V_t^{\text{var}} + V_t^{\text{cor}}$, with

$$V_t^{\text{pos}} = S_{t-1}^\top W \bar{P} W S_{t-1}, \quad V_t^{\text{var}} = \sum_i w_i^2 S_{i,t-1}^2 \big(\Sigma_{t,ii} - 1\big), \quad V_t^{\text{cor}} = \sum_{i \neq k} w_i w_k S_{i,t-1} S_{k,t-1} \big(\Sigma_{t,ik} - \bar{P}_{ik}\big). \tag{11}$$

Proposition 2 (Channels of convexity)**.** *Under Assumption 2:*

(a) $\mathrm{Var}[E_T] = \sum_{t=1}^{T} \mathbb{E}[V_t]$ *and*

$$\kappa_3(E_T) = 3 \sum_{1 \le s < t \le T} \mathrm{Cov}(V_t, e_s) = \mathcal{C}_T^{\mathrm{pos}} + \mathcal{C}_T^{\mathrm{var}} + \mathcal{C}_T^{\mathrm{cor}}, \qquad \mathcal{C}_T^{x} = 3 \sum_{1 \le s < t \le T} \mathrm{Cov}(V_t^{x}, e_s). \tag{12}$$

(b) $\kappa_3(F_T) = \kappa_3(E_T) + \Delta_T$, *with* $\Delta_T = 3\,\mathbb{E}[E_T^2 \tilde{D}_T] + 3\,\mathbb{E}[E_T \tilde{D}_T^2] + \mathbb{E}[\tilde{D}_T^3]$ *and* $\tilde{D}_T = D_T - \mathbb{E}[D_T]$.

(c) For the classic CTA under Assumption 1 and with $\bar{P} = P$: $\Delta_T = 0$, $\mathcal{C}_T^{\mathrm{var}} = \mathcal{C}_T^{\mathrm{cor}} = 0$, *and* $\mathcal{C}_T^{\mathrm{pos}} = 6 \sum_{h=1}^{T-1} (T-h) \beta_h \rho_S(h)\, \mathrm{tr}((WP)^3)$, *the third moment of Proposition 1.*

(d) If (z_t, m_t, Σ_t) *is strictly stationary,* $\kappa_3(E_T) = 3 \sum_{h=1}^{T-1} (T-h)\, \mathrm{Cov}(V_t, e_{t-h})$, *and likewise for each* $\mathcal{C}_T^{x}$.

The proposition says where the convexity of a trend follower comes from: its conditional variance must co-move with its own past P&L. The *position channel* $\mathcal{C}^{\mathrm{pos}}$ is the mechanism of the static benchmark. After a gain, the positions have grown in the direction of the move that produced it, so the variance of the portfolio, at fixed correlations, is higher; Proposition 1 is the special case in which this is the only channel. The *co-movement channel* $\mathcal{C}^{\mathrm{var}} + \mathcal{C}^{\mathrm{cor}}$ adds the dependence of variances and correlations on the past. Its correlation part is positive when, after the portfolio has gained, correlations rise in the direction of its positions; this is what we call co-trending.

Definition 1 (Co-trending)**.** A portfolio of trend followers exhibits co-trending at horizon T if $\mathcal{C}_T^{\mathrm{cor}} > 0$.

The *drift part* Δ_T collects the co-moments of the predictable drift with the martingale part and the third moment of the drift itself. It vanishes without predictability and is negative when the trend P&L is predictably low after large moves, for example when trends reverse after a crisis, as in the momentum crashes of Daniel and Moskowitz (2016). In futures

markets the drift has an observable part: the roll yield, the slope of the futures curve, is known one day ahead and enters m_t (Section 5.1).

Both channels work through the conditional variance V_t. Most CTAs scale their positions to a portfolio volatility target, and how they do so decides whether the channels survive.

Corollary 3 (Risk targeting)**.** *Under Assumption 2, let the positions be scaled by an $\mathcal{F}_{t-1}$-measurable factor $\ell_t > 0$ with $\sup_t \mathbb{E}|\ell_t e_t|^3 < \infty$, so that the portfolio earns $\ell_t g_t$, with innovation $\ell_t e_t$ and martingale part $E_T^\ell = \sum_t \ell_t e_t$. (a) $\kappa_3(E_T^\ell) = 3\sum_{s<t} \mathrm{Cov}(\ell_t^2 V_t, \ell_s e_s)$. (b) If $V_t > 0$ almost surely and the conditional variance is targeted, $\ell_t^2 V_t = \bar{c} > 0$, then $\kappa_3(E_T^\ell) = 0$: both channels vanish, and the third moment of the horizon return is the drift part of the scaled portfolio alone. (c) If instead the variance at a static correlation is targeted, $\ell_t^2 V_t^{\mathrm{pos}} = \bar{c}$, then $\kappa_3(E_T^\ell) = 3\sum_{s<t} \mathrm{Cov}\big(\ell_t^2(V_t^{\mathrm{var}} + V_t^{\mathrm{cor}}), \ell_s e_s\big)$: the position channel vanishes and the co-movement channel remains.*

A rule that scales the current positions by an estimate of their covariance, $\ell_t \propto (S_{t-1}^\top W \hat{\Sigma}_t W S_{t-1})^{-1/2}$, approximates part (b): it removes the position channel and the part of the co-movement channel that $\hat{\Sigma}_t$ tracks; with a static $\hat{\Sigma}_t$ it would remove only the position channel (part (c)). A rule that scales by the realized volatility of past portfolio P&L reacts to V_t only with a lag and leaves part of the covariance in place. Volatility targeting raises Sharpe ratios (Moreira and Muir, 2017; Harvey et al., 2018); the corollary identifies its cost in convexity.

Remark 1 (Estimation). Proposition 2 suggests estimating the martingale part by the sample analog $\hat{\mathcal{C}}^\chi = 3\sum_{h=1}^{T-1}(T-h)\widehat{\mathrm{Cov}}(\chi_t, g_{t-h})$ with $\chi_t = g_t^2$ (total), $\chi_t = V_t^{\mathrm{pos}}$, and $\chi_t = g_t^2 - V_t^{\mathrm{pos}}$ split into its diagonal and off-diagonal parts. Only V_t^{pos} requires a model, the static matrix $\bar{P}$; the co-movement channel is measured from the realized products $f_{i,t} f_{k,t}$ without estimating Σ_t. If $m_t = 0$, then $\mathbb{E}[g_t^2 g_s] = \mathbb{E}[V_t g_s]$ for $s < t$, and all other terms of the third moment of F_T, those with the latest date appearing once or three times, have zero expectation (Lemma A.3 in the Appendix). The difference between the realized third moment and $\hat{\mathcal{C}}^{\mathrm{total}}$ then estimates

zero, and in general it estimates the drift part. Because $\hat{\mathcal{C}}^x$ uses only covariances between a squared and a lagged P&L, it is far less noisy than the sample skewness of overlapping sums (Exhibit A2).

2.4 Cross-sectional filters

So far the cross-section was fixed by the risk weights W; the second half of the research question asks what other cross-sectional choices achieve. The trend filter of Equation (2) acts in time: it turns the past returns of a market into a signal. A second filter acts across markets: it maps the vector of signals into positions. We call a symmetric $N \times N$ matrix C a *cross-sectional filter*; the portfolio holds $x_{t-1} = CS_{t-1}$ and earns $g_t^C = S_{t-1}^\top C z_t$. The classic CTA is $C = W$: each market is traded on its own signal with its own risk weight. Every other choice trades a market partly on the signals of the others. This subsection shows that the spectral law holds for every cross-sectional filter, derives the filter with the highest Sharpe ratio and its convexity, and proposes an estimator.

The premium of a filter. Let

$$\Gamma = \mathbb{E}\big[z_t S_{t-1}^\top\big] = \sum_{j\geq 1} \beta_j \, \mathbb{E}\big[z_t z_{t-j}^\top\big], \qquad \Gamma_s = \tfrac{1}{2}\big(\Gamma + \Gamma^\top\big), \qquad \Pi = W^{1/2}\Gamma_s W^{1/2}, \tag{13}$$

be the trend cross-moment, assumed not to depend on t (for example under strict stationarity), its symmetric part, and its risk-weighted version. The diagonal of Γ holds the filtered autocovariances of each market, its off-diagonal elements the filtered cross-autocovariances: the covariance of one market's return with another market's signal.

Lemma 2 (Premium of a cross-sectional filter)**.** *Under Assumption 2 and Equation (13): (a) a cross-sectional filter C earns $\mathbb{E}[g_t^C] = \operatorname{tr}(C\Gamma) = \sum_i C_{ii}\Gamma_{ii} + \sum_{i\neq k} C_{ik}\Gamma_{ki}$, where the second sum collects cross-serial (lead–lag) premia; (b) for every orthonormal basis (q_k), the trend follower on the portfolio $q_k^\top W^{1/2} z_t$, $\Phi_{k,t} = (q_k^\top W^{1/2} S_{t-1})(q_k^\top W^{1/2} z_t)$, has the premium*

$\mathbb{E}[\Phi_{k,t}] = q_k^\top \Pi q_k =: \pi_k$, *and* $\sum_k \pi_k = \operatorname{tr}(\Pi) = \sum_i w_i \Gamma_{ii} = \mathbb{E}[g_t]$*; (c) the classic filter* $C = W$ *earns only the own-market premia, and the filters* $C = W^{1/2} Q \operatorname{diag}(\omega) Q^\top W^{1/2}$ *with* $\omega = \mathbf{1}$ *are the same for every basis: a basis matters only through non-uniform eigen-trend weights; (d) with* $\hat{\Gamma} = T_s^{-1} \sum_t z_t S_{t-1}^\top$ *over a sample of* T_s *days in place of* Γ*, (a) to (c) hold exactly for sample means.*

The lemma separates two sources of trend premium, as Lo and MacKinlay (1990) separate own and cross-serial contributions to contrarian profits. A classic CTA earns the time-series momentum of each market, $\sum_i w_i \Gamma_{ii}$. A filter with off-diagonal elements earns, in addition, the lead–lag premia Γ_{ki}, the cross-asset predictability documented by Pitkäjärvi et al. (2020). Which filter is best, and what it costs in convexity, follows from a generalization of the spectral law.

Proposition 3 (Spectral law for cross-sectional filters)**.** *Let* C *be a cross-sectional filter and* $\vartheta_1, \dots, \vartheta_N$ *the eigenvalues of* CP*, which are real because* $CP = (CP^{1/2})P^{1/2}$ *and* $P^{1/2}CP^{1/2}$ *have the same characteristic polynomial;* $F_T^C = \sum_{t=1}^T g_t^C$*. (a) Under Assumption 1,* F_T^C *has the distribution of* $\sum_k \vartheta_k \Psi_k$ *with* Ψ_k *independent copies of* Ψ*, and for every* $n \geq 2$

$$\kappa_n\big(F_T^C\big) = \kappa_n(\Psi) \operatorname{tr}\big((CP)^n\big), \qquad \varsigma\big(F_T^C\big) = \varsigma_1(T) \frac{\operatorname{tr}\big((CP)^3\big)}{\operatorname{tr}\big((CP)^2\big)^{3/2}}. \tag{14}$$

(b) Let (z_t) *instead be stationary Gaussian with* $\mathbb{E}[z_t z_t^\top] = P$ *and* $\mathbb{E}[z_t z_{t-h}^\top] = \varepsilon A_h$ *for* $h \geq 1$*,* $\sum_h \|A_h\| < \infty$*, so that* $\Gamma = \varepsilon \sum_{j \geq 1} \beta_j A_j$*. Then* $\mathbb{E}[F_T^C] = T \operatorname{tr}(C\Gamma)$ *exactly,* $\operatorname{Var}[F_T^C] = T \operatorname{tr}((CP)^2)(1 + O(\varepsilon))$*, and the Sharpe ratio of the* T*-day return is*

$$\operatorname{SR}(C) = \sqrt{T} \frac{\operatorname{tr}(C\Gamma)}{\operatorname{tr}((CP)^2)^{1/2}} + O(\varepsilon^2)$$

uniformly over filters with $\operatorname{tr}((CP)^2) = 1$*; the cumulants of order* $n \geq 3$ *differ from (a) by* $O(\varepsilon)$*. We call the leading term the first-order Sharpe ratio.*

Part (a) applies the cumulants of Gaussian quadratic forms (Mathai and Provost, 1992)

to the T-day sum of a trend-following P&L; Proposition 1 is the case $C = W$, for which the eigenvalues of CP are the μ_k. The general case adds one fact: a filter for which CP has negative eigenvalues trades some directions against their own trend, and these directions contribute negatively to the third moment. With $\beta_j \geq 0$, so that $\varsigma_1(T) > 0$, positive weight on every direction keeps the filter convex. The classic filter is one choice; the eigen-trends can be traded with any weights.

Corollary 4 (Eigen-trend tilting)**.** *For weights $\omega_k \geq 0$ let $C^\omega = W^{1/2} Q \operatorname{diag}(\omega)\, Q^\top W^{1/2}$, with $Q = (q_1, \ldots, q_N)$ the eigenvectors of M. (a) For every realization, $g_t^{C^\omega} = \sum_k \omega_k \Phi_{k,t}$; $\omega = \mathbf{1}$ gives the classic portfolio. (b) Under Assumption 1, $\kappa_n(F_T^\omega) = \kappa_n(\Psi) \sum_k (\omega_k \mu_k)^n$. For $\omega_k \propto \mu_k^{\alpha-1}$, $\alpha \geq 0$ (with $\omega_k = 0$ if $\mu_k = 0$), the skewness of the T-day return is $\varsigma_1(T)/\sqrt{K_3(\mu^\alpha)}$, with $K_3(\mu^\alpha)$ the convexity breadth of Corollary 1 computed from μ_k^α in place of μ_k. (c) If, in addition, every eigen-trend earns the same mean per unit of eigenvalue, $\mathbb{E}[\Phi_{k,t}] = \bar{\pi}\mu_k$, the first-order Sharpe ratio is $\mathrm{SR}_1 \sqrt{K_2(\mu^\alpha)}$.*

The tilt exponent α is a dial between the Sharpe ratio and convexity. At $\alpha = 0$ every eigen-trend has the same risk and the Sharpe breadth reaches the rank of P; as α grows, risk moves to the leading eigen-trends, both breadths fall monotonically towards the multiplicity of μ_1, one if μ_1 is simple, and skewness rises towards its single-market level. The classic CTA, $\alpha = 1$, is one point on this dial, chosen by the convention of equal risk per market rather than by a preference between Sharpe ratio and convexity. Part (c) assumes that the premium is proportional to eigen-trend risk. Without that assumption, the best filter follows from the premium matrix.

Proposition 4 (Sharpe-optimal cross-sectional filter)**.** *Under the assumptions of Proposition 3(b), let $P \succ 0$, $\Gamma_s \neq 0$, and $\tilde{\Gamma} = P^{-1/2} \Gamma_s P^{-1/2} = \sum_k \lambda_k \tilde{q}_k \tilde{q}_k^\top$, with $\lambda_1 \geq \cdots \geq \lambda_N$ the generalized eigenvalues of (Γ_s, P) and $\tilde{\Gamma}^+ = \sum_k \max(\lambda_k, 0)\, \tilde{q}_k \tilde{q}_k^\top$; $\|\cdot\|$ is the Euclidean norm. (a) Over all cross-sectional filters, the first-order Sharpe ratio is maximized by $C^* = cP^{-1}\Gamma_s P^{-1}$, $c > 0$, with maximum $\sqrt{T}\|\lambda\|$ and skewness $\varsigma_1(T) \sum_k \lambda_k^3 / (\sum_k \lambda_k^2)^{3/2}$.*

(b) If $\lambda_1 > 0$, then over the filters that trade every direction as a trend follower, $C \succeq 0$, the first-order Sharpe ratio is maximized by $C^+ = cP^{-1/2}\tilde{\Gamma}^+P^{-1/2}$, $c > 0$, the projection of C^* onto $\{C \succeq 0\}$ in the metric $\mathrm{tr}(((C - C^*)P)^2)$, with maximum $\sqrt{T}\|\lambda^+\|$ and skewness $\varsigma_1(T)/\sqrt{K_3(\lambda^+)}$, which is positive whenever $\varsigma_1(T) > 0$, in particular if $\beta_j \geq 0$ for all j and $\beta_h\rho_S(h) > 0$ for some $h < T$. (c) The classic filter $C = W$ is Sharpe-optimal if and only if $\Gamma_s = cPWP$ for some $c > 0$.

The unconstrained optimum C^* trades the directions with negative λ_k, in which trends revert, against their trend; it gains Sharpe ratio and loses convexity there. The constrained optimum C^+ keeps every direction a trend follower and is convex in the benchmark. The classic CTA is optimal only in the knife-edge case $\Gamma_s \propto PWP$, in which the premium is aligned with the correlation structure. All Sharpe statements in this subsection are first order in the premium; the skewness statements are exact in the benchmark. Part (a) is the trend-following counterpart of the principal portfolios of Kelly et al. (2023), whose prediction matrix plays the role of Γ; the convexity statements are new. Restricted to filters that share the eigenvectors of M, the proposition reduces to a statement about eigen-trend weights.

Corollary 5 (Sharpe-optimal eigen-trend weights)**.** *Let the eigen-trends of Lemma 1 be distributed as under Assumption 1 except for their means, $\mathbb{E}[\Phi_{k,t}] = \pi_k$ (a location shift of their P&L), and let $\mathrm{SR}_k = \pi_k/\mu_k$ be the daily Sharpe ratio of eigen-trend k ($\mu_k > 0$). For the filters C^ω of Corollary 4, the T-day return has the Sharpe ratio and skewness*

$$\mathrm{Sh}(\omega) = \sqrt{T}\,\frac{\sum_k \omega_k\mu_k\,\mathrm{SR}_k}{\left(\sum_k \omega_k^2\mu_k^2\right)^{1/2}}, \qquad \varsigma(\omega) = \varsigma_1(T)\,\frac{\sum_k(\omega_k\mu_k)^3}{\left(\sum_k(\omega_k\mu_k)^2\right)^{3/2}}. \tag{15}$$

Over $\omega \geq 0$, the Sharpe ratio is maximized by $\omega_k \propto \mathrm{SR}_k^+/\mu_k = \pi_k^+/\mu_k^2$, with maximum $\sqrt{T}\,\|\mathrm{SR}^+\|$ and skewness $\varsigma_1(T)/\sqrt{K_3(\mathrm{SR}^+)}$. The classic portfolio is optimal among these filters if and only if $\mathrm{SR}_k^+ = c\mu_k$ for some $c > 0$; with $\pi_k = \bar{\pi}\mu_k$, as in Corollary 4(c), the optimum is the tilt $\alpha = 0$.

Estimation: a risk-premium PCA. The optimal filter needs P^{-1} and Γ_s. Both are estimated with error, and the plug-in $\hat{C}^+$ inverts the smallest eigenvalues of $\hat{P}$, where the error is largest, and weights directions by noisy premia. Lettau and Pelger (2020b,a) face the same problem in factor models of returns and propose a risk-premium PCA (RP-PCA): the eigenvectors of $X^\top X/T_s + \gamma\,\bar{x}\bar{x}^\top$, the second-moment matrix with extra weight γ on the means, which finds factors that explain both the variation and the mean of returns. For a trend follower the premium of a direction is the quadratic form $q^\top \Pi q$ (Lemma 2(b)), so Π takes the place of $\bar{x}\bar{x}^\top$. The RP basis restricts the filter to directions with large variance or large premium, where estimates are reliable, and weights them by Corollary 5: a regularized estimator of C^+.

Proposition 5 (Risk-premium PCA for trend followers)**.** *Let* $\mathrm{tr}(\Pi) > 0$ *and, for* $\gamma \geq 0$*, let* Q_γ *hold the orthonormal eigenvectors, in descending order of eigenvalues, of*

$$M_\gamma = \frac{M}{\mathrm{tr}(M)} + \gamma\,\frac{\Pi}{\mathrm{tr}(\Pi)}. \tag{16}$$

For an orthonormal $N \times m$ *frame* U *let* $\mathcal{V}(U) = \mathrm{tr}(U^\top M U)/\mathrm{tr}(M)$ *and* $\mathcal{R}(U) = \mathrm{tr}(U^\top \Pi U)/\mathrm{tr}(\Pi)$ *be the share of the variance of the eigen-portfolio returns and the share of the trend premium of the eigen-trends spanned by* U*. (a) Lemmas 1(a) and 2 hold for the basis* Q_γ*, and* $\gamma = 0$ *gives the eigen-trends of Lemma 1. (b) The first* m *columns* U_γ *of* Q_γ *maximize* $\mathcal{V}(U) + \gamma\mathcal{R}(U)$ *over all orthonormal* $N \times m$ *frames. (c)* $\mathcal{R}(U_\gamma)$ *is non-decreasing and* $\mathcal{V}(U_\gamma)$ *non-increasing in* γ*.*

Algorithm 1 (Premium-weighted RP eigen-trends)**.** In a training sample, estimate $\hat{M} = W^{1/2}\hat{P}W^{1/2}$ and $\hat{\Pi}$ from $\hat{\Gamma}$, and compute the basis Q_γ of Equation (16), the premia $\hat{\pi}_k = q_k^\top \hat{\Pi} q_k$ (the training means of $\hat{\Phi}_k$, Lemma 2(d)), and the training variances $\hat{v}_k$ of $\hat{\Phi}_k$. Trade the filter $C = W^{1/2} Q_\gamma\,\mathrm{diag}(\omega)\,Q_\gamma^\top W^{1/2}$ with $\omega_k \propto \hat{\pi}_k^+/\hat{v}_k$.

Remark 2 (Estimation). The weights replace μ_k^2 of Corollary 5 by the estimated variance of each eigen-trend and ignore the correlations between eigen-trends, which are zero in the

benchmark only for $\gamma = 0$. The premia are estimated with noise: an eigen-trend with an annual Sharpe ratio of 0.5 has, over ten years of daily data, a t-statistic of about 1.6. Whether the premium information survives estimation, and whether regularization improves on the plug-in $\hat{C}^+$, are out-of-sample questions (Section 6).

3 Data and Methodology

This section describes the data, the construction of the trend-following portfolios, the estimators that map the theory of Section 2 to the data, the out-of-sample protocols, and the inference. All quantities are point in time: a quantity used on day t is computed from data up to day t, and a position decided at the close of day t earns the return of day $t + 1$.

3.1 Data

The main sample is a panel of 84 liquid futures markets in seven asset classes (equity indices 21, government bonds 16, short-term interest rates (STIR) 4, currencies 10, energy 7, metals 6, agriculture 20), daily from July 1959 to July 2026. Each market is a continuous excess-return series built from generic first contracts, so that returns contain no roll gaps and no collateral return (Carver, 2023). The main tests use the 73 markets with data before mid-1999 and evaluate them from January 2001, so that every market has a full history for the volatility and signal estimates; the analysis of skewness over time uses all markets available in each window. Transaction costs follow the one-way, volume-based schedule of Hurst et al. (2017, Exhibit B1).

The second sample is a panel of Bloomberg generic first and second contracts and ratio-adjusted first contracts of 52 commodities (Section D of the Appendix). It was held back until the design of Section 5.1 was fixed and serves as a separately constructed test sample (its markets overlap with the commodities of the main panel), for the decomposition of

returns along the futures curve, and as a second test sample in Section 6. The tests use the 37 commodities with returns before July 1999, evaluated from January 2001 to July 2024, the end of the adjusted series.

3.2 Trend-following systems and portfolios

Normalized returns and signals. Daily returns $r_{i,t}$ are divided by an EWMA volatility with a span of 33 days, estimated from returns up to day $t-1$,

$$z_{i,t} = \frac{r_{i,t}}{\hat{\sigma}_{i,t-1}}, \qquad \hat{\sigma}^2_{i,t} = \nu_\sigma \hat{\sigma}^2_{i,t-1} + (1-\nu_\sigma)\, r^2_{i,t}, \qquad \nu_\sigma = 1 - \frac{2}{34}, \tag{17}$$

which gives every market approximately unit variance in every period. We study the three designs of Sepp and Lucic (2026). The *European* system holds a position proportional to the EWMA signal of Equation (2); the main tests use a single filter with a span of 100 days, and the long–short variant LS(L,S), with weights $\beta_j \propto \nu_L^{j-1} - \nu_S^{j-1} \geq 0$ normalized to $\sum_j \beta_j^2 = 1$, enters with LS(250,20), the reference system of Sepp and Lucic (2026). The *time-series momentum* (TSMOM) system averages the signs of normalized returns over ten periods of ten days (Moskowitz et al., 2012). The *American* system takes a long (short) position when a 20-day price average exceeds (falls below) a 250-day average by more than five average true ranges (ATR) and exits on a trailing stop of five average true ranges (Faith, 2007).

Positions and portfolio. In the European and TSMOM systems the position of market i, per unit of capital, is

$$H_{i,t} = \frac{\sigma^*}{\hat{\sigma}_{i,t}}\, S_{i,t}, \tag{18}$$

so that every market contributes the same risk per unit of signal; the reference system caps the signal at ± 3 and the weights as in Sepp and Lucic (2026), and the American system

fixes its size at entry in proportion to price over average true range. The portfolio return is $r_t^p = \sum_i H_{i,t-1} r_{i,t}$, net of costs $\sum_i \mathrm{tc}_i |H_{i,t} - H_{i,t-1}|$ when stated. The tests of Sections 4, 5, and 6, except the implementable portfolios of Sections 5.2 and 6, use the scale-free P&L $f_{i,t} = S_{i,t-1} z_{i,t}$ of the single-filter European system and the equal-risk portfolio $g_t = \sum_i w_i f_{i,t}$, $w_i = 1/N$, which is the object of Equation (3). Our implementation of the three systems reproduces the reference implementation of Sepp and Lucic (2026) to within 10^{-12} in daily gross and net P&L on all 84 markets (Section C of the Appendix).

Portfolio risk rules. Most CTAs scale their portfolio to a volatility target σ^p. We distinguish two rules, which Corollary 3 separates. The *ex-ante* rule scales the positions H_t decided on day t, which earn the return of day $t+1$, by an EWMA estimate $\hat{\Sigma}_t^r$ (span 60) of the covariance of returns up to day t,

$$\ell_{t+1}^{\mathrm{ante}} = \frac{\sigma^p}{\sqrt{260\, H_t^\top \hat{\Sigma}_t^r H_t}}, \tag{19}$$

and so holds the conditional variance of the portfolio approximately constant. The *ex-post* rule scales the portfolio P&L of day t by $\ell_t^{\mathrm{post}} = \sigma^p / \hat{\sigma}_{t-1}^p$, with $\hat{\sigma}_{t-1}^p$ the EWMA volatility (span 100) of portfolio P&L up to day $t-1$. Without a portfolio rule, risk is targeted only at the instrument level, Equation (18).

3.3 Estimators

Let $G_\tau = \sum_{t=\tau-T+1}^{\tau} g_t$ denote the overlapping T-day sums of a daily P&L series and $\hat{m}_n$ their n-th sample central moment. The main horizon is $T = 55$ days, where the single-market skewness of a 100-day filter peaks (Sepp and Lucic, 2026).

Skewness and the spectral law. Convexity is measured by the sample skewness of horizon returns,

$$\hat{\varsigma}_T = \hat{m}_3/\hat{m}_2^{3/2}. \tag{20}$$

The prediction of Proposition 1 for a portfolio of markets I is the mean single-market skewness of its members times the spectral factor of their sample correlation matrix $\hat{P}_I$,

$$\hat{\varsigma}_T^{\mathrm{SL}} = \Big(\frac{1}{|I|}\sum_{i\in I}\hat{\varsigma}_{T,i}\Big)\frac{\mathrm{tr}\big((W\hat{P}_I)^3\big)}{\mathrm{tr}\big((W\hat{P}_I)^2\big)^{3/2}}, \tag{21}$$

and the breadths $\hat{K}_2$, $\hat{K}_3$, $\hat{K}_4$ are those of Corollary 1 with the eigenvalues of $W^{1/2}\hat{P}_I W^{1/2}$. For a cross-sectional filter C the prediction replaces the spectral factor by $\mathrm{tr}((C\hat{P})^3)/\mathrm{tr}((C\hat{P})^2)^{3/2}$ (Proposition 3); its ex-ante version takes $\hat{P}$ and the single-market skewness from the training sample only. We also report the skewness of calendar-quarter returns, the measure of Sepp and Lucic (2026).

Channels of convexity. Following Remark 1, the channels of Proposition 2 are estimated by

$$\hat{\mathcal{C}}^{\chi} = \frac{3}{\hat{m}_2^{3/2}}\sum_{h=1}^{T-1}(T-h)\,\widehat{\mathrm{Cov}}(\chi_t, g_{t-h}), \qquad \chi_t \in \{g_t^2,\ \hat{V}_t^{\mathrm{pos}},\ \hat{V}_t^{\mathrm{var}},\ \hat{V}_t^{\mathrm{cor}}\}, \tag{22}$$

with $\hat{V}_t^{\mathrm{pos}} = S_{t-1}^{\top} W\hat{P}WS_{t-1}$ for the sample correlation $\hat{P}$, and $\hat{V}_t^{\mathrm{var}}$ and $\hat{V}_t^{\mathrm{cor}}$ the diagonal and off-diagonal parts of $g_t^2 - \hat{V}_t^{\mathrm{pos}} = \sum_{i,k} w_i w_k\big(f_{i,t}f_{k,t} - S_{i,t-1}S_{k,t-1}\hat{P}_{ik}\big)$. The total, $\chi_t = g_t^2$, estimates the martingale part; the difference $\hat{\varsigma}_T - \hat{\mathcal{C}}^{g^2}$ estimates the drift part. Division by $\hat{m}_2^{3/2}$ makes every term a contribution to skewness.

Attribution of convexity. For components G_τ^c that add up to G_τ, the contribution of component c to skewness is

$$\hat{a}_c = \frac{\widehat{\mathrm{Cov}}\big(G_\tau^c, (G_\tau - \bar{G})^2\big)}{\hat{m}_2^{3/2}}, \qquad \sum_c \hat{a}_c = \hat{\varsigma}_T, \tag{23}$$

the sample version of Equation (A1) in Section E.3 of the Appendix. We apply it to three exact decompositions: the eigen-trends $\hat{\Phi}_k$ of Lemma 1 computed with the eigenvectors of $\hat{P}$ (Section 4); the spot and roll parts of commodity returns (Section 5.1); and, for the third central moment, the markets' own moments, the co-moments within asset classes, and the co-moments across asset classes (Section 5). The spot–roll split uses the exact identity

$$\ln \frac{\mathsf{F}_1(t)}{\mathsf{F}_{\text{held}}(t-1)} = \underbrace{\ln \mathsf{F}_1(t) - \ln \mathsf{F}_1(t-1)}_{\text{spot}} + \underbrace{\mathbf{1}\{\text{roll at } t\}\big(\ln \mathsf{F}_1(t-1) - \ln \mathsf{F}_2(t-1)\big)}_{\text{roll, known at } t-1}, \tag{24}$$

with $\mathsf{F}_{\text{held}}(t-1) = \mathsf{F}_2(t-1)$ on roll days and $\mathsf{F}_1(t-1)$ otherwise; roll days are the days on which the ratio-adjusted series changes relative to the unadjusted one. With the normalized parts $z_{i,t}^{\text{spot}}$ and $z_{i,t}^{\text{roll}}$ of Equation (17) and the signal of the total return, $g_t = g_t^{\text{spot}} + g_t^{\text{roll}}$.

Premium matrix and RP basis. The trend cross-moment and the premium matrix of Equation (13) are estimated by

$$\hat{\Gamma} = \frac{1}{T_s} \sum_t z_t S_{t-1}^\top, \qquad \hat{\Pi} = W^{1/2}\, \tfrac{1}{2}\big(\hat{\Gamma} + \hat{\Gamma}^\top\big)\, W^{1/2}, \tag{25}$$

over the T_s days of a training sample; the RP basis is the eigenvectors of $\hat{M}/\mathrm{tr}(\hat{M}) + \gamma\, \hat{\Pi}/\mathrm{tr}(\hat{\Pi})$ with $\hat{M} = W^{1/2}\hat{P}W^{1/2}$, and by Lemma 2(d) the premia $\hat{\pi}_k = q_k^\top \hat{\Pi} q_k$ equal the training means of the eigen-trend P&L. The lead–lag part of the mean of a portfolio with position matrix C is $\sum_{i \neq k} C_{ik}\hat{\Gamma}_{ki}$.

3.4 Portfolio designs and out-of-sample protocols

Breadth and composition. To separate the effect of breadth from that of composition, we form equal-risk portfolios of the 73 markets in four designs: random subsets of N markets (150 for each N); markets from a single asset class; markets drawn round-robin across asset classes; and greedy portfolios that add, at each step, the market that raises $\hat{K}_3$ most. Each

portfolio's realized skewness (20) is compared with its prediction (21); regressions use cells of design and size. The analysis over time uses rolling ten-year windows with all markets available in each window.

Eigen-trend tilting. The tilt of Corollary 4 is implemented point in time. On the first trading day of each month, the correlation matrix of the normalized returns over the preceding 756 days, shrunk by 10% towards the identity, gives the eigenvectors $\hat{Q}$ and eigenvalues $\hat{\mu}_k$; until the next month, the instrument positions are $H_{i,t} = \sigma^* x_{i,t}/\hat{\sigma}_{i,t}$ with

$$x_t = W^{1/2}\hat{Q}\,\mathrm{diag}(\omega)\,\hat{Q}^\top W^{1/2} S_t, \qquad \omega_k = (\hat{\mu}_k/\bar{\mu})^{\alpha-1}, \qquad \alpha \in \{0.5, 1, 1.5, 2\}, \tag{26}$$

where $\bar{\mu}$ is the mean eigenvalue and $\alpha = 1$ is the classic CTA. Markets without a full window are left out for that month.

Risk-premium PCA. Algorithm 1 is estimated in a training sample, 2001–2012 or 2013–2026, and its position matrix is applied without change in the other period. The primary specification is $\gamma = 1$, all basis vectors, and $\omega_k \propto \hat{\pi}_k^+/\hat{v}_k$, normalized so that $\sum_k(\omega_k\hat{\mu}_k^\gamma)^2 = \sum_k(\hat{\mu}_k^\gamma)^2$ with $\hat{\mu}_k^\gamma = q_k^\top \hat{M} q_k$, which gives the weighted portfolio the benchmark variance of the classic one if the correlations between RP eigen-trends are ignored (exactly for $\gamma = 0$); scale does not affect Sharpe ratios or skewness. It is compared with the classic CTA, premium weights on the principal basis ($\gamma = 0$), the tilt $\alpha = 1.5$, and the plug-in optimal filter $\hat{C}^+$ of Proposition 4, computed with $\hat{P}$ shrunk by 10% towards the identity and scaled to the same benchmark variance. The implementable versions re-estimate the filters on the first trading day of every month from the preceding 756 days, as in Equation (26), and trade them net of costs; they include the capped reference system LS(250,20) and the risk-premium PCA applied to its signal.

Benchmarks and performance measures. The backtest of the tilt (Section E of the Appendix) uses as benchmarks LS(250,20); equal risk per asset class with the single filter; the SG Trend Index of Société Générale, an index of large trend-following CTAs, net of fees and including interest on collateral; and a 60/40 portfolio of S&P 500 (ES) and ten-year US Treasury (TY) futures. All trend strategies are scaled to $\sigma^p = 10\%$ with the ex-ante rule (19) unless stated and evaluated from January 2001 to July 2026. Besides Sharpe ratios gross and net of costs and the skewness of monthly, quarterly, and 55-day returns, we report the mean quarterly return in the worst 10% of quarters of the S&P 500 futures and the convexity coefficient c of

$$r_q = a + b\, r_{\mathrm{ES},q} + c\, r_{\mathrm{ES},q}^2 + \epsilon_q, \tag{27}$$

with Newey–West standard errors (Newey and West, 1987). Investor value is measured for a 60/40 portfolio with a 20% sleeve, rebalanced monthly, by its Sharpe ratio, skewness, maximum drawdown, and the annualized certainty equivalent of monthly returns r_m at constant relative risk aversion $\psi = 5$, $\mathrm{CE} = 12\left[(\overline{(1+r_m)^{1-\psi}})^{1/(1-\psi)} - 1\right]$.

3.5 Inference

Sample skewness is noisy: over 25 years, 55-day skewness rests on about 115 non-overlapping periods. We report medians and interquartile ranges across many portfolios and use four resampling devices. A *moving-block bootstrap* of days with blocks of one year (252 or 260 days) preserves the dependence of overlapping windows and across markets; it is used for skewness, the channels, and the spot–roll attribution (300 to 500 draws, as stated in each note). A *stationary block bootstrap* (Politis and Romano, 1994) with a mean block of 63 days is used for differences in Sharpe ratio and skewness between strategies (1,000 draws). *Phase-randomized surrogates* (Prichard and Theiler, 1994) test for state-dependent co-movement: the returns are transformed to the frequency domain, every frequency receives the same random phase shift in all markets, and the series are transformed back, which keeps every

auto- and cross-covariance, and hence volatility, autocorrelation, correlation, and lead–lag relations, but destroys dependence on the state of the trends (200 surrogates). Under Proposition 1, the P&L correlation of two trend followers equals the squared return correlation ρ_{ik}^2, so the excess of the data over the surrogates measures co-trending beyond linear dependence. Finally, 200 *simulations of the static benchmark* with the sample correlation matrix give the sampling distribution of the estimators (22) under Assumption 1 (Exhibit A2).

3.6 Pre-specification and holdout samples

The designs of the channel estimators, the eigen-trend attribution, the backtest, the curve decomposition, and the risk-premium PCA, including primary specifications, metrics, decision rules, and hypotheses, were recorded in a dated research log and committed to version control before they were run; the research question in its present form and the three confirmatory tests of Section B of the Appendix were stated after the results of the risk-premium PCA were known, and the second of them, on skewness differences whose point estimates were known, is treated as descriptive; analyses added after a primary result are labelled post hoc. The results of Section 4 and the P&L correlations of Section E.1 of the Appendix had been computed before the log was started and are exploratory, so the period 2001–2026 of the main panel is not a clean holdout for them. The Sharpe-optimal filter (Proposition 4) and its plug-in estimate were added under a new logged design after the results of the risk-premium PCA were known. Earlier studies documented in the log, a forecast of co-trending and a supervised convexity filter, are not counted among the trials of Section 7.1. The commodity panel was a holdout sample, opened only under a logged entry, and a deviation from its pre-specified roll rule, made before any strategy statistic was computed, is reported in Section B of the Appendix. Results are reported whatever their sign, including hypotheses that the data reject.

4 Diversification and Convexity

The first result answers the question for the classic filter $C = W$: how the number and the composition of its markets move the Sharpe ratio and the skewness of the portfolio, and whether the spectral law predicts both. Exhibit 2 reports skewness and Sharpe ratios of random subsets of N markets, and Exhibit 3 plots them.

	Skewness of 55-day return				
Portfolio	Realized	IQR	Formula	Sharpe ratio	Subsets
Panel A: Random subsets of N markets					
$N = 1$	2.13	1.69–2.48	2.13	0.17	150
$N = 2$	1.77	1.46–2.18	1.55	0.25	150
$N = 3$	1.54	1.25–1.87	1.30	0.39	150
$N = 4$	1.36	1.19–1.74	1.29	0.40	150
$N = 6$	1.25	1.05–1.52	1.14	0.47	150
$N = 8$	1.20	1.00–1.43	1.08	0.48	150
$N = 12$	1.17	0.95–1.35	1.03	0.56	150
$N = 16$	1.07	0.94–1.31	1.03	0.63	150
$N = 24$	1.08	0.93–1.24	1.03	0.63	150
$N = 32$	1.07	0.97–1.19	1.05	0.67	150
$N = 48$	1.06	0.99–1.14	1.06	0.71	150
$N = 64$	1.06	1.02–1.11	1.08	0.72	150
All 73 markets	1.06		1.10	0.72	1
Panel B: Six markets, by asset-class breadth					
One asset class	1.50		1.53	0.27	150
Six asset classes	1.35		1.06	0.62	150

Exhibit 2: **Diversification and the Skewness of Trend-Following Portfolios.** 73 futures with data before mid-1999, January 2001–July 2026. Each market carries the scale-free P&L $S_{t-1}z_t$ of a single EWMA filter (span 100 days) on volatility-normalized returns, gross of costs, with equal risk and no portfolio volatility targeting. Realized: median across subsets of the sample skewness of overlapping 55-day sums; IQR: interquartile range. Formula: mean single-market skewness of the subset times $\mathrm{tr}(P^3)/\mathrm{tr}(P^2)^{3/2}$ (Proposition 1), with P the correlation of the volatility-normalized returns over the sample. The median single-market skewness is 2.13. Rank correlation of realized and predicted skewness across the random subsets: 0.62. Sharpe breadth $K_2 = (\sum \mu)^2 / \sum \mu^2$ of all 73 markets: 14.5; convexity breadth $K_3 = (\sum \mu^2)^3/(\sum \mu^3)^2$: 3.8 (Corollary 1).

A single market has a median skewness of 2.13, close to the closed form of 2.35 for white noise. Skewness falls quickly as markets are added, to 1.36 with four markets and 1.20 with eight, and then stops falling: from 16 markets on, the median stays between 1.06 and 1.08, and the portfolio of all 73 markets has a skewness of 1.06. Independent markets would imply 0.27 at 64 markets. The Sharpe ratio, in contrast, keeps rising, from 0.17 for a single market

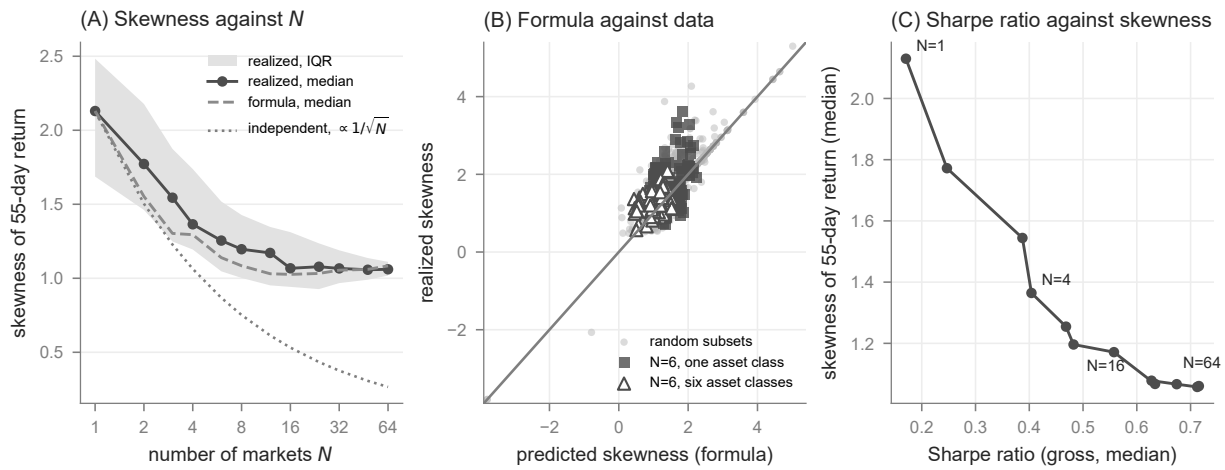


Exhibit 3: **The Convexity Cost of Diversification.** Panel A: median and interquartile range of the realized skewness of the 55-day return across 150 random subsets of N markets, the median prediction of Proposition 1, and the $1/\sqrt{N}$ benchmark for independent markets. Panel B: realized against predicted skewness for every subset, with the six-market portfolios drawn from one or from six asset classes. Panel C: median Sharpe ratio against median skewness by N. Data and design as in Exhibit 2.

to 0.63 at 16 markets and 0.72 at 64. Proposition 1 predicts the plateau: 1.10 for the full universe. The prediction is the level the common trends allow; Corollary 1 translates it into breadth. The 73 markets offer a Sharpe breadth of 14.5 but a convexity breadth of only 3.8; the first-order Sharpe ratio $\mathrm{SR}_1\sqrt{K_2} = 0.17\sqrt{14.5} = 0.65$ matches the Sharpe ratios of 0.63 at 16 and 0.72 at 64 markets, and the ratio of portfolio to single-market skewness of about one half is that of four independent trends.

Proposition 1 also implies that the ratio of portfolio to single-market skewness does not depend on the horizon. At horizons from 5 to 252 days the slope of the ratio on log horizon is not significant, and at the main horizon the all-market ratio (0.49) is close to the formula (0.51) (Exhibit A12 in Section E.3 of the Appendix).

If breadth worked through the number of markets, portfolios with the same N would have the same skewness. Exhibit 4 shows that they do not. Six markets from one asset class keep 76% of single-market skewness; six markets chosen to maximize convexity breadth keep 47%. Across the 34 cells of design and size, convexity breadth explains 80% of the variation of the ratio and the number of markets 52%; with both, the coefficient on $1/\sqrt{K_3}$ is 0.60

(95% interval 0.27 to 0.94) and that on $1/\sqrt{N}$ is 0.15 (−0.23 to 0.52). The greedy portfolios make the point sharply: after about 12 markets only correlated markets remain to be added, convexity breadth falls, and skewness rises again although the portfolios grow (Exhibit A13, Panel B, of the Appendix). Portfolios within one asset class lie on the prediction; portfolios spread across classes lie above it, a first sign of the co-trending studied in Section 5. The coefficient on $1/\sqrt{K_3}$ is below one for the same reason.

Panel A: Ratio of portfolio to single-market skewness on $1/\sqrt{N}$ and $1/\sqrt{K_3}$ (cell medians)

	Estimate	95% interval	Theory
Intercept	0.24	[-0.05, 0.56]	0
$1/\sqrt{N}$	0.15	[-0.23, 0.52]	0
$1/\sqrt{K_3}$	0.60	[0.27, 0.94]	1
R^2: N only / K_3 only / both	0.52 / 0.80 / 0.82		

Panel B: Same number of markets, different composition (medians)

Design	N	K_3	Formula $1/\sqrt{K_3}$	Realized ratio
One asset class	4	1.4	0.84	0.82
Random	4	3.3	0.55	0.65
Round-robin across classes	4	3.4	0.54	0.66
Greedy maximum K_3	4	4.0	0.50	0.52
One asset class	6	1.5	0.82	0.76
Random	6	3.4	0.55	0.62
Round-robin across classes	6	4.3	0.48	0.58
Greedy maximum K_3	6	5.9	0.41	0.47
One asset class	8	1.5	0.82	0.77
Random	8	4.1	0.49	0.58
Greedy maximum K_3	8	7.6	0.36	0.45

Exhibit 4: **Composition, Not Count.** 73 markets, design and sample as in Exhibit 2. Portfolios: random subsets; markets from a single asset class; markets drawn round-robin across asset classes; and greedy portfolios that add, at each step, the market that raises the convexity breadth $K_3 = (\sum \mu^2)^3/(\sum \mu^3)^2$ most. Dependent variable: skewness of the 55-day portfolio return divided by the mean single-market skewness of the members. Panel A regresses its median in each cell of design and N, because the skewness of a single portfolio is noisy (on single portfolios the R^2 of both regressors is 0.09). Under Proposition 1 with equal risk the ratio equals $1/\sqrt{K_3}$. 95% intervals: moving-block bootstrap of the daily P&L (blocks of 260 days, 300 replications), portfolios held fixed. 1,640 portfolios in 34 cells.

The same holds over time. Exhibits 5 and 6 follow rolling 10-year windows from 1984 to 2026, with all markets available in each window. The universe grows from 31 to 81 markets, but convexity breadth does not grow with it: it rises to about 8 in the windows ending around 2001, falls to 2.6 in 2016, when correlations across asset classes were high, and recovers to 4.5 (Exhibit 6). Portfolio skewness moves inversely: it is below 0.6 in nine of

the ten windows ending between 1998 and 2007, the most diversified ones, with a low of 0.33. Across windows, the correlation of portfolio skewness with convexity breadth is −0.70 and with the prediction 0.67, while single-market skewness stays between 1.4 and 1.9 throughout. Over time, as across portfolios, skewness follows the composition of the universe, not its size.

				Skewness of 55-day return		
10-year window ending	N	K_2	K_3	Single (median)	Portfolio	90% band
Dec 1984	31	10.4	3.7	1.82	0.99	[0.54, 1.24]
Dec 1990	38	12.7	4.7	1.81	0.86	[0.36, 1.25]
Dec 1995	43	14.7	5.4	1.81	1.28	[0.51, 1.28]
Dec 2000	63	22.0	7.7	1.74	0.54	[0.19, 0.64]
Dec 2005	71	19.1	5.7	1.67	0.56	[0.19, 0.78]
Dec 2010	73	13.6	3.5	1.47	0.82	[0.21, 1.19]
Dec 2015	74	12.3	2.8	1.67	1.18	[0.56, 1.53]
Dec 2020	81	14.8	3.3	1.86	1.27	[0.44, 1.64]
Jul 2026	81	16.1	4.5	1.79	1.28	[0.61, 1.79]

Growing universe, all available markets: 1975–2026 0.75; 2001–2026 1.08. Across windows: corr(K_3, portfolio) -0.70; corr(formula, por

Exhibit 5: **Skewness over Time.** Rolling 10-year windows; in each window the markets with trend P&L on at least 95% of the days (single EWMA filter, span 100 days, gross, equal risk, no portfolio volatility targeting). $K_2 = (\sum \mu)^2 / \sum \mu^2$ and $K_3 = (\sum \mu^2)^3 / (\sum \mu^3)^2$ from the window correlation of volatility-normalized returns. Formula: mean single-market skewness times $\mathrm{tr}(P^3)/\mathrm{tr}(P^2)^{3/2}$ (Proposition 1). 90% band: moving-block bootstrap of the portfolio P&L (blocks of 260 days, 300 replications). The table shows selected windows; the figure shows all.

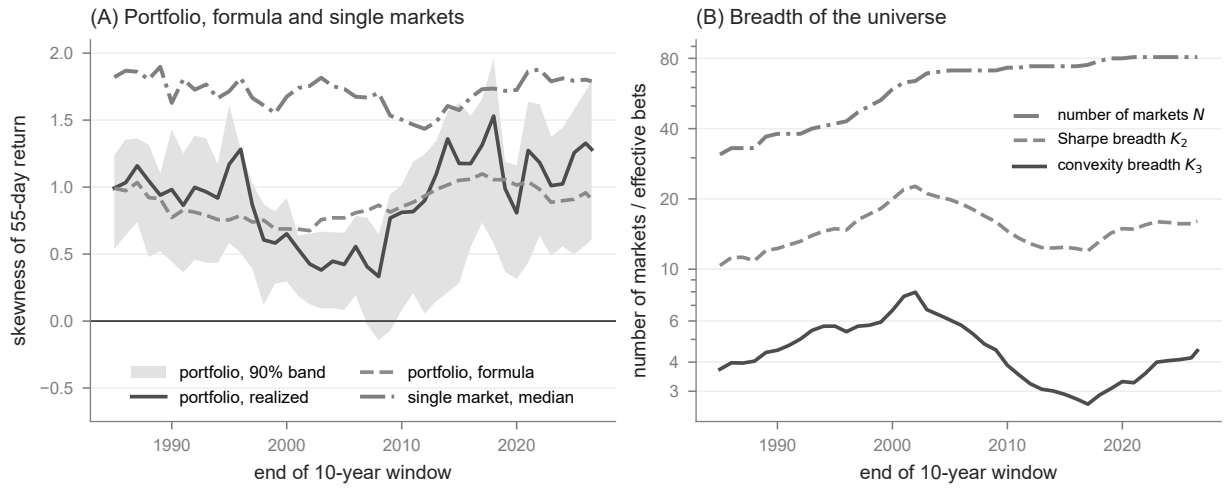


Exhibit 6: **Skewness and Breadth over Time.** Rolling 10-year windows. Panel A: realized skewness of the equal-risk portfolio with a 90% block-bootstrap band, the prediction of Proposition 1, and the median single-market skewness. Panel B: number of markets, Sharpe breadth K_2, and convexity breadth K_3. Design as in Exhibit 5.

One market is an outlier: the Swiss franc has a skewness of −3.8, the result of a single

day, 15 January 2015, when the Swiss National Bank removed its exchange-rate floor and the franc moved 37 standard deviations against the prevailing trend. Proposition 1 assumes Gaussian returns and does not cover such jumps.

For the classic filter the answer is thus that diversification buys Sharpe ratio at the price of convexity until the convexity breadth is exhausted, at about 16 markets here, and costs little convexity thereafter. Convexity sits in the strongest eigen-trends (Corollary 2): the first carries 51% of the variance but 71% of the convexity, all eigen-trends beyond the fifth 12% of the variance but 2% of the convexity, and realized contributions follow the prediction (Exhibit A14 in Section E.3 of the Appendix). One observation does not fit the benchmark: portfolios spread across asset classes are more skewed than predicted. Section 5 explains it and asks what decides whether this convexity reaches the investor.

5 What Breaks the Benchmark: Co-Trending and Risk Targeting

The second result concerns the departures from the benchmark that decide whether the convexity of a filter reaches the investor. Six markets from six asset classes have a skewness of 1.35, well above the prediction of 1.06 (Exhibit 2, Panel B), and portfolios spread across classes lie above the prediction in Exhibit 4. Proposition 1 uses the unconditional correlation of returns; if correlations rise when trends align, as in a crisis in which bonds rally, equities fall, and a diversified trend follower is long the one and short the other, P&L co-moves more than the unconditional correlation implies, and exactly when P&L is large. Two descriptive facts point in this direction (Section E of the Appendix). Across asset classes, the P&L correlation of two trend followers is 0.054, against 0.021 implied by their return correlations, and phase-randomized surrogates, which keep all linear dependence, reproduce only 0.006 of the excess (Exhibit A6). And three quarters of the skewness of the 73-market portfolio

is co-skewness across asset classes, concentrated in 2008 and 2020, although in the third moment this part cannot be separated from sampling noise (Exhibit A8). Proposition 2 turns the description into a measurement.

Panel A: Channels of the skewness of the 55-day portfolio return

Component	Estimate	Bootstrap (90%)	Surrogate mean	Surrogates (95%)	p
Realized skewness (overlapping 55-day sums)	1.06	[0.63, 1.35]	1.30	[0.71, 2.28]	0.705
Martingale part (Proposition 2)	1.79	[1.22, 2.23]	0.86	[0.69, 1.06]	<0.005
Position channel	0.74	[0.58, 0.86]	0.78	[0.64, 0.94]	0.675
Co-movement channel	1.05	[0.60, 1.42]	0.08	[-0.01, 0.19]	<0.005
of which variance	0.06	[0.04, 0.08]	0.01	[0.00, 0.01]	<0.005
of which correlation	0.98	[0.54, 1.37]	0.08	[-0.01, 0.18]	<0.005
Drift part (realized minus martingale part)	-0.73	[-1.14, -0.24]	0.44	[-0.26, 1.45]	1.000
Same-day third moments	0.00	[-0.06, 0.07]	0.01	[-0.00, 0.02]	0.900
Squared P&L followed by P&L	-0.07	[-0.44, 0.39]	0.09	[-0.05, 0.23]	0.985
P&L on three distinct days	-0.67	[-1.24, -0.11]	0.34	[-0.27, 1.27]	1.000
Proposition 1 (static formula)	1.10				

Panel B: Years with the largest correlation channel

Year	Correlation	Position	Variance
2008	0.30	0.14	0.01
2020	0.28	0.11	0.01
2002	0.06	0.05	0.00
2011	0.05	0.04	0.00
2022	0.04	0.03	0.00

Exhibit 7: **The Two Channels of Convexity.** 73 markets, January 2001–July 2026, equal-risk portfolio of single-filter trend P&L (span 100 days), 55-day horizon. Proposition 2 writes the third moment of the horizon return as $3\sum_h (T-h)\,\mathrm{Cov}(\chi_t, g_{t-h})$, with g_t the daily portfolio P&L. Total: $\chi_t = g_t^2$. Position channel: $\chi_t = V_t^{\mathrm{pos}} = S_{t-1}^\top W \bar{P} W S_{t-1}$, the conditional variance with the static correlation $\bar{P}$ of the sample. Co-movement channel: $\chi_t = g_t^2 - V_t^{\mathrm{pos}}$, split exactly into its diagonal (variance) and off-diagonal (correlation) parts. All terms are divided by the cubed standard deviation of overlapping 55-day sums. Drift part: the remaining terms of the third moment, which vanish when the P&L is a martingale difference (Proposition 2). Bootstrap: moving blocks of 252 days, 500 draws. Surrogates: 200 multivariate phase-randomized versions of the returns, which keep all linear dependence and remove state-dependent co-movement; p: share of surrogates with at least the estimate (a value of one: below all surrogates). Panel B: contributions by the calendar year of the later date t.

Exhibits 7 and 8 measure the channels of the 73-market portfolio. The martingale part of the third moment amounts to a skewness of 1.79 (90% bootstrap interval 1.22 to 2.23). The position channel contributes 0.74 (0.58 to 0.86) and is reproduced by the surrogates (mean 0.78, $p = 0.68$): it is the mechanical convexity of trend following, which needs nothing beyond the linear dependence of returns. The co-movement channel contributes 1.05 (0.60 to 1.42), almost all of it through correlations (0.98, against 0.06 through variances, which

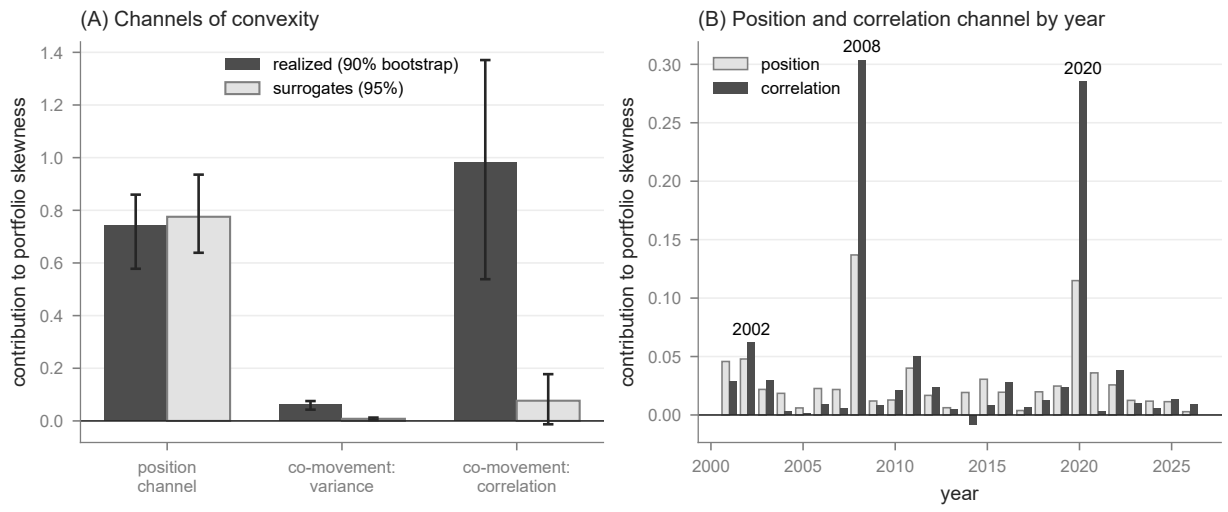


Exhibit 8: **The Two Channels of Convexity.** Panel A: position channel and the variance and correlation parts of the co-movement channel of Proposition 2, with 90% block-bootstrap intervals, and their means and 95% ranges in 200 phase-randomized surrogates. Panel B: position and correlation channel by calendar year. Design as in Exhibit 7.

the volatility normalization keeps small). In none of the 200 surrogates does the correlation part reach the realized value; because phase randomization also makes the margins close to Gaussian, the excess reflects joint large moves as well as state-dependent correlation; the surrogates average 0.08, and in the static benchmark the co-movement channel averages 0.01 with a standard deviation of 0.09 (Exhibit A2). By Definition 1, the portfolio co-trends: after it has gained, correlations rise in the direction of its positions. The correlation channel comes from crises, with 0.30 from 2008 and 0.28 from 2020, while the position channel is spread over the sample.

The drift part is -0.73 (bootstrap interval -1.14 to -0.24) and comes almost entirely from products of P&L on three distinct days (-0.67): the trend P&L is predictably weak after large moves, consistent with trend reversals after crises. This estimate is imprecise. In the static benchmark, where the true drift part is zero, the same estimator has a mean of -0.19, reflecting the small-sample bias of skewness, and a 90% range of -0.79 to 0.65 (Exhibit A2). The realized skewness of 1.06 is therefore close to the static formula (1.10) not because the static model holds, but because co-trending adds convexity that a negative,

noisy drift part offsets. Net of the drift part, the realized skewness is not above that of the surrogates (mean 1.30, $p = 0.70$): co-trending raises the martingale part, but in this panel trend reversals after crises give back what it adds. On the commodity panel, where the drift part is small (-0.07), realized skewness exceeds the static law (1.65 against 1.36). The martingale part is also the more reliable measure: in the static benchmark its standard deviation is 0.15, against 0.39 for the sample skewness of overlapping sums.

5.1 Predictable drift: the futures curve

The drift part of Proposition 2 has an observable component in commodity futures, where the return of a rolled futures position splits exactly into the change of the front price, the spot part, and the roll yield, the gap between the expiring and the next contract, which is known one day before it is earned (Erb and Harvey, 2006; Gorton and Rouwenhorst, 2006; Koijen et al., 2018). By Proposition 2 the roll yield moves the mean of the trend P&L but enters the third moment only through the drift part. On a separately constructed panel of 37 commodities, January 2001 to July 2024, the roll yield earns 61% of the mean P&L of the classic trend portfolio (90% interval 42% to 92%) but contributes -0.02 (-0.11 to 0.05) to its skewness of 1.65, which comes from the spot part (Exhibit A16 in Section E.4 of the Appendix). Premium and convexity can therefore sit in different places, so a filter chosen for premium need not be chosen for convexity, which Section 6 examines. Before turning to the filter, we ask whether the convexity of the classic portfolio survives the way CTAs target risk.

5.2 Risk targeting

Both channels work through the conditional variance of the positions, and Corollary 3 implies that a portfolio rule which holds it constant removes them. Exhibit 9 tests the implication on implementable portfolios, January 2001 to July 2026, net of costs: the classic single filter,

its eigen-trend tilts of Corollary 4, and the capped reference system LS(250,20), each under three portfolio risk rules. The comparison was added after the pre-specified backtest of Section E of the Appendix, in which every strategy was scaled with the ex-ante rule and every skewness was close to zero.

Strategy	Risk rule	Sharpe net	Skewness Month	Skewness Quarter	Skewness 55-day	Crisis (%)	Convexity c (t)
Eigen-trend tilt $\alpha = 0.5$	none	-0.40	0.72	0.85	0.80	4.3	1.64 (3.7)
	ex post (P&L)	-0.35	0.56	0.77	0.60	4.3	1.65 (4.2)
	ex ante (positions)	-0.58	0.02	0.10	0.00	3.6	1.32 (2.5)
Classic equal-risk CTA ($\alpha = 1$)	none	0.22	1.08	1.43	1.35	8.8	2.26 (4.8)
	ex post (P&L)	0.28	0.76	0.93	0.79	7.9	1.83 (4.5)
	ex ante (positions)	0.10	0.03	0.21	0.04	7.3	1.67 (3.0)
Eigen-trend tilt $\alpha = 1.5$	none	0.29	1.46	2.02	1.90	10.1	2.40 (5.1)
	ex post (P&L)	0.36	1.01	0.91	1.01	9.0	1.87 (4.9)
	ex ante (positions)	0.25	0.07	0.22	0.07	8.3	1.70 (3.0)
Eigen-trend tilt $\alpha = 2$	none	0.22	1.69	2.40	2.18	9.9	2.30 (4.9)
	ex post (P&L)	0.30	1.17	0.94	1.20	9.2	1.88 (5.3)
	ex ante (positions)	0.20	0.10	0.17	0.06	8.1	1.65 (3.1)
Reference system LS(250,20)	none	0.63	0.13	0.49	0.41	7.2	1.19 (2.0)
	ex post (P&L)	0.81	0.18	0.45	0.35	7.2	1.00 (1.8)
	ex ante (positions)	0.87	0.06	0.20	0.07	7.7	1.02 (2.0)

Exhibit 9: **Risk Targeting and Convexity.** Strategies, sample, costs, and measures as in Exhibit A10, with three portfolio risk rules. None: instrument risk targeting only, the return series scaled to 10% volatility over the sample. Ex post: the portfolio P&L scaled by its own lagged EWMA volatility (span 100) to 10%. Ex ante: the rule of Exhibit A10, which scales the current positions by an EWMA estimate (span 60) of their covariance and so holds the conditional variance V_t of Proposition 2 approximately constant (Corollary 3). Added after the primary result (Section B of the Appendix).

Without a portfolio rule, the skewness of the 55-day return rises with the tilt as Corollary 4(b) predicts, from 0.80 at $\alpha = 0.5$ to 1.35, 1.90, and 2.18 at $\alpha = 1$, 1.5, and 2; relative to $\alpha = 1$, the realized ratios of 0.59, 1.40, and 1.61 compare with predicted ratios of 0.41, 1.42, and 1.63. Crisis returns rise with it, from 4.3% at $\alpha = 0.5$ to about 10% a quarter at $\alpha \geq 1.5$ in the worst decile of equity quarters. The ex-ante rule, which scales the current positions by an estimate of their covariance and so approximates the conditional-variance target of Corollary 3(b), removes this convexity: the classic CTA keeps a 55-day skewness of 0.04 of its 1.35, the reference system 0.07 of 0.41. The ex-post rule, which scales by the realized volatility of past P&L and reacts to the conditional variance only with a lag, keeps

more than half (0.79 for the classic CTA, 1.20 for $\alpha = 2$) and gives the highest Sharpe ratio of the three rules for every single-filter portfolio. The convexity coefficient of the quarterly smile tells the same story (2.26 without targeting, 1.83 ex post, 1.67 ex ante, for $\alpha = 1$). How a program targets risk thus decides whether it delivers the convexity its market set offers.

Because the comparison was added after the primary backtest, we repeated it with predictions fixed in advance on the commodity panel (Exhibit A18 in Section E.5 of the Appendix). Without a rule the classic commodity trend follower has a 55-day skewness of 1.65; the ex-ante rule lowers it to 0.43 (difference -1.22, 90% interval -1.75 to -0.46) and the ex-post rule to 0.72 (-0.93, -1.35 to -0.17), at Sharpe ratios of 1.13, 1.15, and 1.23. The ordering is as predicted, but both thresholds are narrowly missed: the ex-ante rule keeps 26% of the skewness against a predicted at most 25%, slightly more than predicted, and the ex-post rule 43% against a predicted at least 50%, less than predicted. The answer to the second part of the question is that co-trending adds to the martingale part of convexity, trend reversals after crises take part of it back, and the portfolio risk rule decides how much survives: a rule that targets the risk of the current positions removes most of it.

6 Choosing the Cross-Sectional Filter

The third result asks whether a filter other than the classic one can raise the Sharpe ratio without giving up convexity. The simplest alternative, the eigen-trend tilt of Corollary 4, moves convexity as predicted (Section 5.2) but not the Sharpe ratio, because its condition (c), a premium proportional to risk, fails: gross of costs, the Sharpe ratio of the tilted single filter is 0.59, 0.80, 0.79, and 0.69 for $\alpha = 0.5$, 1, 1.5, and 2, against 1.43, 0.80, 0.48, and 0.35 if the condition held, and the five leading eigen-trends earn 98% of the mean P&L with 77% of its variance (Exhibit A3 of the Appendix). In the pre-specified backtest the effect of the tilt on skewness could not be seen, because its ex-ante volatility target removes the convexity

of every strategy (Section E of the Appendix). The premium has its own structure, and Propositions 4 and 5 say how to use it. Exhibit 10 tests them on a design fixed in advance, and Exhibit 11 places every filter on the map of Sharpe ratio and skewness that the question is about.

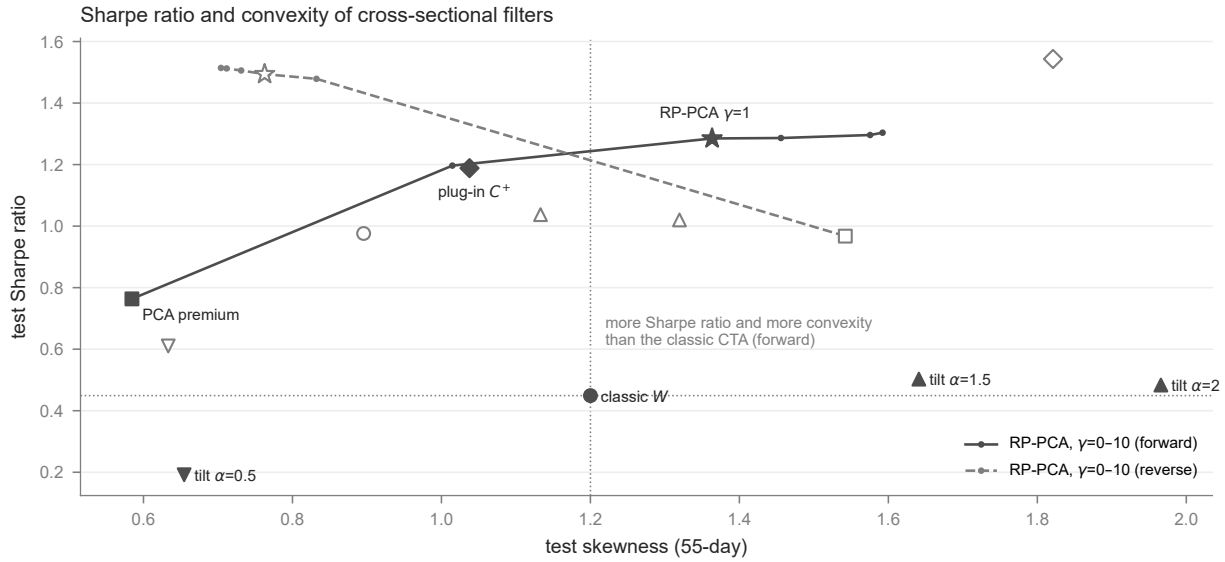


Exhibit 11: **Sharpe Ratio and Convexity of Cross-Sectional Filters.** Test-sample Sharpe ratio against test-sample skewness of the 55-day return for every filter of Exhibit 10, the tilts $\alpha = 0.5$ and 2, and the premium-weighted RP eigen-trends for γ from 0 to 10; filled markers: train 2001–2012, test 2013–2026; open markers: the reverse split; dotted lines: the classic CTA in the forward split, so that filters above and to the right of both lines have a higher Sharpe ratio and a higher skewness; open markers compare with the open circle, the classic CTA in the reverse split. Design as in Exhibit 10.

By construction the RP basis trades variance for premium (Proposition 5(c)): estimated in 2001–2012, its first five vectors carry 39% of the variance of the eigen-portfolio returns and 176% of the premium, against 50% and 98% for principal components. Out of sample the concentration persists in 2013–2026 (151% against 99%) but not in the reverse split (Exhibit A19 of the Appendix).

Panel B of Exhibit 10 reports the filters. Premium weighting on the principal basis (S1) raises the test Sharpe ratio of the classic CTA (S0) from 0.45 to 0.76 in 2013–2026 and leaves it unchanged in 2001–2012 (0.97 against 0.98). Premium weighting on the RP basis (S2) raises it to 1.28 and 1.49. The differences to the classic CTA, 0.84 and 0.52, and to S1,

0.52 and 0.53, have 90% bootstrap intervals above zero in both splits; by the rule fixed in advance, the risk-premium PCA improves on principal components and on the classic CTA, and every γ from 0.5 to 10 gives a similar result (Exhibit A4). The plug-in optimal filter $\hat{C}^{+}$ (S4) reaches 1.19 and 1.54. Out of sample it cannot be told apart from S2 (differences 0.10 and -0.05, intervals -0.27 to 0.45 and -0.57 to 0.43), although its in-sample Sharpe ratios of 5.8 and 5.2 show how much of the training sample it fits; the point estimates favour the regularized filter in one split and the plug-in in the other, and neither difference is significant; with yearly re-estimation the plug-in does better before costs (Section 7.1). The tilt (S3) changes the Sharpe ratio little.

Lemma 2 identifies where the gain comes from. The classic CTA earns only the own-market premia; its lead–lag share is zero by construction. Of the test-sample mean of S2, 61% (forward) and 25% (reverse) are lead–lag premia, earned by trading each market partly on the signals of the others: the RP basis finds directions in which trends in one market predict returns in another, the cross-asset predictability of Pitkäjärvi et al. (2020). Principal components do not; on the principal basis the lead–lag premia of the weighted portfolio are negative.

Panel C of Exhibit 10 and Exhibit 11 answer the second half of the question. The premium-aware filters lie above the classic CTA on the map. Their convexity cost is not measurable with this sample: the test-sample skewness of S2 is 1.36 against 1.20 for the classic CTA in 2013–2026 and 0.76 against 0.90 in 2001–2012 (differences 0.16, 90% interval -0.81 to 0.66, and -0.13, -0.44 to 0.38), and that of S4 is 1.04 and 1.82 (differences -0.16, -0.80 to 0.30, and 0.93, 0.08 to 1.34). The intervals are wide enough to admit a cost of up to 0.8 as well as a gain, and premium weighting on the principal basis (S1) halves the skewness in 2013–2026 (0.58 against 1.20) while raising it in 2001–2012 (1.54 against 0.90). The spectral law of Proposition 3 predicts the skewness of the risk-based filters, the classic CTA and the tilt, to within 0.3 in both splits, and its ex-ante version, computed from the training sample alone, differs from it by at most 0.15. For the premium-weighted filters it is

too low in three of six cases, by 0.69 to 1.07; across all filters, realized skewness is at least the prediction in eight of ten cases. This is the direction Section 5 implies: a filter chosen for its trend premium loads on the directions in which trends pay, and the static law, which assumes no predictability, leaves out the channels through which they add convexity; we have not estimated the channels filter by filter, so this reading is consistent with Section 5 rather than tested. The test of the law on these filters was fixed in advance and fails as formulated (Section B of the Appendix); the spectrum gives the convexity of a filter in the benchmark, not a forecast for filters selected on premia. Corollary 5 adds why the sign of the difference to the classic CTA is not fixed: the skewness of a premium-weighted filter depends on the dispersion of the eigen-trend Sharpe ratios, which differs between periods.

Which diversification does the better filter choose? The ex-post attribution in Exhibit A20 (Section E.7 of the Appendix) answers this. In the spectrum of CP, the premium-aware filter S2 holds slightly fewer Sharpe bets than the classic CTA (13.1 against 15.2 in 2013–2026, 12.3 against 13.0 in 2001–2012) but about twice as many convexity bets (8.8 against 4.3, 6.7 against 3.2); the tilt is the concentrated filter, with 2.1 and 1.9 convexity bets, and in 2013–2026 the most skewed. Among the risk-based filters, skewness thus falls with convexity breadth, as the first result implies; the premium-weighted filters are more skewed than their convexity breadth implies, the departure found above. The better filter also changes where the P&L comes from: in 2013–2026 equities turn from -23% of the mean P&L of the classic CTA to $+18\%$, and the share of energy in the third moment rises from 26% to 41%. The premium-aware filter does not buy its Sharpe ratio by concentrating; it trades different directions.

Panel E of Exhibit 10 implements the filters with monthly re-estimation from the preceding three years, net of costs. On the single filter, RP-PCA earns a Sharpe ratio of 1.23 gross and 0.53 net without a portfolio target, and 1.31 and 0.49 with the ex-ante target, against 0.68 and 0.22, and 0.80 and 0.10, for the classic CTA. The plug-in $\hat{C}^+$ earns as much before costs (1.28 and 1.34) but trades 2.6 times as much as RP-PCA without and 1.5 times as

much with the ex-ante target, and keeps 0.29 and 0.24 after costs: in implementation, regularization pays through the stability of the estimated filter. The comparison that matters to practitioners is with the capped reference system LS(250,20), which earns 0.63 net without and 0.87 net with the ex-ante target. Applied to the same signal, RP-PCA earns 0.84 and 0.90 net (1.04 and 1.14 gross); its skewness without a portfolio target is 0.98, against 0.41. The gain without a portfolio target is 0.21 (90% bootstrap interval 0.01 to 0.41; two-sided Ledoit–Wolf $p = 0.071$); with the ex-ante target it is 0.03 (-0.14 to 0.20), so a portfolio volatility target on the reference system captures most of what the filter adds. Without a portfolio target the skewness of the reference system rises by a point estimate of 0.57, from 0.41 to 0.98, with an interval that touches zero (-0.01 to 0.83). For practice this is the useful comparison: RP-PCA on the reference signal without a portfolio target earns about the net Sharpe ratio that the reference system reaches only with the ex-ante target (0.84 against 0.87), but keeps a skewness of 0.98 instead of 0.07; on the single filter its skewness is 0.99 against 1.35 for the classic CTA (-0.36, -0.77 to 0.11).

The commodity panel bounds the result (Panel D). There, premium weighting does not help: S2 earns test Sharpe ratios of 0.67 and 1.14, against 0.96 and 1.27 for the classic CTA, and its lead–lag share is close to zero (-5% and 0%). Commodity trends rarely lead each other, so the filter has only own-market premia to work with, and their estimates are noisy. The risk-premium PCA improves the cross-sectional filter where the premium has cross-sectional structure, as in a diversified universe of financial and commodity futures; it is not a free lunch in a single asset class.

The answer to the third part of the question is therefore yes for the Sharpe ratio; for convexity, the data rule out only large losses. A filter that weights directions by trend premium rather than by variance raises the Sharpe ratio of the classic CTA significantly, and its effect on convexity is not distinguishable from zero in either direction; it does so by trading different directions, with more convexity bets, not by concentrating. Implemented net of costs on the signal of a capped reference system, it delivers the Sharpe ratio that the

reference system reaches only with an ex-ante volatility target while keeping the skewness that this target removes. The gain needs cross-sectional structure in the premium, which commodities alone lack, and an ex-ante volatility target removes both the convexity and most of the gain. It is not an artefact of the two splits: with yearly re-estimation the filter beats the classic CTA in 21 of 23 years, and in simulated data without predictability the same estimators produce no comparable gain (Section 7.1).

Panel A: Shares of the first five basis vectors (%)	Train 2001–2012, test 2013–2026				Train 2013–2026, test 2001–2012			
	Mean		Variance		Mean		Variance	
	Train	Test	Train	Test	Train	Test	Train	Test
PCA ($\gamma = 0$)	98	99	94	88	128	107	91	93
RP-PCA ($\gamma = 1$)	176	151	94	85	343	77	84	65
Panel B: Strategies with positions fixed in the training sample								
	Sharpe ratio		Skewness (55-day)		Sharpe ratio		Skewness (55-day)	
	Train	Test	Train	Test	Train	Test	Train	Test
S0 Classic equal-risk CTA	0.98	0.45	0.90	1.20	0.45	0.98	1.20	0.90
S1 PCA, premium-weighted ($\gamma = 0$)	2.13	0.76	1.03	0.58	1.36	0.97	1.01	1.54
S2 RP-PCA, premium-weighted ($\gamma = 1$)	3.00	1.28	0.46	1.36	2.97	1.49	0.46	0.76
S3 Eigen-trend tilt $\alpha = 1.5$	0.97	0.50	1.23	1.64	0.55	1.04	1.51	1.13
S4 Plug-in optimal filter C^+	5.79	1.19	0.63	1.04	5.16	1.54	1.12	1.82
S2 − S0, test Sharpe (90%)		0.84	[0.54, 1.12]			0.52	[0.28, 0.79]	
S2 − S1, test Sharpe (90%)		0.52	[0.18, 0.87]			0.53	[0.06, 1.01]	
S2 − S4, test Sharpe (90%)		0.10	[-0.27, 0.45]			-0.05	[-0.57, 0.43]	
Lead–lag share of the test mean (%), S0 / S1 / S2 / S3 / S4	0 / -36 / 61 / 75 / 27				0 / -237 / 25 / 75 / -84			
Panel C: Convexity of the filters, test-sample skewness of the 55-day return								
	Law	Ex ante	Realized		Law	Ex ante	Realized	
S0 Classic equal-risk CTA	0.95	0.98	1.20		0.98	0.95	0.90	
S1 PCA, premium-weighted ($\gamma = 0$)	0.58	0.49	0.58		0.56	0.60	1.54	
S2 RP-PCA, premium-weighted ($\gamma = 1$)	0.67	0.59	1.36		0.68	0.65	0.76	
S3 Eigen-trend tilt $\alpha = 1.5$	1.36	1.34	1.64		1.28	1.27	1.13	
S4 Plug-in optimal filter C^+	0.89	0.75	1.04		0.75	0.85	1.82	
S2 − S0, realized skewness (90%)			0.16	[-0.81, 0.66]			-0.13	[-0.44, 0.38]
S4 − S0, realized skewness (90%)			-0.16	[-0.80, 0.30]			0.93	[0.08, 1.34]
Panel D: Commodity panel (train 2001–2012 / test 2013–2024, and reverse)								
S0 Classic equal-risk CTA	1.27	0.96	1.70	1.25	0.96	1.27	1.25	1.70
S1 PCA, premium-weighted ($\gamma = 0$)	2.33	0.55	0.86	1.06	1.60	0.67	0.55	4.24
S2 RP-PCA, premium-weighted ($\gamma = 1$)	2.92	0.67	0.58	0.67	2.63	1.14	0.80	2.27
S3 Eigen-trend tilt $\alpha = 1.5$	1.03	0.93	2.58	1.63	0.95	1.04	1.61	2.42
S4 Plug-in optimal filter C^+	4.59	0.85	1.21	1.16	3.76	1.36	1.42	1.61
Lead–lag share of the test mean (%), S0 / S1 / S2 / S3 / S4	0 / -115 / -5 / 60 / -37				-0 / -0 / 0 / 46 / -20			
Panel E: Implementable version, monthly re-estimation, January 2001–July 2026								
	Sharpe ratio		Skewness	Turnover				
	Gross	Net	55-day					
Classic equal-risk CTA, no portfolio target	0.68	0.22	1.35	8				
Classic equal-risk CTA, ex-ante target	0.80	0.10	0.04	276				
RP-PCA, premium-weighted, no portfolio target	1.23	0.53	0.99	14				
RP-PCA, premium-weighted, ex-ante target	1.31	0.49	0.10	320				
Plug-in optimal filter C^+, no portfolio target	1.28	0.29	1.26	37				
Plug-in optimal filter C^+, ex-ante target	1.34	0.24	-0.10	490				
Reference system LS(250,20), capped, no portfolio target	0.75	0.63	0.41	53				
Reference system LS(250,20), capped, ex-ante target	1.03	0.87	0.07	68				
RP-PCA on the LS(250,20) signal, no portfolio target	1.04	0.84	0.98	6				
RP-PCA on the LS(250,20) signal, ex-ante target	1.14	0.90	0.16	103				
RP-PCA on LS minus LS, net Sharpe, no portfolio target (90%)		0.21	[0.01, 0.41]					
RP-PCA on LS minus LS, net Sharpe, ex-ante target (90%)		0.03	[-0.14, 0.20]					
RP-PCA minus classic, skewness, no portfolio target (90%)			-0.36	[-0.77, 0.11]				
RP-PCA on LS minus LS, skewness, no portfolio target (90%)			0.57	[-0.01, 0.83]				

Exhibit 10: **Choosing the Cross-Sectional Filter.** 73 markets, scale-free single-filter P&L (span 100, equal risk), daily. M and Π (Lemma 2) are estimated in the training sample; the basis is the eigenvectors of $M/\mathrm{tr}(M) + \gamma\,\Pi/\mathrm{tr}(\Pi)$ (Proposition 5). Panel A: shares of the first five eigen-trends of the basis in the mean and in the sum of the variances of the eigen-trend P&L. Panel B: positions $x_t = CS_t$ fixed in the training sample; premium-weighted: $\omega_k \propto \max(\hat{\pi}_k, 0)/\hat{v}_k$ (Algorithm 1); intervals: stationary block bootstrap of the test-sample P&L (mean block 63 days, 1,000 draws). Panel C: skewness predicted by Proposition 3, the mean single-market skewness times $\mathrm{tr}((CP)^3)/\mathrm{tr}((CP)^2)^{3/2}$, with P and the single-market skewness from the test sample (law) or from the training sample only (ex ante), and the realized test skewness; intervals: moving-block bootstrap with blocks of 252 days (500 draws). Panel D: the same design on the commodity panel. Panel E: monthly re-estimation from the preceding 756 days, instrument weights, costs, and risk rules as in Exhibit 9; turnover: annual notional turnover in multiples of capital, comparable within a risk rule; intervals as in Panels B and C, on daily net P&L. Positions in Panels A to D are scale-free and gross of costs. The reverse split trains on the later period and is a cross-validation, not a real-time test. Designs fixed before the run (Section B of the Appendix).

7 Robustness and Discussion

7.1 Statistical robustness

Each of the three results was tested further with designs fixed in advance (Section B of the Appendix). Exhibit 12 reports the tests of the third result, the least standard; Exhibit A21 in Section E.8 of the Appendix reports those of the first two.

Panel A: Walk-forward, yearly re-estimation on an expanding window, test years 2004–2026			
	Classic (S0)	RP-PCA (S2)	Plug-in (S4)
Sharpe ratio	0.60	1.33	1.59
Skewness (55-day)	1.24	1.34	0.89
S2 − S0, Sharpe (90%)	0.72 [0.52, 0.93]		Ledoit–Wolf p = <0.001
S2 − S4, Sharpe (90%)	-0.27 [-0.55, -0.01]		
S2 − S0, skewness (90%)	0.10 [-0.51, 0.59]		
Years with S2 > S0	91% of 23		
Panel B: Overfitting, multiple testing, and power (positions fixed in the training sample)			
	Train 2001–2012, test 2013–2026		Train 2013–2026, test 2001–2012
S2 − S0, test Sharpe	0.84		0.52
Null Monte Carlo: mean, 95th percentile	0.00, 0.31		-0.01, 0.35
Null Monte Carlo p-value	0.003		0.007
S2 − S0: Ledoit–Wolf p (Holm)	<0.001 (<0.001)		0.002 (0.009)
S2 − S1: Ledoit–Wolf p (Holm)	0.012 (0.050)		0.045 (0.135)
S2 − S4: Ledoit–Wolf p (Holm)	0.682 (1.000)		0.862 (1.000)
Deflated Sharpe ratio of S2 (29 trials)	0.998		1.000
Minimum detectable skewness difference (5%, power 80%)	1.06		1.07

Exhibit 12: **Statistical Robustness of the Filter Choice.** Filters as in Exhibit 10 (73 markets, scale-free P&L, gross). Panel A: for every test year from 2004, the filters are estimated on all data from January 2001 to the end of the previous year and held fixed during the year; statistics of the concatenated test P&L; Sharpe intervals: stationary bootstrap (mean block 63 days, 1,000 draws), skewness: moving blocks of 252 days (500 draws); Ledoit–Wolf: HAC test of the Sharpe difference (Ledoit and Wolf, 2008). Panel B: null Monte Carlo: 600 simulated panels of Gaussian returns with the sample correlation matrix and no predictability, with the same signals, splits, and estimators, so that every gain is fitted noise; RP-PCA needs a positive training premium, so the null distribution is conditional on it (287 and 283 simulations); p-value: share of simulations at least as large. Holm: adjusted for the six tests in the panel. Deflated Sharpe ratio: probability that the true Sharpe ratio of S2 exceeds the expected maximum of all filters and grid points evaluated (Bailey and López de Prado, 2014). Minimum detectable difference: $(z_{0.95} + z_{0.80})$ times the standard deviation of the test-sample skewness difference of S2 and S0 in the simulations. Design fixed before the run (Section B of the Appendix).

The filter gain is not an artefact of two sample splits. Re-estimated every year on an expanding window and tested on the following year, from 2004 to 2026, RP-PCA earns a Sharpe ratio of 1.33 against 0.60 for the classic CTA (difference 0.72, 90% interval 0.52 to 0.93, Ledoit–Wolf $p < 0.001$) and beats it in 21 of 23 years. Nor is it fitted noise: in

simulated panels with the sample correlation matrix and no predictability (the 287 and 283 of 600 draws with a positive training premium, for which the estimator is defined), the same estimators produce test-sample gains with a mean of zero and a 95th percentile of 0.31 and 0.35, against realized gains of 0.84 and 0.52 ($p = 0.003$ and 0.007). It survives multiple testing: across the 29 filters and grid points evaluated in each split, the deflated Sharpe ratio of RP-PCA is 0.998 and 1.000, and its difference to the classic CTA remains significant after a Holm correction for the six tests of Panel B ($p < 0.001$ and 0.009), while its differences to the PCA filter (0.050 and 0.135) and to the plug-in do not. With yearly re-estimation, the plug-in filter does better than RP-PCA before costs (1.59 against 1.33, difference 0.27, interval 0.01 to 0.55); its turnover, which costs it more than RP-PCA gains (Section 6), makes the regularized estimator preferable in implementation, not before costs.

The convexity comparison has little power. Under the static benchmark, the test-sample skewness difference of two filters has a standard deviation of 0.43, so only differences larger than about 1.06 could be detected at the 5% level with a power of 80%. That premium-aware filters keep their convexity is therefore an absence of evidence of a loss: the point estimates are close to zero in both splits and in the walk-forward test (0.10, interval −0.51 to 0.59), but a loss of half the skewness of the classic CTA could not be ruled out, and the sign of the point estimate varies with the measure: −0.86 at a 21-day horizon in the forward split, −0.04 and 0.07 under the quantile measure (Exhibit A21). The first two results are robust in the dimensions we checked (Exhibit A21). Leaving out any one asset class, the convexity breadth stays between 3.1 and 4.0, less than a third of the Sharpe breadth, the ratio of portfolio to single-market skewness (0.45 to 0.57) stays close to its prediction, and the filter gain stays positive in all 14 cases. The ratio lies between 0.47 and 0.59 under a quantile measure of skewness, with winsorized P&L, and at horizons of 21 and 126 days. The ex-ante risk rule removes more than one unit of skewness at spans of 30, 60, and 100 days, every difference significant, and the ex-post rule 0.57 (interval −0.96 to −0.15).

7.2 Trend-following systems

Exhibit 13 repeats the main measurements for the three reference designs of Sepp and Lucic (2026), net of costs, and with portfolio volatility targeting.

	Skewness of 55-day return				All / single			Quarterly skewness	
System	Single	$N = 4$	$N = 16$	All	Realized	Formula	Sharpe	Full history	Since 2001
Panel A: Systems									
European, single filter (span 100)	2.07	1.44	1.28	1.33	0.65	0.51	0.69	1.23	1.57
European, long-short LS(250,20)	1.27	0.86	0.56	0.41	0.29	0.51	0.75	0.55	0.48
TSMOM (10 periods of 10 days)	1.06	0.68	0.34	0.16	0.16	0.51	0.85	0.57	0.35
American (250/20, ATR 5)	0.80	0.58	0.41	0.35	0.44	0.51	0.68	0.65	0.72
LS(250,20), net of costs	1.25	0.86	0.55	0.41	0.30	0.51	0.63	0.58	0.48
LS(250,20), portfolio vol targeting	1.27	–	–	0.36	0.25	0.51	0.95	0.65	0.46
Panel B: Ratio of all-market to single-market skewness by filter speed and horizon									
Filter	$T = 21$	$T = 55$	$T = 126$	$T = 252$					
Single EWMA, span 100	0.59	0.49	0.47	0.43					
Long-short LS(100,10)	0.32	0.46	0.50	0.42					
Single EWMA, span 250	0.11	0.27	0.36	0.35					
Long-short LS(250,20)	-0.37	0.24	0.38	0.32					

Exhibit 13: **Robustness across Trend-Following Systems.** Columns 2–8: 73 markets, January 2001–July 2026. Panel A uses the engine implementation of each system (instrument weights with the signal cap of ±3 and the weight caps of Sepp and Lucic (2026), log returns), so its single-filter row differs from the scale-free, uncapped P&L of Exhibit 2 (Panel B of this exhibit). Single: median skewness of overlapping 55-day returns across markets; $N = 4$, $N = 16$: median over 100 random subsets; All: equal-weighted portfolio of all markets. All / single: skewness of the all-market portfolio divided by the mean single-market skewness, realized and predicted by the spectral factor of Proposition 1. Sharpe: annualized, all-market portfolio. Quarterly skewness: calendar-quarter returns of the portfolio of all markets available on each day (growing universe), over the full history from 1959 and since 2001, the comparison of Sepp and Lucic (2026, Section 7.4). Systems: positions as in Sepp and Lucic (2026); American: binary positions with entry band and trailing stop of five average true ranges; net of costs: one-way volume-based costs of Hurst et al. (2017, Exhibit B1); portfolio vol targeting: daily P&L scaled to 15% annualized volatility with a lagged EWMA estimate (span 100 days). Panel B: scale-free trend P&L with equal risk as in Exhibit A12, for filters of different speed; Proposition 1 predicts the same ratio at every horizon (0.51).

The ordering by breadth holds for every system, but the size of the convexity cost depends on the design (Panel A). In the engine implementation, with the signal and weight caps of the reference systems, the single filter keeps more of its single-market skewness than the formula predicts (span 100: ratio 0.65, against 0.49 for the uncapped scale-free P&L of Panel B and Exhibit A12), whereas the slow long–short system LS(250,20) keeps 0.29, TSMOM 0.16, and the American system 0.44, against a prediction of 0.51. Costs leave skewness unchanged, and targeting the realized volatility of the portfolio P&L lowers it slightly (from 0.41 to 0.36 for

LS(250,20) with the volatility targeting of the reference implementation), as Sepp and Lucic (2026) and Harvey et al. (2018) anticipate; targeting the covariance of the current positions removes it (Section 5.2). The quarterly skewness of the LS(250,20) portfolio since 2001 is 0.48, the value Sepp and Lucic report; over the full history it is 0.55, so in our data the low post-2000 skewness of the reference system reflects its design rather than the period.

Panel B shows why slow systems lose more. For filters of medium speed, the ratio of portfolio to single-market skewness is close to constant across horizons, as Proposition 1 implies (span 100: 0.43 to 0.59). For slow filters, the separation breaks down at short horizons (LS(250,20): -0.37 at 21 days, 0.38 at 126 days). Slow systems hold the same common exposures for years, such as long bonds or long equities, and at short horizons their portfolios inherit the negative skewness of these markets, a departure from the white-noise benchmark. Convexity then builds only over horizons long relative to the filter, and a quarterly measure of a 250-day system sits below the peak of its skewness profile.

The spectral law rests on three simplifications. Returns are Gaussian and serially independent: autocorrelation adds terms of the order of the autocorrelations (Sepp and Lucic, 2026), drift and persistent exposures matter for slow filters, and jumps against the trend, as for the Swiss franc in 2015, add negative skewness. Correlations are constant. And there is no portfolio volatility targeting. Proposition 2 replaces Gaussianity by zero conditional third moments and allows predictability and time-varying correlations, but still excludes skewed innovations such as the 2015 franc jump; Corollary 3 addresses risk targeting; the factor attribution (Proposition 6 in Section E.3 of the Appendix) needs no distributional assumption beyond the independence of idiosyncratic P&L. The spectral law should therefore be read as the benchmark from which co-trending, drift, filter speed, and risk targeting are measured, not as a forecast of realized skewness. It is of the right order on the separately constructed commodity panel (1.36 predicted against 1.65 realized, Exhibit A16); an independently constructed panel of financial futures would extend this check to the asset classes in which the correlation channel is strongest.

The spectral law gives the convexity of a filter in the benchmark; for filters selected on trend premia it understates realized skewness, and the bootstrap intervals of skewness differences are wide. The 84 markets are those of a published data set assembled ex post; contracts that were delisted are absent, which affects all filters alike but may flatter absolute Sharpe ratios. The implementable filters have three limitations. They are estimated from three years of data and re-estimated every month; a longer window or stronger shrinkage would lower turnover, but we have not searched over these choices. The Sharpe-optimal filter is derived under the benchmark covariance, and the premium estimates that it and the risk-premium PCA use are noisy: out of sample, the plug-in and the regularized filter cannot be told apart statistically, and the gain of either is confirmed in a walk-forward test but rests on one panel of futures and one implementable path. And the comparison of portfolio risk rules was added after the primary backtest.

8 Conclusion

This article asked how the cross-sectional filter of a trend-following portfolio determines its Sharpe ratio and its convexity, and whether a filter can raise the Sharpe ratio without giving up convexity. Both quantities are set by the spectrum of one matrix, CP, the product of the filter and the correlation matrix: in the Gaussian benchmark every filter is a weighted sum of independent single-market trend followers with the eigenvalues of CP as weights. For the classic CTA this makes diversification cheap for the Sharpe ratio and costly for convexity: a broad futures universe offers about fifteen independent bets on the Sharpe ratio but only about four on convexity, skewness stops falling at about 16 markets, and it follows the composition of the universe rather than its size. Whether this convexity reaches the investor depends on the markets and on the risk rule: correlations that rise in the direction of the positions in crises add to the martingale part of convexity, trend reversals after crises take part of it back, and a rule that targets the risk of the current positions removes most

of it, in the main panel and in a confirmatory test on commodities. The gap between the skewness of a single contract and that of a diversified program, which Sepp and Lucic (2026) attribute to filter design and period, is explained by the spectrum, which halves the skewness of a single market, and by the slow design of the reference system, not by the period. And a filter can do better than the classic one. The Sharpe-optimal filter weights directions by trend premium rather than by variance and earns lead–lag premia that equal-risk trend following leaves aside, a quarter to three fifths of its premium in our data. A risk-premium PCA estimates it as a regularized version of the plug-in: re-estimated every year, it raises the Sharpe ratio of the classic CTA from 0.60 to 1.33 before costs and beats it in 21 of 23 years, a gain far beyond what the same estimators fit in data without predictability ($p \leq 0.007$) and robust to the 29 variants evaluated. It does so with twice as many convexity bets as the classic CTA and without a detectable loss of convexity, although only large losses would be detectable. Its value as an estimator is stability rather than precision: before costs the plug-in does as well or better, but it trades two to three times as much and keeps less after costs. The gain is small against a capped reference system with a portfolio volatility target and absent on commodities alone, whose trends rarely lead one another; the static spectrum understates the convexity of premium-weighted filters.

For an investor in trend following, four implications follow. A program need not trade few markets to keep its convexity; beyond moderate breadth, additional markets raise the Sharpe ratio at no further cost in skewness, and breadth across asset classes is worth more than breadth within one. Convexity breadth, which can be computed from a correlation matrix in a line of code, is a better guide to the protection a program offers than the number of markets it trades. The risk rule matters as much as the market set: targeting the covariance of current positions sells the convexity the program is held for. And the cross-sectional filter is a design choice with a measurable payoff: trading markets on each other's signals earns premia the classic construction cannot; the spectrum of CP gives its convexity in the benchmark, and in the data premium-weighted filters were at least as skewed

as that benchmark implies. The most useful combination in our data is such a filter without a portfolio volatility target: on the reference signal it nearly matches the net Sharpe ratio of the targeted reference system (0.84 against 0.87) and keeps the convexity that targeting removes (skewness 0.98 against 0.07). A further extension is a test of whether positioning data on speculators (Uhl, 2025; Boos and Grob, 2023) help explain when the correlation channel is strong.

Conflict of Interest

The author is employed by Quintet Private Bank. The views expressed are those of the author and do not necessarily reflect those of Quintet Private Bank; the author reports no other competing interests.

References

Bailey, D. H. and M. López de Prado (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management 40*(5), 94–107.

Baltas, N. and R. Kosowski (2020). Demystifying time-series momentum strategies: Volatility estimators, trading rules and pairwise correlations. In *Market Momentum: Theory and Practice*. Wiley. SSRN 2140091.

Benichou, R., Y. Lempérière, E. Sérié, J. Kockelkoren, P. Seager, J.-P. Bouchaud, and M. Potters (2017). Agnostic risk parity: Taming known and unknown-unknowns. *Journal of Investment Strategies 6*(3), 1–12.

Boos, D. and L. Grob (2023). Tracking speculative trading. *Journal of Financial Markets 64*, 100774.

Capital Fund Management (2018). The convexity of trend following. Technical note, Capital Fund Management.

Carver, R. (2023). *Advanced Futures Trading Strategies*. Harriman House.

Daniel, K. and T. J. Moskowitz (2016). Momentum crashes. *Journal of Financial Economics 122*(2), 221–247.

Dao, T.-L., T.-T. Nguyen, C. Deremble, Y. Lempérière, J.-P. Bouchaud, and M. Potters (2017). Tail protection for long investors: Trend convexity at work. *Journal of Investment Strategies 7*(1), 61–84.

Erb, C. B. and C. R. Harvey (2006). The strategic and tactical value of commodity futures. *Financial Analysts Journal 62*(2), 69–97.

Faith, C. (2007). *Way of the Turtle: The Secret Methods that Turned Ordinary People into Legendary Traders*. McGraw-Hill.

Firoozye, N. and A. Koshiyama (2019). Optimal dynamic strategies on Gaussian returns. Working paper, arXiv:1906.01427.

Fung, W. and D. A. Hsieh (2001). The risk in hedge fund strategies: Theory and evidence from trend followers. *Review of Financial Studies 14*(2), 313–341.

Giglio, S., D. Xiu, and D. Zhang (2024). Test assets and weak factors. *Journal of Finance*. Forthcoming; early view December 2024.

Gorton, G. and K. G. Rouwenhorst (2006). Facts and fantasies about commodity futures. *Financial Analysts Journal 62*(2), 47–68.

Grebenkov, D. S. and J. Serror (2014). Following a trend with an exponential moving average: Analytical results for a Gaussian model. *Physica A 394*, 288–303.

Grebenkov, D. S. and J. Serror (2015). Optimal allocation of trend following strategies. *Physica A 433*, 107–125.

Hamill, C., S. Rattray, and O. Van Hemert (2016). Trend following: Equity and bond crisis alpha. Working paper, SSRN 2831926.

Harvey, C. R., E. Hoyle, R. Korgaonkar, S. Rattray, M. Sargaison, and O. Van Hemert (2018). The impact of volatility targeting. *Journal of Portfolio Management 45*(1), 14–33.

Hoffman, B. and K. Kaminski (2016). The taming of the skew. Research note, Campbell & Company.

Hurst, B., Y. H. Ooi, and L. H. Pedersen (2017). A century of evidence on trend-following investing. *Journal of Portfolio Management 44*(1), 15–29.

Jusselin, P., E. Lezmi, H. Malongo, C. Masselin, T. Roncalli, and T.-L. Dao (2017). Understanding the momentum risk premium: An in-depth journey through trend-following strategies. Working Paper WP-64-2017, Amundi Asset Management.

Kelly, B., S. Malamud, and L. H. Pedersen (2023). Principal portfolios. *The Journal of Finance 78*(1), 347–387.

Koijen, R. S. J., T. J. Moskowitz, L. H. Pedersen, and E. B. Vrugt (2018). Carry. *Journal of Financial Economics 127*(2), 197–225.

Ledoit, O. and M. Wolf (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance 15*(5), 850–859.

Lettau, M. and M. Pelger (2020a). Estimating latent asset-pricing factors. *Journal of Econometrics 218*(1), 1–31.

Lettau, M. and M. Pelger (2020b). Factors that fit the time series and cross-section of stock returns. *Review of Financial Studies 33*(5), 2274–2325.

Lo, A. W. and A. C. MacKinlay (1990). When are contrarian profits due to stock market overreaction? *Review of Financial Studies 3*(2), 175–205.

Martin, R. J. (2023). Design and analysis of momentum trading strategies. Working paper, arXiv:2101.01006.

Masturzo, J. (2025). Walking the tightrope: Trend following's tricky tradeoffs. Research note, Research Affiliates.

Mathai, A. M. and S. B. Provost (1992). *Quadratic Forms in Random Variables: Theory and Applications*. New York: Marcel Dekker.

Moore, H., R. Goodall, and E. Sanchez Martin (2026). A trend following deep dive: The optimal market mix for a trend follower. Research note, Man Group.

Moreira, A. and T. Muir (2017). Volatility-managed portfolios. *Journal of Finance 72*(4), 1611–1644.

Moskowitz, T. J., Y. H. Ooi, and L. H. Pedersen (2012). Time series momentum. *Journal of Financial Economics 104*(2), 228–250.

Newey, W. K. and K. D. West (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica 55*(3), 703–708.

Pitkäjärvi, A., M. Suominen, and L. Vaittinen (2020). Cross-asset signals and time series momentum. *Journal of Financial Economics 136*(1), 63–85.

Politis, D. N. and J. P. Romano (1994). The stationary bootstrap. *Journal of the American Statistical Association 89*(428), 1303–1313.

Potters, M. and J.-P. Bouchaud (2006). Trend followers lose more often than they gain. *Wilmott Magazine 26*, 58–63.

Prichard, D. and J. Theiler (1994). Generating surrogate data for time series with several simultaneously measured variables. *Physical Review Letters 73*(7), 951–954.

Safari, S. A. and C. Schmidhuber (2026). Trends, volatility, correlations, and critical phenomena in financial markets. Working paper, arXiv:2606.20145.

Sepp, A. and V. Lucic (2026). The science and practice of trend-following systems. Working paper, arXiv:2607.19497. Earlier version: Sepp (2018), SSRN 3167787.

Uhl, B. (2025). Speculators and time series momentum in commodity futures markets. *Review of Financial Economics 43*(2), 213–230.

Valeyre, S. (2024). Optimal trend following portfolios. *Journal of Investment Strategies 12*(3).

Appendix

This Appendix proves all results of Section 2 and checks them by simulation (Section A), documents the pre-specification of the convexity analysis and its sensitivity (Section B), describes the trend-following systems and their validation (Section C), documents the data (Section D), and reports additional results (Section E). Exhibits carry the prefix A. Throughout, $(\mathcal{F}_t)$ is the filtration of Section 2.1, $\|\cdot\|_p$ is the L^p norm of a random variable, and sums over j run from one to infinity.

A. Proofs

A.1 Preliminaries

Moments. Under Assumption 2(iv) and $\sum_j |\beta_j| < \infty$, Minkowski's inequality gives $\|S_{i,t-1}\|_6 \le \sum_j |\beta_j| \sup_t \|z_{i,t}\|_6 < \infty$. By Hölder's inequality, $\|f_{i,t}\|_3 \le \|S_{i,t-1}\|_6 \|z_{i,t}\|_6$, so every product of at most three daily P&L terms is integrable, and so are V_t (since $\|V_t\|_{3/2} \le \|e_t\|_3^2$ by the conditional Jensen inequality) and $V_t e_s$. Under Assumption 1 all moments exist, because S_{t-1} and z_t are Gaussian.

Lemma A.1 (single market). *Let z_t be independent standard normal, $\phi_t = (\sum_j \beta_j z_{t-j}) z_t$, and $\Psi = \sum_{t=1}^T \phi_t$. Then $\mathrm{Var}[\Psi] = T$ and $\kappa_3(\Psi) = \mathbb{E}[\Psi^3] = 6 \sum_{h=1}^{T-1} (T-h) \beta_h \rho_S(h)$.*

Proof. Write $\tilde{S}_{t-1} = \sum_j \beta_j z_{t-j}$, which is $\mathcal{F}_{t-1}$-measurable, standard normal, and independent of z_t. Then $\mathbb{E}[\phi_t] = 0$, $\mathbb{E}[\phi_t^2] = \mathbb{E}[\tilde{S}_{t-1}^2]\mathbb{E}[z_t^2] = 1$, and for $s < t$, $\mathbb{E}[\phi_t \phi_s] = \mathbb{E}[z_t]\,\mathbb{E}[\tilde{S}_{t-1}\phi_s] = 0$; hence $\mathrm{Var}[\Psi] = T$. For the third moment, expand $\mathbb{E}[\Psi^3] = \sum_{t,u,v} \mathbb{E}[\phi_t \phi_u \phi_v]$ and let τ be the largest of the three dates. If τ occurs once, z_τ enters linearly and is independent of all other factors, so the term vanishes. If τ occurs three times, $\mathbb{E}[\phi_\tau^3] = \mathbb{E}[\tilde{S}_{\tau-1}^3]\mathbb{E}[z_\tau^3] = 0$. If $\tau = t$ occurs twice and the third date is $s < t$, which happens for three of the orderings of (t, u, v), then $\mathbb{E}[\phi_t^2 \phi_s] = \mathbb{E}[\tilde{S}_{t-1}^2\, \tilde{S}_{s-1} z_s]$. With $h = t - s$, write $\tilde{S}_{t-1} = \beta_h z_s + R$, where $R = \sum_{j \ne h} \beta_j z_{t-j}$ is independent of z_s. Expanding,

$$\mathbb{E}[\tilde{S}_{t-1}^2 \tilde{S}_{s-1} z_s] = \beta_h^2\, \mathbb{E}[z_s^3]\, \mathbb{E}[\tilde{S}_{s-1}] + 2\beta_h\, \mathbb{E}[z_s^2]\, \mathbb{E}[R\, \tilde{S}_{s-1}] + \mathbb{E}[R^2 \tilde{S}_{s-1}]\, \mathbb{E}[z_s] = 2\beta_h\, \mathbb{E}[R\, \tilde{S}_{s-1}],$$

where we used that z_s is independent of R and of $\tilde{S}_{s-1}$. Only the terms of R with dates before s correlate with $\tilde{S}_{s-1} = \sum_j \beta_j z_{s-j}$; these are $\sum_j \beta_{h+j} z_{s-j}$, so $\mathbb{E}[R\, \tilde{S}_{s-1}] = \sum_j \beta_{h+j}\beta_j = \rho_S(h)$. Each lag h occurs for $T - h$ pairs $s < t$ in the horizon, and the three orderings give the factor

three: $\mathbb{E}[\Psi^3] = 3\sum_{h=1}^{T-1}(T-h)\,2\beta_h\rho_S(h)$. $\square$

Lemma A.2 (pairing for many markets). *Under Assumption 1, for $s < t$ and $h = t-s$,*

$$\mathbb{E}\big[S_{i,t-1}S_{k,t-1}S_{l,s-1}z_{l,s}\big] = \beta_h\rho_S(h)\,\big(P_{il}P_{kl} + P_{kl}P_{il}\big) = 2\beta_h\rho_S(h)\,P_{il}P_{kl}.$$

Proof. Write $S_{i,t-1} = \beta_h z_{i,s} + R_i$, with $R_i = \sum_{j\neq h}\beta_j z_{i,t-j}$ independent of z_s, and likewise for $S_{k,t-1}$; the signal $S_{l,s-1}$ contains no z_s. Terms with an odd number of factors from the Gaussian vector z_s vanish: those with one factor ($R_iR_kS_{l,s-1}z_{l,s}$) and with three ($\beta_h^2 z_{i,s}z_{k,s}z_{l,s}S_{l,s-1}$). The two cross terms remain:

$$\beta_h\big(\mathbb{E}[z_{i,s}z_{l,s}]\,\mathbb{E}[R_kS_{l,s-1}] + \mathbb{E}[z_{k,s}z_{l,s}]\,\mathbb{E}[R_iS_{l,s-1}]\big),$$

and $\mathbb{E}[R_kS_{l,s-1}] = \sum_{j\geq 1}\beta_{h+j}\beta_j P_{kl} = \rho_S(h)P_{kl}$. $\square$

A.2 Eigen-trends and the spectral law

Proof of Lemma 1. (a) By linearity, $s_{k,t-1} = q_k^\top W^{1/2}\sum_j \beta_j z_{t-j} = \sum_j \beta_j q_k^\top W^{1/2} z_{t-j} = \sum_j \beta_j y_{k,t-j}$. Since $\{q_k\}$ is an orthonormal basis, $\sum_k q_kq_k^\top = I$, and

$$g_t = S_{t-1}^\top W z_t = \big(W^{1/2}S_{t-1}\big)^\top\Big(\sum_k q_kq_k^\top\Big)\big(W^{1/2}z_t\big) = \sum_k\big(q_k^\top W^{1/2}S_{t-1}\big)\big(q_k^\top W^{1/2}z_t\big) = \sum_k \Phi_{k,t}.$$

No assumption on the returns is used. (b) Let $Q = (q_1,\dots,q_N)$. Under Assumption 1, $y_t = Q^\top W^{1/2}z_t$ is Gaussian with covariance $Q^\top W^{1/2}PW^{1/2}Q = Q^\top MQ = \mathrm{diag}(\mu_1,\dots,\mu_N)$, independent over time. Uncorrelated jointly Gaussian variables are independent, so the components $y_{k,t}$ are independent over k and t. For $\mu_k > 0$ set $\xi_{k,t} = y_{k,t}/\sqrt{\mu_k}$; for $\mu_k = 0$, $y_{k,t} = 0$ almost surely and $\xi_{k,t}$ can be taken as independent standard normal variables independent of everything else, without changing $\Phi_{k,t} = 0$. Then $s_{k,t-1} = \sqrt{\mu_k}\sum_j \beta_j\xi_{k,t-j}$ and $\Phi_{k,t} = \mu_k\,(\sum_j \beta_j\xi_{k,t-j})\,\xi_{k,t} = \mu_k\phi_{k,t}$. Each $(\phi_{k,t})_t$ is a measurable function of $(\xi_{k,t})_t$ alone, the same function for every k; since the sequences $(\xi_{k,t})_t$ are independent and identically distributed over k, so are the processes $(\phi_{k,t})_t$. $\square$

Proof of Proposition 1. By Lemma 1, $F_T = \sum_t\sum_k \mu_k\phi_{k,t} = \sum_k \mu_k\Psi_k$ with $\Psi_k = \sum_{t=1}^T \phi_{k,t}$. The Ψ_k are independent, and each has the distribution of the T-day P&L Ψ of a single-market trend follower on white noise. Cumulants are additive over independent summands and homogeneous of degree n, $\kappa_n(cX) = c^n\kappa_n(X)$; hence $\kappa_n(F_T) = \sum_k \mu_k^n\kappa_n(\Psi)$,

and $\sum_k \mu_k^n = \operatorname{tr}(M^n) = \operatorname{tr}((WP)^n)$ because $M = W^{1/2}PW^{1/2}$ is similar to WP. Lemma A.1 gives $\kappa_2(\Psi) = T$ and $\kappa_3(\Psi)$. Dividing, $\varsigma_P = \kappa_3(F_T)/\kappa_2(F_T)^{3/2} = \varsigma_1 \operatorname{tr}((WP)^3)/\operatorname{tr}((WP)^2)^{3/2}$ and $\delta_P = \kappa_4(F_T)/\kappa_2(F_T)^2 = \delta_1 \operatorname{tr}((WP)^4)/\operatorname{tr}((WP)^2)^2$. For a single market ($N = 1$, $w = 1$) the trace factors equal one and the expressions reduce to Lemma D.1 and Proposition D.2 of Sepp and Lucic (2026). □

A direct proof. The third moment also follows without the eigen-trends. With $g_t = \sum_i w_i f_{i,t}$, the argument of Lemma A.1 applies date by date: terms in which the largest date occurs once vanish, as do those in which it occurs three times, because third moments of the Gaussian vector z_t are zero. The surviving terms pair a later date t twice with an earlier date s once, and conditioning on $\mathcal{F}_{t-1}$ gives $\mathbb{E}[f_{i,t}f_{k,t}f_{l,s}] = P_{ik}\,\mathbb{E}[S_{i,t-1}S_{k,t-1}S_{l,s-1}z_{l,s}] = 2\beta_h\rho_S(h)P_{ik}P_{il}P_{kl}$ by Lemma A.2. Summing over markets, $\sum_{i,k,l} w_i w_k w_l P_{ik}P_{kl}P_{li} = \operatorname{tr}((WP)^3)$: correlation enters through closed cycles. Summing over dates gives $\mathbb{E}[F_T^3] = 6\sum_h (T-h)\beta_h\rho_S(h)\operatorname{tr}((WP)^3)$, and the variance follows as in Lemma A.1, $\operatorname{Var}[g_t] = \sum_{i,k} w_i w_k P_{ik}^2 = \operatorname{tr}((WP)^2)$.

The spectral factor is at most one. With $\mu_k \geq 0$, $\sum_k \mu_k^3 \leq \mu_1 \sum_k \mu_k^2 \leq (\sum_k \mu_k^2)^{1/2} \sum_k \mu_k^2$, so $\operatorname{tr}((WP)^3) \leq \operatorname{tr}((WP)^2)^{3/2}$, with equality if and only if a single eigenvalue is nonzero, that is, if WP and hence P have rank one. Aggregation cannot raise skewness above the single-market level under Assumption 1. For equal risk, $P = I$ gives $N^{-1/2}$, and k equal blocks of perfectly correlated markets give $k^{-1/2}$.

Simulation. The simulations of Exhibit A1 compare Equation (6) with the skewness of simulated portfolios for equicorrelated, block-structured, and random correlation matrices, with equal and unequal risk weights.

Markets and correlation	Formula	Monte Carlo	s.e.
Single market	2.109	2.139	0.017
10 independent markets	0.667	0.670	0.005
10 markets, $\rho = 0.3$	1.472	1.486	0.015
20 markets, $\rho = 0.1$	0.695	0.694	0.006
Two blocks of 5, $\rho_{\text{in}} = 0.6$, $\rho_{\text{out}} = 0$	1.384	1.368	0.010
6 markets, random P, equal risk	1.133	1.154	0.009
6 markets, random P, unequal risk	1.211	1.205	0.012

Exhibit A1: **Monte Carlo Check of Proposition 1.** Skewness of the 10-day return of a portfolio of EWMA trend followers (span 20 days) on Gaussian white-noise markets with correlation matrix P and risk weights W. Formula: $\varsigma_1(T)\operatorname{tr}((WP)^3)/\operatorname{tr}((WP)^2)^{3/2}$ with the single-market skewness $\varsigma_1(T) = 2.109$. Monte Carlo: 400,000 paths after a burn-in of 120 days; s.e.: batch-means standard error over 20 batches.

Proof of Corollary 1. *Sharpe ratio.* If every market's P&L has mean $\bar{\pi}$ per day and, to first order, the covariances of Proposition 1, the portfolio has mean $\bar{\pi}T\sum_i w_i = \bar{\pi}T\sum_k \mu_k$ (because $\sum_k \mu_k = \mathrm{tr}(WP) = \sum_i w_i P_{ii} = \sum_i w_i$) and variance $T\sum_k \mu_k^2$, so its first-order Sharpe ratio is $\bar{\pi}\sqrt{T}\sum_k \mu_k/(\sum_k \mu_k^2)^{1/2} = \mathrm{SR}_1\sqrt{K_2}$. *Skewness and kurtosis.* By Proposition 1, $\varsigma_P/\varsigma_1 = \sum \mu^3/(\sum \mu^2)^{3/2} = K_3^{-1/2}$ and $\delta_P/\delta_1 = \sum \mu^4/(\sum \mu^2)^2 = K_4^{-1}$. *Inequalities.* All sums run over the $r = \mathrm{rank}(P)$ nonzero eigenvalues; $\mathrm{rank}(WP) = \mathrm{rank}(P)$ because W is invertible. (i) $K_4 \geq 1$: $(\sum \mu^2)^2 = \sum \mu^4 + \sum_{k\neq l} \mu_k^2\mu_l^2 \geq \sum \mu^4$, with equality if and only if $r = 1$. (ii) $K_4 \leq K_3$: by the Cauchy–Schwarz inequality, $(\sum \mu^3)^2 = (\sum \mu\cdot\mu^2)^2 \leq \sum \mu^2 \sum \mu^4$; dividing by $(\sum \mu^2)^3$ gives $K_3^{-1} \leq K_4^{-1}$. (iii) $K_3 \leq K_2$: by the Cauchy–Schwarz inequality, $(\sum \mu^2)^2 = (\sum \mu^{1/2}\mu^{3/2})^2 \leq \sum \mu \sum \mu^3$; squaring and dividing by $\sum \mu^2 (\sum \mu^3)^2$ gives $(\sum \mu^2)^3/(\sum \mu^3)^2 \leq (\sum \mu)^2/\sum \mu^2$. (iv) $K_2 \leq r$: $(\sum \mu)^2 = (\sum 1\cdot\mu)^2 \leq r\sum \mu^2$. Equality in the Cauchy–Schwarz inequality holds if and only if the two vectors are proportional; in (ii), (iii), and (iv) this means that μ_k is constant over the nonzero eigenvalues. Conversely, if all r nonzero eigenvalues are equal, $K_2 = K_3 = K_4 = r$. □

Proof of Corollary 2. The shares follow from Proposition 1: $\mathrm{Var}[\mu_k\Psi_k] = T\mu_k^2$ and $\kappa_3(\mu_k\Psi_k) = \mu_k^3\kappa_3(\Psi)$, and the eigen-trends are independent. Fix p and let $A_n = \sum_{k\leq p}\mu_k^n$ and $B_n = \sum_{k>p}\mu_k^n$. Inequality (9) reads $A_3/(A_3+B_3) \geq A_2/(A_2+B_2)$, which is equivalent to $A_3B_2 \geq A_2B_3$, that is,

$$\sum_{k\leq p<l} \mu_k^2\mu_l^2\,(\mu_k - \mu_l) \;\geq\; 0.$$

Every term is nonnegative because the eigenvalues are ordered, $\mu_k \geq \mu_l$ for $k \leq p < l$. The sum is zero if and only if $\mu_k = \mu_l$ for all $k \leq p < l$ with $\mu_l > 0$; for $p < r$ there is such an l, and then all nonzero eigenvalues are equal. □

Proof of Corollary 4. (a) With $A = W^{1/2}Q\,\mathrm{diag}(\omega)Q^\top W^{1/2}$, which is symmetric,

$$g_t^\omega = S_{t-1}^\top A\, z_t = \sum_k \omega_k\,(q_k^\top W^{1/2}S_{t-1})(q_k^\top W^{1/2}z_t) = \sum_k \omega_k\, s_{k,t-1}\, y_{k,t} = \sum_k \omega_k\Phi_{k,t}.$$

For $\omega = \mathbf{1}$, $QQ^\top = I$ gives $A = W$ and $x_{t-1} = WS_{t-1}$. (b) By Lemma 1(b), $\omega_k\Phi_{k,t} = \omega_k\mu_k\phi_{k,t}$ with independent and identically distributed processes ϕ_k, so $F_T^\omega = \sum_k \omega_k\mu_k\Psi_k$ and, as in the proof of Proposition 1, $\kappa_n(F_T^\omega) = \kappa_n(\Psi)\sum_k(\omega_k\mu_k)^n$. With $\omega_k = c\,\mu_k^{\alpha-1}$ the factor c cancels in the skewness, and $\omega_k\mu_k \propto \mu_k^\alpha$ gives $\varsigma(F_T^\omega) = \varsigma_1(T)\sum_k \mu_k^{3\alpha}/(\sum_k \mu_k^{2\alpha})^{3/2} = \varsigma_1(T)/\sqrt{K_3(\mu^\alpha)}$. (c) The mean is $\bar{\pi}T\sum_k \omega_k\mu_k \propto \bar{\pi}T\sum_k \mu_k^\alpha$ and the variance $T\sum_k(\omega_k\mu_k)^2 \propto T\sum_k \mu_k^{2\alpha}$ (to first order, as in Corollary 1), so the first-order Sharpe ratio is $\bar{\pi}\sqrt{T}\sum_k \mu_k^\alpha/(\sum_k \mu_k^{2\alpha})^{1/2} = \mathrm{SR}_1\sqrt{K_2(\mu^\alpha)}$. For $\alpha = 0$ all nonzero terms

are equal and $K_2 = K_3 = \operatorname{rank}(P)$; as $\alpha \to \infty$, both breadths tend to the multiplicity of the largest eigenvalue. Both fall monotonically: $\frac{d}{d\alpha} \ln K_2(\mu^\alpha) = 2(\mathbb{E}_\alpha - \mathbb{E}_{2\alpha})[\ln \mu] \le 0$ and $\frac{d}{d\alpha} \ln K_3(\mu^\alpha) = 6(\mathbb{E}_{2\alpha} - \mathbb{E}_{3\alpha})[\ln \mu] \le 0$, where $\mathbb{E}_b$ averages over the nonzero μ_k with weights $\mu_k^b / \sum_l \mu_l^b$ and is nondecreasing in b. □

A.3 Channels of convexity

Proof of Proposition 2. Integrability of all terms follows from Section A.1. (a) The innovation $e_t = S_{t-1}^\top W \eta_t$ is a martingale difference, $\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] = S_{t-1}^\top W \, \mathbb{E}[\eta_t \mid \mathcal{F}_{t-1}] = 0$, because S_{t-1} is $\mathcal{F}_{t-1}$-measurable. Hence $\mathbb{E}[E_T] = 0$ and, for $s < t$, $\mathbb{E}[e_t e_s] = \mathbb{E}[e_s \, \mathbb{E}[e_t \mid \mathcal{F}_{t-1}]] = 0$, so $\operatorname{Var}[E_T] = \sum_t \mathbb{E}[e_t^2] = \sum_t \mathbb{E}[V_t]$. For the third moment, $\kappa_3(E_T) = \mathbb{E}[E_T^3] = \sum_{t,u,v} \mathbb{E}[e_t e_u e_v]$; let τ be the largest of the three dates.

- If τ occurs once, the product of the other two factors is $\mathcal{F}_{\tau-1}$-measurable, and $\mathbb{E}[e_\tau \cdot (\cdot)] = \mathbb{E}[(\cdot) \, \mathbb{E}[e_\tau \mid \mathcal{F}_{\tau-1}]] = 0$.
- If τ occurs three times, $\mathbb{E}[e_\tau^3 \mid \mathcal{F}_{\tau-1}] = \sum_{i,k,l} x_{i,\tau-1} x_{k,\tau-1} x_{l,\tau-1} \, \mathbb{E}[\eta_{i,\tau} \eta_{k,\tau} \eta_{l,\tau} \mid \mathcal{F}_{\tau-1}] = 0$ with $x_{\tau-1} = W S_{\tau-1}$, by Assumption 2(iii).
- If $\tau = t$ occurs twice and the third date is $s < t$, $\mathbb{E}[e_t^2 e_s] = \mathbb{E}[e_s \, \mathbb{E}[e_t^2 \mid \mathcal{F}_{t-1}]] = \mathbb{E}[V_t e_s] = \operatorname{Cov}(V_t, e_s)$, because $\mathbb{E}[e_s] = 0$. Three of the orderings of (t, u, v) have this form.

Hence $\kappa_3(E_T) = 3 \sum_{s<t} \operatorname{Cov}(V_t, e_s)$. Since $\bar{P}$ has a unit diagonal, $\Sigma_t = \bar{P} + \operatorname{diag}(\Sigma_t - \bar{P}) + \operatorname{off}(\Sigma_t - \bar{P})$, and the quadratic form $V_t = S_{t-1}^\top W \Sigma_t W S_{t-1}$ splits into the three terms of Equation (11); the covariance is linear in its first argument. (b) Write $F_T - \mathbb{E}[F_T] = E_T + \tilde{D}_T$. Expanding the cube, $\kappa_3(F_T) = \mathbb{E}[(E_T + \tilde{D}_T)^3] = \mathbb{E}[E_T^3] + 3\mathbb{E}[E_T^2 \tilde{D}_T] + 3\mathbb{E}[E_T \tilde{D}_T^2] + \mathbb{E}[\tilde{D}_T^3]$. (c) Under Assumption 1, $m_t = \mathbb{E}[z_t \mid \mathcal{F}_{t-1}] = 0$, so $d_t = 0$, $D_T = 0$, and $\Delta_T = 0$; $\Sigma_t = P = \bar{P}$, so $V_t^{\mathrm{var}} = V_t^{\mathrm{cor}} = 0$. Finally, $e_s = g_s$ and, by Lemma A.2,

$$\mathbb{E}[V_t^{\mathrm{pos}} g_s] = \sum_{i,k,l} w_i w_k w_l P_{ik} \, \mathbb{E}[S_{i,t-1} S_{k,t-1} S_{l,s-1} z_{l,s}] = 2\beta_h \rho_S(h) \sum_{i,k,l} w_i w_k w_l P_{ik} P_{kl} P_{li} = 2\beta_h \rho_S(h) \operatorname{tr}((WP)^3),$$

so $\mathcal{C}_T^{\mathrm{pos}} = 3 \sum_{h=1}^{T-1} (T-h) \, 2\beta_h \rho_S(h) \operatorname{tr}((WP)^3)$, the third moment of Proposition 1. (d) Under strict stationarity, $\operatorname{Cov}(V_t, e_s)$ depends only on $h = t - s$, and there are $T - h$ pairs with lag h in the horizon. □

Lemma A.3 (no drift). *Under Assumption 2 with $m_t = 0$ for all t: for $s < u < t$, $\mathbb{E}[g_t^2 g_s] = \mathbb{E}[V_t g_s]$, $\mathbb{E}[g_t^3] = 0$, $\mathbb{E}[g_t g_s^2] = 0$, and $\mathbb{E}[g_t g_u g_s] = 0$. Hence $\kappa_3(F_T) = 3 \sum_{s<t} \mathbb{E}[g_t^2 g_s]$:*

the realized third moment and the sum of forward terms $g_t^2 g_s$ have the same expectation, and all other terms of the sample third moment have expectation zero.

Proof. With $m_t = 0$, $g_t = e_t$ and $D_T = 0$. The four statements are the three cases of the proof of Proposition 2(a): the products in which the latest date occurs once ($g_t g_s^2$ and $g_t g_u g_s$) and three times (g_t^3) have zero expectation, and $\mathbb{E}[g_t^2 g_s] = \mathbb{E}[V_t g_s]$. Part (b) of the proposition gives $\kappa_3(F_T) = \kappa_3(E_T)$. □

Proof of Corollary 3. Because ℓ_t is $\mathcal{F}_{t-1}$-measurable and $\sup_t \mathbb{E}|\ell_t e_t|^3 < \infty$, $\ell_t e_t$ is a martingale difference with conditional variance $\mathbb{E}[\ell_t^2 e_t^2 \mid \mathcal{F}_{t-1}] = \ell_t^2 V_t$ and conditional third moment $\ell_t^3\,\mathbb{E}[e_t^3 \mid \mathcal{F}_{t-1}] = 0$, and all products of at most three terms are integrable: $\|\ell_t^2 V_t\|_{3/2} \leq \|\ell_t e_t\|_3^2$ and Hölder's inequality. The proof of Proposition 2(a) uses only these three properties; applied to $\ell_t e_t$, it gives (a). If $\ell_t^2 V_t = \bar{c}$, then $\mathrm{Cov}(\bar{c}, \ell_s e_s) = 0$ for every $s < t$, which gives (b); (c) follows from (a) with $\ell_t^2 V_t = \bar{c} + \ell_t^2 (V_t^{\mathrm{var}} + V_t^{\mathrm{cor}})$. □

The estimators of Remark 1 replace $\mathrm{Cov}(V_t^x, e_s)$ by sample covariances of χ_t and g_s over the evaluation period. Under strict stationarity and ergodicity they are consistent for the channels when $m_t = 0$; with a predictable drift, $\mathbb{E}[g_t^2 g_s] = \mathbb{E}[V_t e_s] + \mathbb{E}[V_t d_s] + \mathbb{E}[d_t^2 g_s]$ for $s < t$; the term $\mathbb{E}[V_t d_s]$ is of first and $\mathbb{E}[d_t^2 g_s]$ of second order in the drift, and both are attributed to the martingale part. Exhibit A2 checks the estimators under the static benchmark: the martingale part is centered on Proposition 1 (1.24 against 1.18), the co-movement channel on zero, and its standard deviation (0.15) is less than half that of the sample skewness of overlapping sums (0.39), whose mean (1.05) shows the familiar downward bias of sample skewness.

Statistic	Mean	Standard deviation	90% range
Realized skewness (overlapping sums)	1.05	0.39	[0.59, 1.74]
Martingale part (Proposition 2)	1.24	0.15	[1.02, 1.50]
Position channel	1.23	0.13	[1.03, 1.46]
Co-movement channel	0.01	0.09	[-0.13, 0.14]
Drift part	-0.19	0.47	[-0.79, 0.65]
Proposition 1	1.18		

Exhibit A2: **The Channel Estimator under the Static Benchmark.** 200 simulations of 6,600 days of Gaussian white-noise returns with the sample correlation matrix of the 73 markets (Assumption 1), single filter with a span of 100 days, equal risk, horizon 55 days. The estimators are those of Exhibit 7. Proposition 1: simulated single-market skewness times the spectral factor.

A.4 Cross-sectional filters and risk-premium PCA

Proof of Lemma 2. (b) Under Assumption 2(iv), $\sum_j |\beta_j| < \infty$, and bounded second moments, the series $S_{t-1} = \sum_j \beta_j z_{t-j}$ converges in L^2, so $\mathbb{E}[z_t S_{t-1}^\top] = \sum_j \beta_j \mathbb{E}[z_t z_{t-j}^\top]$ by dominated convergence. Then

$$\mathbb{E}[\Phi_{k,t}] = \mathbb{E}\big[(q_k^\top W^{1/2} S_{t-1})(z_t^\top W^{1/2} q_k)\big] = q_k^\top W^{1/2}\, \mathbb{E}[S_{t-1} z_t^\top]\, W^{1/2} q_k = q_k^\top W^{1/2} \Gamma^\top W^{1/2} q_k = q_k^\top \Pi q_k,$$

because a quadratic form depends only on the symmetric part of its matrix. Summing over an orthonormal basis, $\sum_k q_k^\top \Pi q_k = \operatorname{tr}(Q^\top \Pi Q) = \operatorname{tr}(\Pi) = \sum_i w_i \Gamma_{ii}$, and $\operatorname{tr}(\Pi) = \mathbb{E}[g_t]$ by Lemma 1(a). (a) For a cross-sectional filter C, $\mathbb{E}[S_{t-1}^\top C z_t] = \operatorname{tr}(C\, \mathbb{E}[z_t S_{t-1}^\top]) = \operatorname{tr}(C\Gamma) = \sum_{i,k} C_{ik}\Gamma_{ki}$; splitting the sum into $i = k$ and $i \neq k$ gives the two parts. For $C = W^{1/2} B W^{1/2}$ with B symmetric, $\operatorname{tr}(C\Gamma) = \operatorname{tr}(B W^{1/2} \Gamma_s W^{1/2}) = \operatorname{tr}(B\Pi)$. (c) $Q\,\mathrm{diag}(\mathbf{1})Q^\top = QQ^\top = I$ for every orthonormal Q, so $C = W$ is diagonal and the cross-serial sum vanishes. (d) Every step is linear in the expectation operator and holds with sample means in place of expectations. □

Proof of Proposition 3. (a) Under Assumption 1, write $z_t = P^{1/2}\xi_t$ with ξ_t independent standard normal over t (on an enlarged probability space if P is singular). Because the trend filter is linear and the same in every market, $S_{t-1} = P^{1/2}\tilde{S}_{t-1}$ with $\tilde{S}_{t-1} = \sum_j \beta_j \xi_{t-j}$, the signal vector of the white noise ξ. Then $g_t^C = S_{t-1}^\top C z_t = \tilde{S}_{t-1}^\top \tilde{C}\, \xi_t$ with $\tilde{C} = P^{1/2} C P^{1/2}$ symmetric. Write $\tilde{C} = \sum_k \vartheta_k \tilde{q}_k \tilde{q}_k^\top$ with orthonormal $\tilde{q}_k$. The series $\tilde{q}_k^\top \xi_t$ are independent standard normal over k and t, and $\tilde{q}_k^\top \tilde{S}_{t-1}$ is the trend signal of $\tilde{q}_k^\top \xi$. Hence $g_t^C = \sum_k \vartheta_k (\tilde{q}_k^\top \tilde{S}_{t-1})(\tilde{q}_k^\top \xi_t)$ is a weighted sum of independent single-market trend followers on white noise, exactly as in Lemma 1(b), and $F_T^C = \sum_k \vartheta_k \Psi_k$. Cumulants add over independent summands and are homogeneous of degree n, so $\kappa_n(F_T^C) = \kappa_n(\Psi) \sum_k \vartheta_k^n$, and $\sum_k \vartheta_k^n = \operatorname{tr}(\tilde{C}^n) = \operatorname{tr}((CP)^n)$ because $\tilde{C}$ and CP have the same characteristic polynomial. The skewness follows by division. (b) The mean is linear in the autocovariances and equals $T \operatorname{tr}(C\Gamma)$ by Lemma 2(a). F_T^C is a quadratic form in a Gaussian vector whose covariance is that of (a) plus terms of order ε; by Isserlis' theorem its variance is $T \operatorname{tr}((CP)^2)$ plus terms each containing at least one εA_h, hence $T \operatorname{tr}((CP)^2)(1 + O(\varepsilon))$, and likewise every cumulant differs from (a) by $O(\varepsilon)$. Since the mean is $O(\varepsilon)$, the Sharpe ratio is the stated leading term plus $O(\varepsilon^2)$. □

Proof of Proposition 4. With $\tilde{C} = P^{1/2} C P^{1/2}$, $\operatorname{tr}(C\Gamma) = \operatorname{tr}(C\Gamma_s) = \operatorname{tr}(\tilde{C}\tilde{\Gamma}) = \langle \tilde{C}, \tilde{\Gamma} \rangle$ and $\operatorname{tr}((CP)^2) = \operatorname{tr}(\tilde{C}^2) = \|\tilde{C}\|^2$ in the Frobenius inner product, so by Proposition 3(b) the first-order Sharpe ratio is $\sqrt{T}\langle \tilde{C}, \tilde{\Gamma} \rangle / \|\tilde{C}\|$; $P \succ 0$ makes $C \mapsto \tilde{C}$ a bijection, and by Sylvester's law

of inertia $C \succeq 0$ if and only if $\tilde{C} \succeq 0$. (a) By the Cauchy–Schwarz inequality the ratio is at most $\|\tilde{\Gamma}\| = \|\lambda\|$, with equality if and only if $\tilde{C} \propto \tilde{\Gamma}$, that is, $C \propto P^{-1/2}\tilde{\Gamma}P^{-1/2} = P^{-1}\Gamma_s P^{-1}$. The eigenvalues of C^*P are those of $\tilde{\Gamma}$, and Proposition 3(a) gives the skewness. (b) The set $\mathcal{K}$ of positive semidefinite matrices is a closed convex cone, and the projection of $\tilde{\Gamma}$ onto it is $\tilde{\Gamma}^+$. For every $\tilde{C} \in \mathcal{K}$, $\langle \tilde{C}, \tilde{\Gamma} - \tilde{\Gamma}^+ \rangle \leq 0$, because $\tilde{\Gamma} - \tilde{\Gamma}^+$ is negative semidefinite; hence $\langle \tilde{C}, \tilde{\Gamma} \rangle \leq \langle \tilde{C}, \tilde{\Gamma}^+ \rangle \leq \|\tilde{C}\| \, \|\tilde{\Gamma}^+\|$, with equality throughout for $\tilde{C} \propto \tilde{\Gamma}^+$. The maximum is $\sqrt{T}\|\tilde{\Gamma}^+\| = \sqrt{T}\|\lambda^+\|$, and the eigenvalues of C^+P are $\lambda^+ \geq 0$, so Proposition 3(a) gives the skewness $\varsigma_1(T)\sum_k(\lambda_k^+)^3/(\sum_k(\lambda_k^+)^2)^{3/2} = \varsigma_1(T)/\sqrt{K_3(\lambda^+)}$, which is positive if $\varsigma_1(T) > 0$; with $\beta_j \geq 0$, $\rho_S(h) \geq 0$ and Lemma A.1 give $\varsigma_1(T) > 0$. Since $\mathrm{tr}(((C - C^*)P)^2) = \|\tilde{C} - \tilde{C}^*\|^2$, C^+ is the projection of C^* onto $\{C \succeq 0\}$ in that metric. (c) By (a), W is optimal if and only if $W \propto P^{-1}\Gamma_s P^{-1}$, that is, $\Gamma_s = cPWP$ with $c > 0$; then $\tilde{C} \propto P^{1/2}WP^{1/2} \succeq 0$, so W is also optimal in (b). □

Proof of Corollary 5. By Lemma 1(b), the centred eigen-trend P&L $\Phi_{k,t} - \pi_k = \mu_k \phi_{k,t}$ are independent across k, and by assumption the means do not change their joint distribution. Hence $\mathbb{E}[F_T^\omega] = T\sum_k \omega_k \pi_k$, $\mathrm{Var}[F_T^\omega] = T\sum_k \omega_k^2 \mu_k^2$ (Lemma A.1), and $\kappa_3(F_T^\omega) = \kappa_3(\Psi)\sum_k \omega_k^3 \mu_k^3$, as in the proof of Corollary 4. Writing $\pi_k = \mu_k \mathrm{SR}_k$ gives Equation (15). Let $u_k = \omega_k \mu_k \geq 0$. For every $u \geq 0$, $u \neq 0$, $\sum_k u_k \mathrm{SR}_k \leq \sum_k u_k \mathrm{SR}_k^+ \leq \|u\| \, \|\mathrm{SR}^+\|$ by the Cauchy–Schwarz inequality, with equality if and only if $u \propto \mathrm{SR}^+$ (when $\mathrm{SR}^+ \neq 0$). Hence $\omega_k \propto \mathrm{SR}_k^+/\mu_k$ and $\mathrm{Sh} = \sqrt{T}\|\mathrm{SR}^+\|$. Substituting $u \propto \mathrm{SR}^+$ into the skewness gives $\varsigma_1(T)\sum_k(\mathrm{SR}_k^+)^3/(\sum_k(\mathrm{SR}_k^+)^2)^{3/2} = \varsigma_1(T)/\sqrt{K_3(\mathrm{SR}^+)}$. The classic weights $\omega = \mathbf{1}$ are optimal if and only if $\mathrm{SR}_k^+ = c\mu_k$ with $c > 0$. If $\pi_k = \bar{\pi}\mu_k$ with $\bar{\pi} > 0$, then $\mathrm{SR}_k = \bar{\pi}$ and $\omega_k \propto 1/\mu_k = \mu_k^{\alpha - 1}$ with $\alpha = 0$. □

Proof of Proposition 5. (a) M_γ is symmetric, so Q_γ is orthonormal. The proof of Lemma 1(a) uses only $\sum_k q_k q_k^\top = I$, and Lemma 2 holds for every orthonormal basis. For $\gamma = 0$, $M_0 = M/\mathrm{tr}(M)$ has the eigenvectors of M. (b) For a symmetric matrix B and orthonormal $N \times m$ frames U, $\max_U \mathrm{tr}(U^\top BU)$ equals the sum of the m largest eigenvalues of B and is attained by the corresponding eigenvectors (Ky Fan's maximum principle). With $B = M_\gamma$, $\mathrm{tr}(U^\top M_\gamma U) = \mathcal{V}(U) + \gamma\mathcal{R}(U)$. (c) Let $\gamma < \gamma'$ and let $U = U_\gamma$ and $U' = U_{\gamma'}$ be maximizers. Optimality gives $\mathcal{V}(U) + \gamma\mathcal{R}(U) \geq \mathcal{V}(U') + \gamma\mathcal{R}(U')$ and $\mathcal{V}(U') + \gamma'\mathcal{R}(U') \geq \mathcal{V}(U) + \gamma'\mathcal{R}(U)$. Adding the two inequalities, $(\gamma' - \gamma)(\mathcal{R}(U') - \mathcal{R}(U)) \geq 0$, so $\mathcal{R}(U') \geq \mathcal{R}(U)$; the first inequality then gives $\mathcal{V}(U) - \mathcal{V}(U') \geq \gamma(\mathcal{R}(U') - \mathcal{R}(U)) \geq 0$. The argument holds for any selection of maximizers when eigenvalues are tied. □

B. Pre-Specification

The design of the channel estimators and of the eigen-trend attribution was recorded in a dated research log and committed to version control before the analysis was run. The log also records what had been seen before: on the full sample, the share of the largest eigenvalue in $\sum \mu^3$ and the contribution of the first principal component of the 55-day market P&L to the portfolio skewness. Two further studies, run under separate pre-registered protocols and not part of this article, are documented in the same log: a forecast of co-trending from the state of the trends, and a supervised convexity filter in the spirit of Giglio, Xiu, and Zhang (2024), which did not predict the skewness of portfolios out of sample better than the spectral law.

Backtest and futures curves. The backtest of Section E and the curve decomposition of Section 5.1 were specified in the same log before they were run: the tilt rule, the values $\alpha \in \{0.5, 1, 1.5, 2\}$, the benchmarks, the costs, the portfolio volatility target, the sample, the metrics, the bootstrap, and the hypotheses that the realized Sharpe ratio falls and the skewness rises with α, that $\alpha = 2$ has higher crisis returns and a higher convexity coefficient than $\alpha = 1$, and that the roll yield contributes to the mean but not to the skewness of the trend P&L. The Sharpe part of the first hypothesis is rejected (the gross Sharpe ratio rises from $\alpha = 0.5$ to 1.5); its skewness part could not be identified under the pre-specified ex-ante target; $\alpha = 2$ has higher crisis returns than $\alpha = 1$ (8.1% against 7.3%) but not a higher convexity coefficient (1.65 against 1.67); the roll-yield hypothesis holds. The commodity panel had not been used before. Three changes were made after the specification and are reported as such. First, a data check of the pre-specified roll rule, which compares the front price with the previous day's front and second contracts, found on average 78 roll days per market and year: for flat curves the gap between contracts is smaller than the daily price move, and the rule fires on about every second day. Before any strategy statistic was computed, the roll dates were therefore taken from the ratio-adjusted generic series of the same contracts, which ends in July 2024; on the detected dates the roll term equals the lagged gap between the first two contracts (regression slope 0.96 to 1.04 for 35 of 37 markets). Second, after the primary result, the gross Sharpe ratio, the tilt of the reference signal, and the eigen-trend P&L by rank were added (Exhibit A3). Third, after the reconciliation of the backtest with Section 4, the comparison of risk rules in Exhibit 9 and Corollary 3 were added.

Panel A: Eigen-trend tilting of the reference signal LS(250,20)

Strategy	Sharpe net	Sharpe gross	Skewness Quarter	Skewness 55-day	Cost (%)	Turnover
Reference system LS(250,20)	0.87	1.03	0.20	0.07	1.7	68
LS signal, eigen-trend tilt $\alpha = 1$ (no caps)	0.88	1.04	0.23	0.08	1.7	72
LS signal, tilt $\alpha = 1.5$	0.71	0.84	0.21	0.05	1.4	59
LS signal, tilt $\alpha = 2$	0.52	0.64	0.16	0.01	1.3	56

Panel B: Where the trend premium sits: eigen-trend P&L by eigenvalue rank

Eigenvalue rank	Share of mean (%)	Share of variance (%)	Sharpe	Skewness 55-day
1	31.6	49.0	0.31	2.44
2	41.2	16.8	0.68	1.73
3–5	25.6	11.4	0.52	1.80
6–10	-9.3	2.9	-0.37	1.28
11–20	14.4	2.6	0.61	0.74
21–73	-3.5	1.3	-0.21	0.48
All	100.0	100.0	0.68	1.32

Exhibit A3: **Backtest: Post-Hoc Diagnostics.** Not part of the pre-specified design (declared in the research log after the primary result and before these numbers). Panel A: the tilt of Exhibit A10 applied to the signal of the reference system (long span 250, short span 20), without the signal and weight caps of the reference system; same costs, volatility target and sample. Panel B: daily P&L $\Phi_{k,t}$ of the trend follower on eigen-portfolio k of the equal-risk single-filter portfolio (Lemma 1), basis re-estimated monthly from the preceding 756 days, before costs, January 2001–July 2026, grouped by eigenvalue rank. Shares of the variance need not sum to 100% because the realized eigen-trends are correlated.

Risk-premium PCA. The design of Section 6 was recorded in the log before any computation: the estimators, the primary specification ($\gamma = 1$, all basis vectors, premium weights), the two sample splits, the four strategies, the bootstrap, the decision rule, the hypotheses, and the two secondary tests (commodity panel, implementable version). The hypotheses were (H-Q1) that the RP basis concentrates more premium in its first five vectors out of sample, which holds in the forward split only, and (H-Q2) that premium weighting lowers skewness relative to the classic CTA, which the data neither confirm nor reject (higher forward, lower reverse and in the implementable version). A second design, fixed before the plug-in filter was computed, stated that the risk-premium PCA does at least as well as the plug-in $\hat{C}^+$ out of sample in both splits, which holds in the forward split only (neither difference significant), and that the implementable risk-premium PCA has a higher gross Sharpe ratio than LS(250,20), which holds (1.23 and 1.31 against 0.75 and 1.03). After the first run, three bugs in the exhibit script were fixed (table formatting; a missing correlation for a market without returns in a training sample, which affects only the reverse split of the commodity panel; a turnover column whose level is not comparable across risk rules), and the decomposition of the test-sample means into own and lead–lag premia (Lemma 2(a)) was added as a diagnostic. The primary results were unchanged. Exhibit A4 reports the sensitivity to γ and to the number of basis vectors traded; it was not used for selection.

		Test Sharpe ratio		Test skewness (55-day)	
γ	m	Forward	Reverse	Forward	Reverse
0	3	0.73	1.20	1.23	2.00
0	5	0.71	1.20	1.23	2.00
0	10	0.93	1.03	1.02	1.80
0	all	0.76	0.97	0.58	1.54
0.5	3	1.12	1.19	1.45	2.05
0.5	5	1.05	1.26	2.03	1.41
0.5	10	1.14	1.43	1.01	0.87
0.5	all	1.20	1.48	1.01	0.83
1	3	1.15	1.20	1.46	2.07
1	5	1.11	1.28	2.33	1.39
1	10	1.19	1.44	1.32	0.83
1	all	1.28	1.49	1.36	0.76
2	3	1.16	1.20	1.46	2.07
2	5	1.14	1.29	2.43	1.39
2	10	1.21	1.45	1.49	0.81
2	all	1.29	1.51	1.46	0.73
5	3	1.17	1.20	1.46	2.07
5	5	1.15	1.30	2.45	1.39
5	10	1.23	1.45	1.56	0.80
5	all	1.30	1.51	1.58	0.71
10	3	1.17	1.20	1.46	2.08
10	5	1.16	1.30	2.44	1.39
10	10	1.23	1.45	1.59	0.80
10	all	1.30	1.51	1.59	0.70

Exhibit A4: **Risk-Premium PCA: Sensitivity.** Premium-weighted eigen-trends (Algorithm 1) for the weight γ on the premium matrix and the number m of basis vectors traded. Forward: train 2001–2012, test 2013–2026; reverse: the other way round. Design and data as in Exhibit 10; the grid was not used to choose the primary specification ($\gamma = 1$, all vectors).

Research question and confirmatory tests. After the results above were known, the research question was stated in the log as it appears in the introduction, together with three tests fixed before they were run: (i) the spectral law predicts the test-sample skewness of the filters S0 to S4, with the same ranking up to one transposition and errors below 0.5 for S0, S1, S2, and S4; (ii) no 90% bootstrap interval of the skewness difference between a premium-aware filter and the classic CTA lies entirely below zero; (iii) on the commodity panel, the ex-ante risk rule keeps at most a quarter and the ex-post rule at least half of the skewness without a rule. Test (i) fails: the law predicts the risk-based filters but understates the skewness of premium-weighted ones (Section 6). Test (ii) holds, but because its point estimates were known when it was stated, we treat it as descriptive; its intervals are wide. Test (iii) confirms the ordering but misses both thresholds narrowly (26% and 43%). Two further analyses were specified in the log before they were run: the descriptive attribution of Section E.7 and the robustness package of Section 7.1 and Section E.8 (walk-forward, null Monte Carlo, deflated Sharpe ratio, Ledoit–Wolf tests with Holm correction, risk-rule intervals, measures of skewness, block lengths, leaving out asset classes, and power). In the

null Monte Carlo, RP-PCA is undefined when the training premium is not positive; this was found when the first run stopped, before any null result was seen, and the null distribution was then made conditional on a positive premium, with 600 panels drawn instead of 300. Two outcomes differ from the stated expectations: the plug-in filter beats RP-PCA in the walk-forward test before costs, and the sign of the skewness difference between S2 and S0 varies across measures, as the power analysis leads one to expect.

C. Trend-Following Systems and Validation

The European system holds $H_{i,t} = (\sigma^*/\hat{\sigma}_{i,t})\,\min(\max(S_{i,t}, -3), 3)$ in the capped reference implementation (the single filter of Sections 4 to 6 is uncapped), with $\sigma_{i,t}$ an exponentially weighted volatility (span 33 days), σ^* an instrument volatility target, and $S_{i,t}$ the variance-preserving single or long–short EWMA filter of normalized returns. The time-series momentum system averages the signs of normalized returns over ten periods of ten days (Moskowitz et al., 2012); the American system trades the crossover of two price averages with an entry band and a trailing stop measured in average true ranges. All estimates are point in time: the volatility used to normalize the return of day t is estimated up to day $t-1$, and positions decided at the close of day t earn the return of day $t+1$.

The implementation was validated against the published reference implementation of Sepp and Lucic (2026): for five configurations of the three systems on the 84 markets, the daily gross and net P&L and the gross exposure agree to within 10^{-12}. Two defaults differ deliberately: our volatility estimate is seeded point in time, whereas the reference seeds it with the full-sample variance, and the cost rate changes on the day after the period boundaries of the cost schedule.

D. Data

Exhibit A5 lists the 84 markets of the main panel with asset class, first date, and transaction cost. The series are continuous excess-return series built from generic first contracts (Carver, 2023).

Ticker	Market	Class	Start	Cost	Ticker	Market	Class	Start	Cost
ES1 Index	S&P 500	Equities	1962-01	6	SF1 Curncy	CHF/USD	FX	1971-09	3
DM1 Index	Djia	Equities	1962-01	6	CD1 Curncy	CAD/USD	FX	1971-09	3
GX1 Index	Dax	Equities	1962-01	6	JY1 Curncy	JPY/USD	FX	1971-09	3
TP1 Index	Topix	Equities	1962-01	6	SE1 Curncy	SEK/USD	FX	1971-09	3
NO1 Index	Nikkei225	Equities	1970-01	6	NO1 Curncy	NOK/USD	FX	1971-09	3
HI1 Index	Hang Seng	Equities	1970-01	6	EC1 Curncy	EUR/USD	FX	1978-05	3
RTY1 Index	Rus2000	Equities	1978-12	6	AD1 Curncy	AUD/USD	FX	1978-05	3
PT1 Index	S&P/tsx60	Equities	1982-01	6	NV1 Curncy	NZD/USD	FX	1978-05	3
EO1 Index	Aex	Equities	1983-01	6	PE1 Curncy	MXN/USD	FX	1990-01	3
Z 1 Index	Ftse100	Equities	1983-12	6	HO1 Comdty	Nyh Ulsd	Energy	1986-07	10
NQ1 Index	NAS/DAQ	Equities	1985-02	6	CO1 Comdty	Brent	Energy	1988-06	10
VG1 Index	Estoxx50	Equities	1986-12	6	CL1 Comdty	Wti	Energy	1989-01	10
QC1 Index	Omx30	Equities	1986-12	6	QS1 Comdty	GAS/OIL	Energy	1989-07	10
CF1 Index	Cac	Equities	1987-07	6	NG1 Comdty	Nat Gas	Energy	1990-04	10
MES1 Index	Msci Em	Equities	1987-12	6	XB1 Comdty	Gasoline	Energy	2005-10	10
JGS1 Index	Nifty 50	Equities	1990-07	6	CUA1 Comdty	Ethanol	Energy	2007-09	10
XP1 Index	Asx200	Equities	1992-05	6	GC1 Comdty	Gold	Metals	1975-01	10
HC1 Index	Hscei	Equities	1993-07	6	SI1 Comdty	SIL/VER	Metals	1975-01	10
SM1 Index	Smi	Equities	1998-09	6	PL1 Comdty	Platinum	Metals	1986-04	10
TWT1 Index	Taiex	Equities	2000-06	6	PA1 Comdty	Palladium	Metals	1986-04	10
XU1 Index	China A50	Equities	2007-01	6	HG1 Comdty	COP/PER	Metals	1988-12	10
TU1 Comdty	Ust 2y	Bonds	1962-01	1	SCO1 Comdty	Tsi Iron Ore	Metals	2016-01	10
FV1 Comdty	Ust 5y	Bonds	1962-01	1	C 1 Comdty	Corn	Agriculture	1959-07	10
TY1 Comdty	Ust 10y	Bonds	1962-01	1	W 1 Comdty	Wheat Chicagoe	Agriculture	1959-07	10
UXY1 Comdty	Ust 10y Ultra	Bonds	1962-01	1	S 1 Comdty	Soybeans	Agriculture	1959-07	10
US1 Comdty	Ust Bond	Bonds	1962-01	1	SM1 Comdty	Soyb Meal	Agriculture	1964-01	10
RX1 Comdty	Ger 10y	Bonds	1962-03	1	BO1 Comdty	Soyb Oil	Agriculture	1964-01	10
G 1 Comdty	Gilt	Bonds	1962-03	1	CC1 Comdty	Cocoa	Agriculture	1965-01	10
CN1 Comdty	Can 10y	Bonds	1962-03	1	CT1 Comdty	Cotton 2	Agriculture	1967-06	10
JB1 Comdty	Jb 10y	Bonds	1985-10	1	LC1 Comdty	Live Cattle	Agriculture	1972-01	10
XM1 Comdty	Aus 10y	Bonds	1987-09	1	SB1 Comdty	Sugar 11	Agriculture	1972-01	10
OE1 Comdty	Ger 5y	Bonds	1991-10	1	KC1 Comdty	Coffee C	Agriculture	1972-08	10
DU1 Comdty	Ger 2y	Bonds	1992-04	1	KW1 Comdty	Wheat Kansas	Agriculture	1973-01	10
UB1 Comdty	Ger 30y	Bonds	1992-04	1	FC1 Comdty	Feeder Cattle	Agriculture	1978-01	10
OAT1 Comdty	Oat 10y	Bonds	1992-04	1	RS1 Comdty	CAN/OLA	Agriculture	1982-01	10
IK1 Comdty	Btp 10y	Bonds	1992-04	1	LH1 Comdty	Lean Hogs	Agriculture	1986-04	10
WN1 Comdty	Ust Ultra	Bonds	2010-01	1	MWE1 Comdty	Wheat Minneapolis	Agriculture	1988-12	10
IR4 Comdty	Aus 90d	STIR	1989-06	1	QW1 Comdty	White Sugar	Agriculture	1989-01	10
ER4 Comdty	Euribor 3m	STIR	1998-12	1	QC1 Comdty	London Cocoa	Agriculture	1989-01	10
SFR5 Comdty	Sofr 3m	STIR	2018-05	1	CA1 Comdty	Wheat Mil	Agriculture	2007-01	10
SFI5 Comdty	Sonia 3m	STIR	2018-06	1	IJ1 Comdty	Rapeseed	Agriculture	2007-01	10
BP1 Curncy	GBP/USD	FX	1971-09	3	DF1 Comdty	Robusta	Agriculture	2008-01	10

Exhibit A5: **The Futures Markets.** The 84 markets of the main panel, built from Bloomberg generic first contracts as continuous excess-return series. Start: first month with a price. Cost: one-way transaction cost in basis points of notional, the volume-based schedule of Hurst et al. (2017, Exhibit B1), at the end of the sample. The main tests use the 73 markets with data before July 1999.

Commodity panel. The separately constructed sample of Section 5.1 consists of Bloomberg generic first and second contracts of 52 commodities (energy, metals, grains, softs, livestock), daily closing prices without adjustment, and the ratio-adjusted generic first contract, which fixes the roll dates. The daily log return of the rolled first contract is $\ln \mathsf{F}_1(t) - \ln \mathsf{F}_1(t-1)$ on days without a roll and $\ln \mathsf{F}_1(t) - \ln \mathsf{F}_2(t-1)$ on roll days; the difference, $\ln \mathsf{F}_1(t-1) - \ln \mathsf{F}_2(t-1)$, is the roll yield earned on that day. A roll day is a day on which $\ln(\mathrm{Adj}_t / F_{1,t})$ changes by more than 10^{-3}. Non-positive prices (WTI crude

oil on 20 April 2020) give missing returns. The tests use the 37 commodities with returns before July 1999 and a detected roll; unleaded gasoline (discontinued) and lumber (no roll in the adjusted series) are excluded. The markets are heating oil, WTI and Brent crude oil, natural gas, propane, and gasoil; oats, canola, rough rice, soybean meal, soybean oil, soybeans, corn, palm oil, and Chicago and Kansas City wheat; feeder cattle, live cattle, pork bellies, and lean hogs; gold, silver, platinum, palladium, copper, and LME aluminium, lead, nickel, tin, and zinc; and coffee, sugar, cocoa, orange juice, cotton, milk, and rubber. They have on average 7.0 roll days per year. Pork bellies end in 2011 and enter with zero P&L thereafter; propane and pork bellies trade thinly, with no price change on most days.

E. Additional Results

E.1 Co-trending in correlations and across asset classes

Panel A: Mean over pairs of markets

Pairs	P&L corr.	ρ^2	Excess	Surrogates (95%)	Co-trending	p
Within asset classes	0.278	0.219	0.059	[0.041, 0.053]	0.012	0.000
Across asset classes	0.054	0.021	0.033	[0.003, 0.010]	0.027	0.000

Panel B: Co-trending by pair of asset classes (largest eight)

Asset classes	Co-trending	p
Equities – Energy	0.048	0.000
Bonds – FX	0.047	0.000
Metals (within)	0.045	0.000
FX – Energy	0.043	0.000
Equities – FX	0.043	0.000
Equities – Bonds	0.039	0.000
Metals – Agriculture	0.036	0.000
Energy – Metals	0.035	0.000

Exhibit A6: **Co-Trending.** 73 markets, January 2001–July 2026; trend P&L of a single EWMA filter (span 100 days) on volatility-normalized returns. Under Proposition 1 the correlation of the trend P&L of two markets equals the squared correlation ρ^2 of their returns. Excess: mean P&L correlation minus mean ρ^2. Surrogates: 200 multivariate phase-randomized versions of the returns (Prichard and Theiler, 1994), which keep volatility, autocorrelation, cross-correlation, and lead-lag relations but remove correlations that depend on the state of the trends; the 95% range of their excess is shown. Co-trending: excess minus the mean surrogate excess. p: share of surrogates with at least the observed excess.

Under Proposition 1, the P&L correlation of two trend followers equals the squared correlation of their returns. Across asset classes, the mean P&L correlation of two trend followers is 0.054, against 0.021 implied by their return correlations (Exhibits A6 and A7). Linear dependence, as reproduced by the surrogates, explains 0.006 of the excess (95% range 0.003 to 0.010); the remaining 0.027 is larger than in any of 200 surrogates. Within

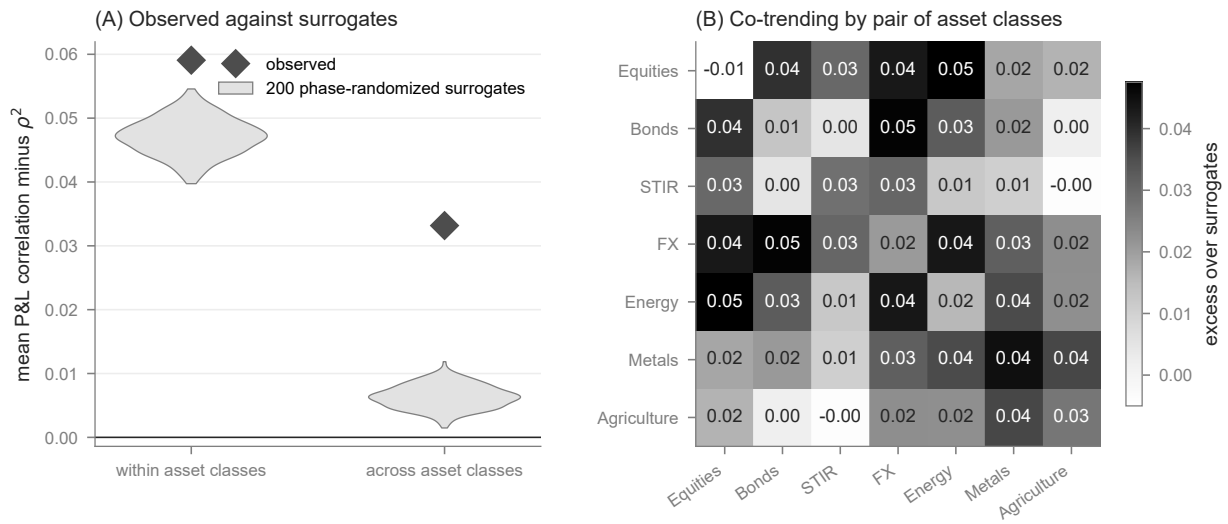


Exhibit A7: **Co-Trending across Asset Classes.** Panel A: mean excess of the trend P&L correlation over ρ^2 within and across asset classes, observed and in 200 phase-randomized surrogates. Panel B: co-trending (observed excess minus mean surrogate excess) by pair of asset classes. Design as in Exhibit A6.

asset classes, the state-dependent excess is less than half as large (0.012). The strongest co-trending occurs between the asset classes that move together in macroeconomic shocks: equities and energy, bonds and currencies, equities and currencies, and equities and bonds. The result confirms for P&L what Moskowitz et al. (2012) document for the correlations of trend strategies and Safari and Schmidhuber (2026) for correlations conditional on trends.

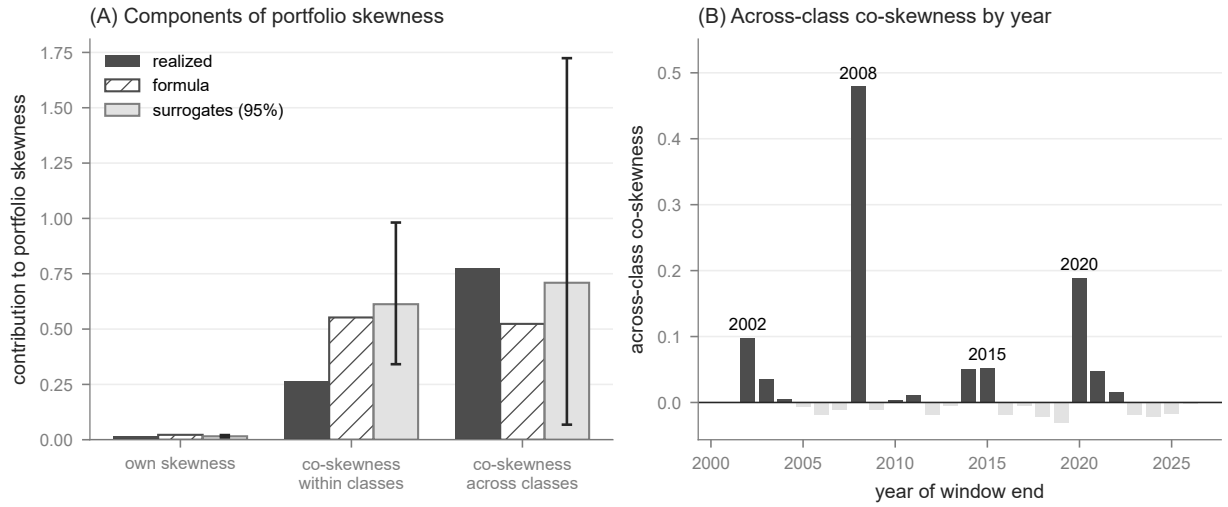


Exhibit A9: **Where the Convexity Comes From.** Panel A: contributions of own skewness, within-class co-skewness, and across-class co-skewness to the skewness of the 55-day portfolio return; realized, Proposition 1, and surrogates with 95% ranges. Panel B: across-class co-skewness of the 55-day windows ending in each year. Design as in Exhibit A8.

Panel A: Contributions to the skewness of the 55-day portfolio return

Component	Realized	Formula	Surrogates (95%)	p
Own skewness of each market	0.02	0.02	[0.01, 0.02]	0.520
Co-skewness within asset classes	0.27	0.55	[0.34, 0.98]	1.000
Co-skewness across asset classes	0.78	0.52	[0.07, 1.72]	0.365
Portfolio skewness	1.06	1.10	[0.78, 2.23]	0.725

Panel B: Years with the largest across-class co-skewness

Year	Contribution	Share of total
2008	0.48	62%
2020	0.19	24%
2002	0.10	13%
2015	0.05	7%
2014	0.05	6%

Exhibit A8: **Where the Convexity Comes From.** 73 markets, January 2001–July 2026, equal-risk portfolio of single-filter trend P&L (span 100 days). The third central moment of overlapping 55-day portfolio returns splits exactly into the markets' own third moments, co-moments of markets in the same asset class, and co-moments across asset classes; divided by the cubed portfolio standard deviation, the three parts add up to the portfolio skewness. Formula: Proposition 1 with the sample correlation matrix, under which the parts are proportional to $\sum \rho_{ij}\rho_{jk}\rho_{ki}$ over own, within-class, and across-class index triples. Surrogates: 200 multivariate phase-randomized versions of the returns (Exhibit A6); p: share of surrogates with at least the realized value. Panel B: across-class contribution of the 55-day windows ending in each calendar year.

Exhibits A8 and A9 split the third moment of the 55-day return of the 73-market portfolio by asset class. The markets' own skewness contributes almost nothing (0.015), co-skewness within asset classes 0.27, less than half of the formula (0.55), and co-skewness across asset classes 0.78 of the total of 1.06, three quarters of the portfolio skewness and half as much again as the formula (0.52). It is concentrated in crises: the 55-day windows ending in 2008 contribute 62% of it and those ending in 2020 a further 24%. Against the surrogates, however, the across-class co-skewness is not significant ($p = 0.37$, 95% range of the surrogates 0.07 to 1.72): in the third moment, the excess over the formula is consistent with persistent trends shared across asset classes, and the state-dependent part that is clear in correlations cannot be separated from sampling noise. The channel estimator of Section 5 is the sharper instrument.

E.2 Eigen-trend tilting against benchmarks

The backtest of the tilt was specified before it was run: the classic single filter and its tilts $\alpha \in \{0.5, 1.5, 2\}$, the reference system LS(250,20), equal risk per asset class, the SG Trend Index, and a 60/40 portfolio, all trend strategies scaled to 10% volatility with the ex-ante rule and net of costs (Exhibits A10 and A11). Before costs, the classic single-filter CTA earns a Sharpe ratio of 0.80 from 2001 to 2026; after costs it earns 0.10. The single filter

Panel A: Performance, January 2001–July 2026 (net of costs)

Strategy	Return (%)	Vol. (%)	Sharpe net	Sharpe gross	Max DD (%)	Skewness Quarter	Skewness 55-day	Crisis (%)	Convexity c (t)
Eigen-trend tilt $\alpha = 0.5$	-5.7	9.9	-0.58	0.59	-84.2	0.10	0.00	3.6	1.32 (2.5)
Classic equal-risk CTA ($\alpha = 1$)	1.0	10.3	0.10	0.80	-55.8	0.21	0.04	7.3	1.67 (3.0)
Eigen-trend tilt $\alpha = 1.5$	2.6	10.4	0.25	0.79	-43.9	0.22	0.07	8.3	1.70 (3.0)
Eigen-trend tilt $\alpha = 2$	2.1	10.4	0.20	0.69	-43.3	0.17	0.06	8.1	1.65 (3.1)
Reference system LS(250,20)	9.3	10.7	0.87	1.03	-22.1	0.20	0.07	7.7	1.02 (2.0)
Asset-class risk parity	-0.8	10.5	-0.08	0.86	-66.1	0.19	0.11	7.7	2.00 (3.9)
SG Trend Index	6.0	12.6	0.48	–	-23.0	0.39	0.55	5.8	1.43 (2.2)
60/40 ES/TY	5.8	10.7	0.54	–	-33.6	-0.38	-0.97	-7.3	0.46 (5.5)

Panel B: Eigen-trend tilting relative to $\alpha = 1$: prediction, realization, bootstrap

	Sharpe ratio Pred.	Sharpe ratio Real.	Skewness (55-day) Pred.	Skewness (55-day) Real.	Δ Sharpe	90% interval	Δ skew. (qu.)	90% interval	Cost (%) / turnover
Eigen-trend tilt $\alpha = 0.5$	1.79	-5.71	0.41	0.04	-0.68	[-0.83, -0.54]	-0.11	[-0.26, 0.22]	11.5 / 453
Classic equal-risk CTA ($\alpha = 1$)	1.00	1.00	1.00	1.00	–		–		7.1 / 276
Eigen-trend tilt $\alpha = 1.5$	0.60	2.46	1.42	1.76	0.15	[0.05, 0.24]	0.01	[-0.16, 0.20]	5.6 / 227
Eigen-trend tilt $\alpha = 2$	0.43	2.00	1.63	1.52	0.10	[-0.07, 0.26]	-0.04	[-0.30, 0.33]	5.0 / 215

Panel C: Investor value, 80% 60/40 and 20% strategy, monthly rebalancing

Sleeve	Sharpe	Skewness	Max DD (%)	CE (%)
Eigen-trend tilt $\alpha = 0.5$	0.48	-0.35	-21.6	2.1
Classic equal-risk CTA ($\alpha = 1$)	0.67	-0.30	-19.9	3.5
Eigen-trend tilt $\alpha = 1.5$	0.72	-0.29	-19.6	3.8
Eigen-trend tilt $\alpha = 2$	0.70	-0.30	-19.7	3.7
Reference system LS(250,20)	0.88	-0.40	-21.2	5.1
Asset-class risk parity	0.62	-0.24	-19.3	3.1
SG Trend Index	0.77	-0.33	-21.9	4.3
60/40 alone	0.62	-0.52	-25.2	3.2

Exhibit A10: **Backtest: Eigen-Trend Tilting against Benchmarks.** 73 futures markets, January 2001–July 2026, daily, net of the costs of Hurst et al. (2017, Exhibit B1). Trend strategies are scaled to 10% volatility with an EWMA estimate (span 60) of the covariance of their current positions. Tilt: $\omega_k \propto \mu_k^{\alpha-1}$ (Corollary 4), eigenvectors re-estimated monthly (Section 3); $\alpha = 1$ is the classic equal-risk CTA, single filter, span 100. LS(250,20): reference system of Sepp and Lucic (2026). SG Trend Index: net of fees, including interest on collateral; all other series are excess returns. Crisis: mean return in the 11 worst quarters of the S&P 500 futures. Convexity: c in $r_q = a + b\, r_{\mathrm{ES},q} + c\, r_{\mathrm{ES},q}^2$, Newey–West t. Panel B: realized: net Sharpe ratio and 55-day skewness under the ex-ante target; predictions $\sqrt{K_2(\mu^\alpha)}$ and $1/\sqrt{K_3(\mu^\alpha)}$ from the average spectrum; differences to $\alpha = 1$ with stationary block-bootstrap intervals (mean block 63 days, 1,000 draws); cost: annual cost drag; turnover: annual notional turnover in multiples of capital. Panel C: CE, annualized certainty-equivalent return at constant relative risk aversion 5. Design fixed before the analysis (Section B of the Appendix).

without signal and weight caps trades 276 times its capital a year, much of it in short-term interest-rate futures that the equal-risk rule leverages, and costs 7.1% a year. The reference system, with the caps of Sepp and Lucic (2026), trades a quarter as much and earns 0.87 after costs; the SG Trend Index earns 0.48 after fees, and the 60/40 portfolio 0.54.

Tilting towards the leading eigen-trends lowers turnover and costs (5.6% at $\alpha = 1.5$, 5.0% at $\alpha = 2$) and raises the net Sharpe ratio to 0.25 at $\alpha = 1.5$ (difference to $\alpha = 1$: 0.15, 90% bootstrap interval 0.05 to 0.24) and 0.20 at $\alpha = 2$; tilting towards the small eigen-trends, $\alpha = 0.5$, lowers it to -0.58. Gross of costs, the Sharpe ratio is 0.59, 0.80, 0.79, and 0.69 for $\alpha = 0.5$, 1, 1.5, and 2, against 1.43, 0.80, 0.48, and 0.35 if the trend premium were proportional to eigen-trend risk (Panel B of Exhibit A11). Exhibit A3 measures the

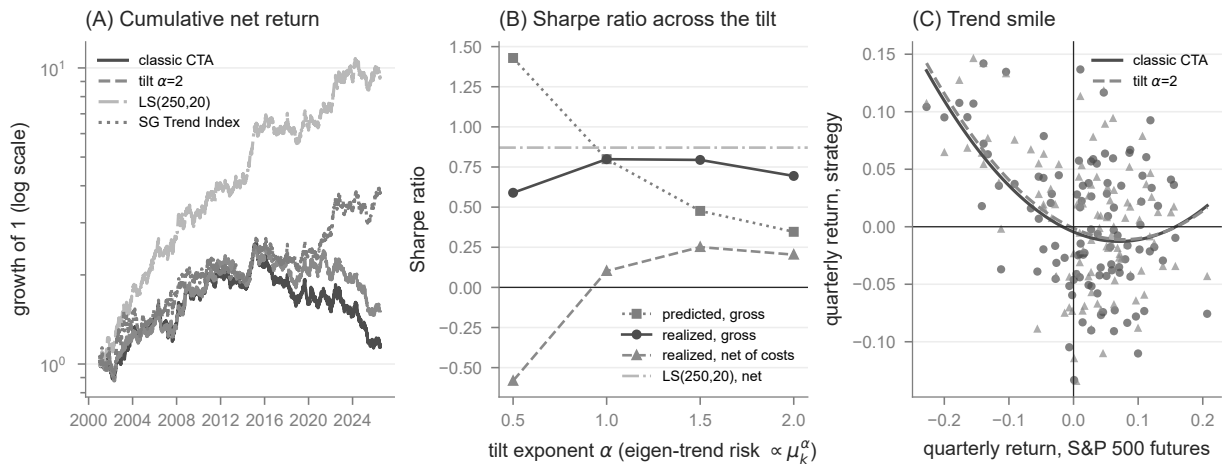


Exhibit A11: **Backtest of Eigen-Trend Tilting.** Panel A: cumulative net return (10% volatility target, log scale) of the classic CTA, the tilt $\alpha = 2$, the reference system LS(250,20), and the SG Trend Index. Panel B: Sharpe ratio of the tilted portfolios against the tilt exponent α, gross and net of costs, and the static prediction of Corollary 4(c), scaled to the gross Sharpe ratio at $\alpha = 1$. Panel C: quarterly returns of the classic CTA and the tilt $\alpha = 2$ against the S&P 500 futures, with quadratic fits. Design as in Exhibit A10.

premium eigen-trend by eigen-trend: the five leading eigen-trends earn 98% of the mean P&L with 77% of its variance, and the eigen-trends beyond the fifth, taken together, 2% of the mean with 7% of the variance. The small eigen-trends are long–short spreads between similar markets; they add Sharpe breadth in the benchmark, but in the data mainly turnover. Under the ex-ante rule, the skewness of all strategies is close to zero (55-day skewness 0.00 to 0.11), which Section 5.2 explains.

For an investor who holds trend following next to a 60/40 portfolio, the sleeve that adds most is the reference system: a 20% allocation raises the Sharpe ratio of the mix from 0.62 to 0.88, its skewness from −0.52 to −0.40, and its certainty-equivalent return at a relative risk aversion of five from 3.2% to 5.1% a year (Exhibit A10, Panel C). The SG Trend Index adds less (0.77 and 4.3%), and the tilt $\alpha = 1.5$ of the uncapped single filter adds 0.72 and 3.8%, more than the classic version of the same system (0.67 and 3.5%). Applied to the signal of the reference system, the tilt does not pay: its net Sharpe ratio falls from 0.88 to 0.71 and 0.52 for $\alpha = 1.5$ and 2 (Exhibit A3).

E.3 The classic filter: horizon, composition, and eigen-trends

The separation of horizon and cross-section is the structural implication of the proposition. Exhibit A12 computes the ratio of portfolio to single-market skewness at horizons from 5 to 252 days. For four markets, the ratio is between 0.65 and 0.69 at every horizon. For 16

markets and for all markets, it drifts down from about 0.6 at short horizons to about 0.45 at a year, but the slope on log horizon is not significant (−0.06, 95% interval −0.18 to 0.21), and at the main horizon the all-market ratio (0.49) is close to the formula (0.51).

	Single market		Ratio portfolio / single market			
T (days)	Closed form	Median	$N = 4$	$N = 16$	All 73	95% interval
5	1.00	1.15	0.65	0.57	0.57	[-0.24, 0.93]
10	1.50	1.67	0.66	0.66	0.69	[0.13, 1.01]
21	2.01	2.01	0.65	0.58	0.59	[0.25, 0.88]
42	2.32	2.10	0.68	0.52	0.49	[0.30, 0.72]
55	2.35	2.13	0.67	0.52	0.49	[0.31, 0.73]
84	2.29	2.00	0.69	0.51	0.46	[0.29, 0.70]
126	2.11	1.75	0.67	0.50	0.47	[0.26, 0.77]
189	1.87	1.49	0.67	0.46	0.42	[0.13, 0.83]
252	1.68	1.36	0.68	0.45	0.43	[0.03, 0.95]
Formula			0.58	0.48	0.51	

Exhibit A12: **The Horizon Separates from the Cross-Section.** Markets with data before mid-1999, January 2001–July 2026; trend P&L of a single EWMA filter (span 100 days), gross, equal risk, no portfolio volatility targeting. Single market: closed-form skewness of the T-day return under white noise (Sepp and Lucic, 2026, Proposition D.2) and the median sample skewness of overlapping T-day sums across markets. Ratio: skewness of the portfolio divided by the mean single-market skewness of its members, median over 100 random subsets for $N = 4$ and $N = 16$. Formula: median spectral factor $\mathrm{tr}(P^3)/\mathrm{tr}(P^2)^{3/2}$ (Proposition 1), which does not depend on T. 95% interval: moving-block bootstrap (blocks of 260 days, 500 replications). Slope of the all-market ratio on $\log T$: -0.057, 95% interval [-0.184, 0.207].

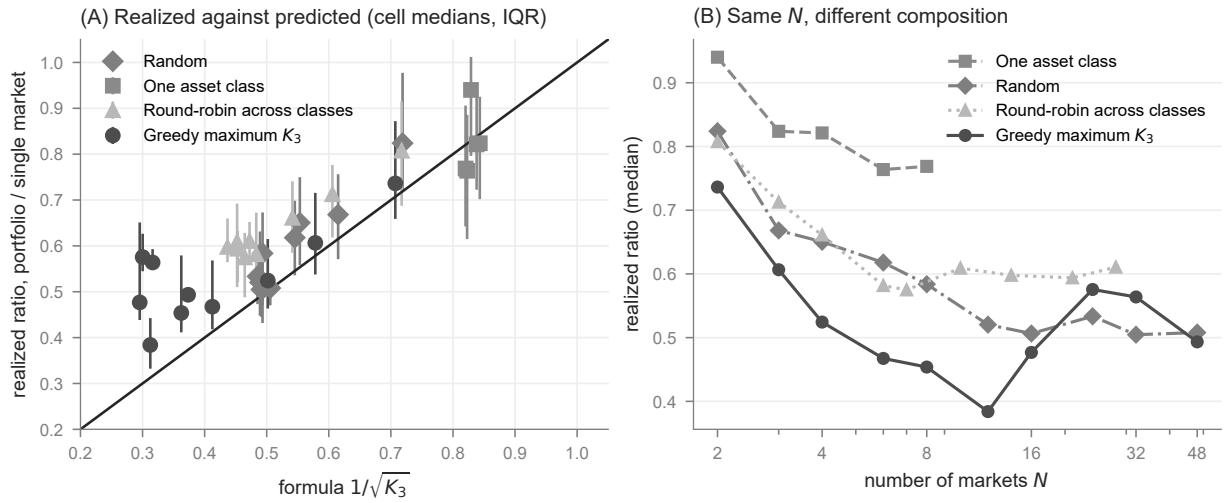


Exhibit A13: **Composition, Not Count.** Panel A: median realized ratio of portfolio to single-market skewness against the formula $1/\sqrt{K_3}$, by portfolio design and number of markets, with interquartile ranges. Panel B: median realized ratio by number of markets. Design as in Exhibit 4.

Exhibits A14 and A15 apply Lemma 1 to the data. The first eigen-trend carries 51% of the variance in the static benchmark but 71% of the convexity, the second 22% and 20%, and

Eigen-trend	Variance share	Convexity share	Static contribution	Own	Total
1	0.51	0.71	0.78	0.67	0.61
2	0.22	0.20	0.21	0.12	0.17
3	0.08	0.04	0.05	0.12	0.24
4	0.04	0.02	0.02	0.01	0.04
5	0.03	0.01	0.01	0.00	-0.01
6–73	0.12	0.02	0.02	0.01	0.02
All	1.00	1.00	1.10	0.92	1.06

Exhibit A14: **Where Convexity Sits.** 73 markets, January 2001–July 2026, equal-risk portfolio of single-filter trend P&L (span 100 days), 55-day horizon. By Lemma 1, the daily portfolio P&L is the sum of the P&L of trend followers on the eigen-portfolios of the sample correlation matrix (eigen-trends, sorted by eigenvalue μ_k). Variance share $\mu_k^2/\sum\mu^2$ and convexity share $\mu_k^3/\sum\mu^3$ (Corollary 2); static contribution: convexity share times the prediction of Proposition 1; own: third moment of the eigen-trend; total: $\mathrm{Cov}(\Phi_k, F^2)/\sigma_F^3$ (Proposition 6 of the Appendix), which adds up to the portfolio skewness; total minus own measures co-trending between eigen-trends.

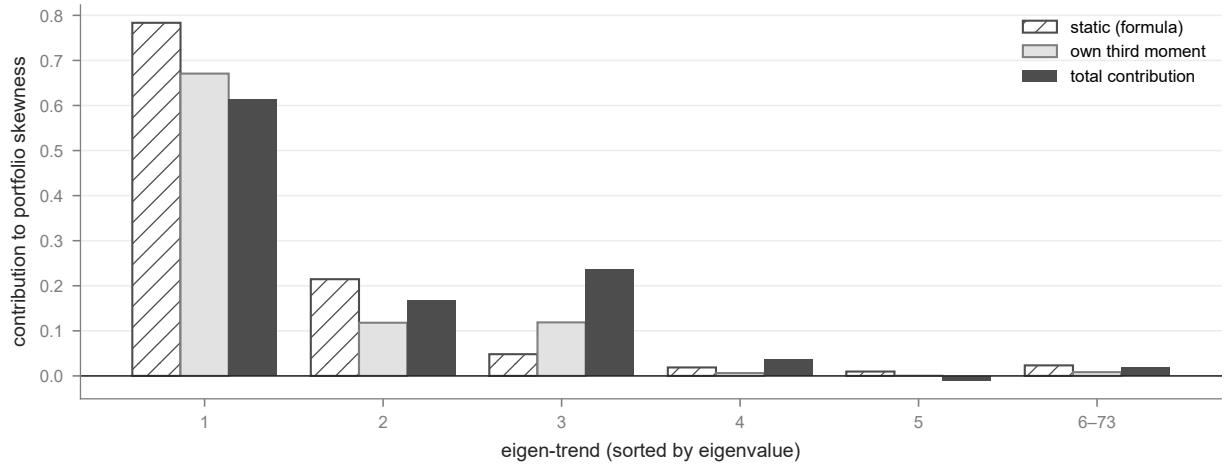


Exhibit A15: **Where Convexity Sits.** Contribution of the five largest eigen-trends to the skewness of the 55-day portfolio return, static prediction (Corollary 2 and Proposition 1), own third moment, and total contribution (Corollary 6). Design as in Exhibit A14.

all eigen-trends beyond the fifth 12% of the variance but 2% of the convexity (Corollary 2). Realized contributions follow the prediction: 0.61 for the first eigen-trend (static 0.78), 0.17 for the second (0.21), and 0.02 for all beyond the fifth (0.02). The exception is the third eigen-trend, which contributes 0.24 against 0.05 predicted. The eigen-trends' own third moments sum to 0.92 of the portfolio skewness of 1.06, so co-movement between eigen-trends adds 0.14. Convexity thus sits in about three dimensions, as a convexity breadth of 3.8 implies, and in the strongest ones, which the spectral law weights through the cubes of the eigenvalues.

Factor attribution. The attribution of convexity to factors used in Section 4 rests on the following results.

Let $X = (X_1, \ldots, X_N)^\top$ be the vector of T-day market P&L, $X_i = \sum_t f_{i,t}$, so that $F = F_T = w^\top X$. Since $\kappa_3(F) = \mathrm{Cov}(F, F^2) - 2\,\mathbb{E}[F]\,\mathrm{Var}[F]$, it is convenient to work with demeaned P&L: for $\mathbb{E}[F] = 0$,

$$\kappa_3(F) = \mathrm{Cov}(F, F^2) = \sum_{i=1}^{N} w_i \,\mathrm{Cov}(X_i, F^2). \tag{A1}$$

Market i contributes in proportion to the covariance of its P&L with the squared portfolio return. A factor model distributes these contributions over a few common factors.

Proposition 6 (Factor attribution of convexity)**.** *Let $X = Bv + u$ with factors $v \in \mathbb{R}^m$, loadings $B \in \mathbb{R}^{N\times m}$, and idiosyncratic P&L u, all with finite third moments, $\mathbb{E}[v] = 0$, $\mathbb{E}[u] = 0$, and u independent of v. For weights w let $a = B^\top w$, the portfolio's exposures to the factors. Then*

$$\kappa_3(F) = \sum_{k=1}^{m} a_k \,\mathrm{Cov}\big(v_k, F^2\big) + \kappa_3(w^\top u), \qquad \mathrm{Cov}\big(v_k, F^2\big) = \mathrm{Cov}\big(v_k, (a^\top v)^2\big). \tag{A2}$$

If, moreover, the components of u are independent with $|\kappa_3(u_i)| \le \bar\kappa$ and $\mathrm{Var}[u_i] \ge \underline{\sigma}^2 > 0$, then

$$\frac{|\kappa_3(w^\top u)|}{\mathrm{Var}[F]^{3/2}} \;\le\; \frac{\bar\kappa \,\sum_i |w_i|^3}{\big(a^\top \mathrm{Var}[v]\, a + \underline{\sigma}^2 \sum_i w_i^2\big)^{3/2}}, \tag{A3}$$

which, with equal weights $w_i = 1/N$, is $O(N^{-1/2})$ in general and $O(N^{-2})$ if $a^\top \mathrm{Var}[v] a$ is bounded away from zero.

Corollary 6 (Convexity factors)**.** *(a) A factor contributes to the skewness of a portfolio only through its covariance with the squared portfolio return: a factor with $\mathrm{Cov}(v_k, F^2) = 0$ contributes nothing, whatever its share in the variance; in particular, a symmetric factor independent of all other factors contributes $a_k^3 \kappa_3(v_k) = 0$. (b) Under Assumption 1, the eigen-trends of Lemma 1 are an exact factor decomposition, $F_T = \sum_k \Phi_{k,T}$ with $\Phi_{k,T} = \sum_t \Phi_{k,t}$, $a_k = 1$, and $u = 0$, and $\mathrm{Cov}(\Phi_{k,T}, F_T^2) = \kappa_3(\Phi_{k,T}) = c_k\, \kappa_3(F_T)$, with the shares c_k of Corollary 2.*

Part (b) gives an exact test of the benchmark that needs no factor estimation. Equation (4) holds in the data for any $\bar P$, so $\kappa_3(F_T) = \sum_k \mathrm{Cov}(\Phi_{k,T}, F_T^2)$ identically. In the benchmark, eigen-trend k contributes its own third moment, c_k of the total; co-movement between eigen-trends, $\mathrm{Cov}(\Phi_{k,T}, F_T^2) - \kappa_3(\Phi_{k,T}) \neq 0$, is co-trending across eigen-portfolios.

Proof of Proposition 6. Write $A = a^\top v$ and $b = w^\top u$, so that $F = A + b$ with $\mathbb{E}[A] = \mathbb{E}[b] = 0$ and A independent of b. Since $\mathbb{E}[F] = 0$, $\kappa_3(F) = \mathbb{E}[F^3] = \mathbb{E}[(A+b)F^2] = \sum_k a_k \mathbb{E}[v_k F^2] + \mathbb{E}[bF^2]$, and $\mathbb{E}[v_k F^2] = \mathrm{Cov}(v_k, F^2)$. Expanding $F^2 = A^2 + 2Ab + b^2$ and using independence and zero means,

$$\mathbb{E}[v_k F^2] = \mathbb{E}[v_k A^2] + 2\,\mathbb{E}[v_k A]\,\mathbb{E}[b] + \mathbb{E}[v_k]\,\mathbb{E}[b^2] = \mathbb{E}[v_k A^2], \qquad \mathbb{E}[bF^2] = \mathbb{E}[b]\,\mathbb{E}[A^2] + 2\,\mathbb{E}[b^2]\,\mathbb{E}[A] + \mathbb{E}[b^3] = \kappa_3($$

This proves Equation (A2). If the u_i are independent, cumulants add: $\kappa_3(b) = \sum_i w_i^3 \kappa_3(u_i)$, so $|\kappa_3(b)| \le \bar{\kappa} \sum_i |w_i|^3$; and $\mathrm{Var}[F] = a^\top \mathrm{Var}[v] a + \sum_i w_i^2 \mathrm{Var}[u_i] \ge a^\top \mathrm{Var}[v] a + \underline{\sigma}^2 \sum_i w_i^2$, which gives Inequality (A3). With $w_i = 1/N$, the numerator is $\bar{\kappa} N^{-2}$; the denominator is at least $(\underline{\sigma}^2/N)^{3/2}$, giving $O(N^{-1/2})$, and at least a positive constant if $a^\top \mathrm{Var}[v] a$ is bounded away from zero, giving $O(N^{-2})$. □

Proof of Corollary 6. (a) The first statement is Equation (A2). If v_k is independent of v_{-k}, write $A = a_k v_k + R$ with R independent of v_k and $\mathbb{E}[R] = 0$; then $\mathbb{E}[v_k A^2] = a_k^2 \mathbb{E}[v_k^3] + 2a_k \mathbb{E}[v_k^2]\mathbb{E}[R] + \mathbb{E}[v_k]\mathbb{E}[R^2] = a_k^2 \kappa_3(v_k)$, and the contribution $a_k^3 \kappa_3(v_k)$ vanishes for a symmetric factor. (b) By Lemma 1(a), $F_T = \sum_k \Phi_{k,T}$ identically, so Equation (A2) holds with $v = (\Phi_{1,T}, \ldots, \Phi_{N,T})$, $a = \mathbf{1}$, and $u = 0$. Under Assumption 1, the $\Phi_{k,T} = \mu_k \Psi_k$ are independent with mean zero, so $\mathrm{Cov}(\Phi_{k,T}, F_T^2) = \mathbb{E}[\Phi_{k,T}(\sum_l \Phi_{l,T})^2] = \mathbb{E}[\Phi_{k,T}^3] = \mu_k^3 \kappa_3(\Psi) = c_k\, \kappa_3(F_T)$. □

E.4 The futures curve

This section details the decomposition of Section 5.1. The return of a rolled futures position splits exactly into the change of the front price, the spot part, and the roll yield, the gap between the expiring and the next contract, which is known one day before it is earned (Erb and Harvey, 2006; Gorton and Rouwenhorst, 2006; Koijen et al., 2018). The roll yield is part of the predictable drift m_t of Assumption 2; by Proposition 2 it moves the mean of the trend P&L but enters the third moment only through the drift part. Exhibits A16 and A17 test this on the commodity panel, 37 markets from January 2001 to July 2024, which had been held back from all earlier analyses; its markets overlap with the commodities of the main panel, so it tests the construction and the holdout discipline rather than new markets. The spectral law gives the right order of magnitude there: the Sharpe breadth is 14.8 and the convexity breadth 3.9, and the predicted portfolio skewness of 1.36 compares with a realized 1.65; the channels look as in the main panel (position 0.72, correlation 0.91, drift −0.07). The roll yield earns 61% of the mean P&L of the trend portfolio (90% interval 42%

Panel A: The spectral law on the separately constructed commodity panel	
Statistic	Value
Markets N	37
Breadths K_2 / K_3	14.8 / 3.9
Mean single-market skewness (55-day)	2.67
Spectral factor $\mathrm{tr}((WP)^3)/\mathrm{tr}((WP)^2)^{3/2}$	0.508
Proposition 1: predicted portfolio skewness	1.36
Realized portfolio skewness	1.65
Panel B: Channels of convexity (Proposition 2)	
Component	Contribution
Realized skewness	1.65
Martingale part	1.72
Position channel	0.72
Co-movement channel	1.00
of which variance	0.09
of which correlation	0.91
Drift part	-0.07

Panel C: Spot and roll			
	Total	Spot	Roll
Share of the mean return (%)	100	39	61
90% interval		[8, 58]	[42, 92]
Sharpe ratio (stand-alone)	1.13	0.40	1.54
90% interval	[0.81, 1.46]	[0.06, 0.76]	[1.17, 2.07]
Contribution to skewness (55-day)	1.65	1.67	-0.02
90% interval	[0.83, 2.16]	[0.83, 2.20]	[-0.11, 0.05]
Skewness of the spot P&L alone		1.61	
Correlation of positions with the curve gap $\ln \mathsf{F}_1 - \ln \mathsf{F}_2$	0.29		

Exhibit A16: **The Futures Curve behind the Classic CTA.** Separately constructed commodity panel (37 markets), January 2001–July 2024, equal-risk portfolio of single-filter trend P&L (span 100), 55-day horizon, scale free. The daily log return of each market splits exactly into spot, $\ln \mathsf{F}_1(t) - \ln \mathsf{F}_1(t-1)$, and roll, $\ln \mathsf{F}_1(t-1) - \ln \mathsf{F}_2(t-1)$ on the roll day, which is known one day ahead and is part of the predictable drift m_t of Assumption 2. Roll days from the ratio-adjusted generic series (7.0 per market and year). Panel A: Proposition 1 with the sample correlation. Panel B: the decomposition of Exhibit 7. Panel C: shares of the mean daily P&L; contributions to skewness $\mathrm{Cov}(G_c, (G - \mathrm{E}G)^2)/\mathrm{sd}(G)^3$ of the 55-day sums, which add up to the total. Intervals: moving-block bootstrap of days (252-day blocks, 500 draws). Design fixed before the run (Section B of the Appendix).

to 92%) and, on its own, a Sharpe ratio of 1.54, against 0.40 for the spot part, but contributes -0.02 (-0.11 to 0.05) to its skewness; the spot part contributes 1.67 of the skewness of 1.65. The trend positions lean towards backwardation: their correlation with the gap between the first and second contract is 0.29 on average. A commodity trend follower earns most of its mean return as carry, which is not convex, and its convexity from spot trends, which in crises align across markets. Diversification adds independent carry and spot premia and raises the Sharpe ratio, but convexity rests on the few common spot trends that the leading eigen-trends capture.

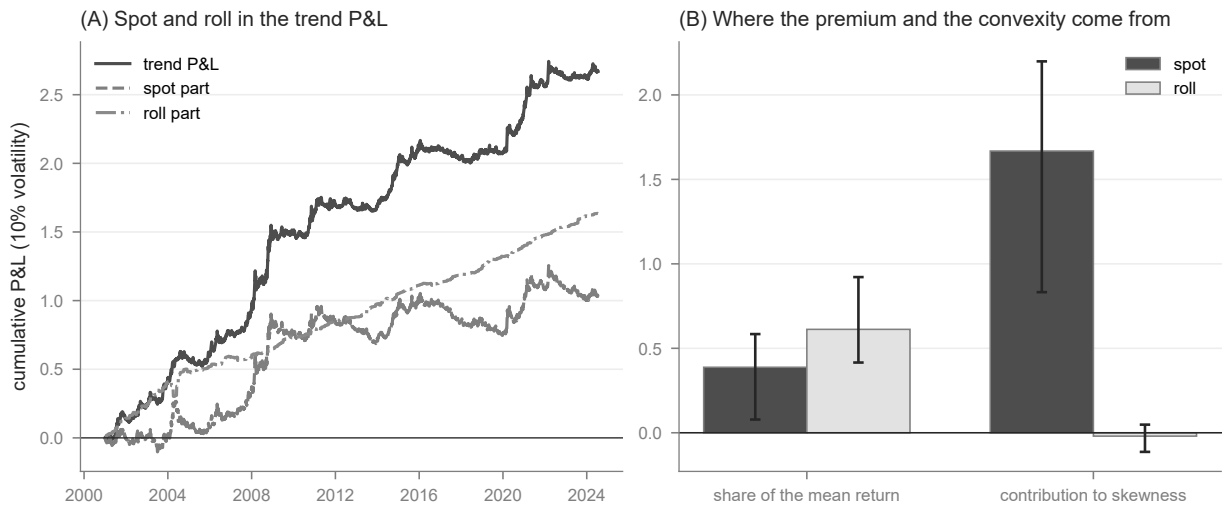


Exhibit A17: **Spot and Roll.** Panel A: cumulative scale-free P&L of the equal-risk commodity trend portfolio and its spot and roll parts, scaled to 10% annualized volatility of the total. Panel B: shares of spot and roll in the mean daily P&L and contributions to the skewness of the 55-day return, with 90% block-bootstrap intervals. Design as in Exhibit A16.

E.5 Risk rules on the commodity panel

Exhibit A18 repeats the comparison of portfolio risk rules of Section 5.2 on the commodity panel, with the predictions fixed in advance (Section B). The ordering is as predicted, and both reductions are significant, but both thresholds are narrowly missed: the ex-ante rule keeps 26% of the skewness without a rule (predicted: at most 25%), the ex-post rule 43% (predicted: at least 50%).

Risk rule	Sharpe ratio	Skewness (55-day)	90% interval	Ratio to none
none	1.13	1.65	[0.83, 2.16]	1.00
ex post (P&L)	1.23	0.72	[0.48, 0.93]	0.43
ex ante (positions)	1.15	0.43	[0.10, 0.71]	0.26
ex post (P&L) − none, skewness		-0.93	[-1.35, -0.17]	
ex ante (positions) − none, skewness		-1.22	[-1.75, -0.46]	

Exhibit A18: **Risk Rules on the Commodity Panel.** Classic single filter (span 100, equal risk) on the 37 markets of the commodity panel, gross of costs, January 2001–July 2024, under the three portfolio risk rules of Exhibit 9: none (constant scale), ex post (daily P&L scaled by the EWMA volatility of past P&L, span 100 days, lagged), and ex ante (positions scaled by the EWMA covariance forecast of the current positions, span 60 days), each with a 10% volatility target. Prediction fixed before the run (Corollary 3): the ex-ante rule keeps at most a quarter of the skewness without a rule, the ex-post rule at least half. Intervals: moving-block bootstrap of daily P&L, blocks of 252 days, 500 draws.

E.6 Variance and premium shares of the risk-premium PCA

Exhibit A19 shows the trade-off of Proposition 5(c). Estimated in 2001–2012, the first five principal eigen-trends carry 50% of the variance of the eigen-portfolio returns ($\mathcal{V}$) and 98% of the trend premium ($\mathcal{R}$); with $\gamma = 1$ the first five RP eigen-trends carry 39% of that variance and 176% of the premium (more than all, because the remaining eigen-trends lose money). Out of sample, in 2013–2026, the RP basis still concentrates more premium in its first five vectors (151% against 99%); in the reverse split it does not (77% against 107%).

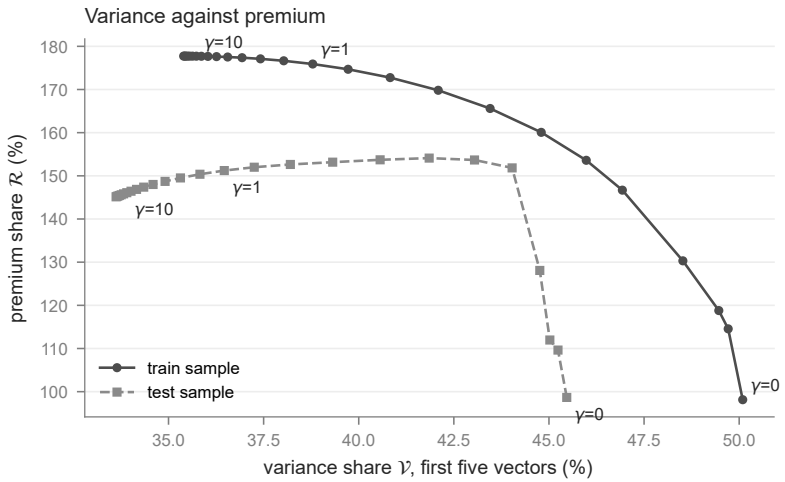


Exhibit A19: **Variance against Premium.** Variance share $\mathcal{V}$ and premium share $\mathcal{R}$ of the first five basis vectors of Proposition 5 as γ rises from 0 to 50, basis estimated in 2001–2012; train: shares in 2001–2012, test: shares in 2013–2026 (by Lemma 2, R in the test sample is the realized share of the mean). Design as in Exhibit 10.

E.7 How much diversification do the filters choose?

Exhibit A20 measures, for the filters of Section 6, the effective numbers of bets implied by the spectrum of CP and the asset classes from which their positions, mean P&L, and third moment come. The analysis was specified in the log before it was run, as descriptive; the stated expectation that premium-aware filters hold fewer bets than the classic CTA holds for Sharpe breadth only. The asset-class shares of the third moment of single filters are noisy and differ between the two splits.

	Train 2001–2012, test 2013–2026				Train 2013–2026, test 2001–2012			
Panel A: Effective bets chosen by the filter								
	$K_2(CP)$	$K_3(CP)$	Sharpe	Skewness	$K_2(CP)$	$K_3(CP)$	Sharpe	Skewness
S0 Classic equal-risk CTA	15.2	4.3	0.45	1.20	13.0	3.2	0.98	0.90
S1 PCA, premium-weighted ($\gamma = 0$)	19.2	11.6	0.76	0.58	15.8	9.9	0.97	1.54
S2 RP-PCA, premium-weighted ($\gamma = 1$)	13.1	8.8	1.28	1.36	12.3	6.7	1.49	0.76
S3 Eigen-trend tilt $\alpha = 1.5$	5.8	2.1	0.50	1.64	5.3	1.9	1.04	1.13
S4 Plug-in optimal filter C^+	9.6	4.9	1.19	1.04	10.6	5.5	1.54	1.82
Panel B: Shares by asset class (%), classic CTA (S0) and RP-PCA (S2)								
	Position		Mean		Position		Mean	
	S0	S2	S0	S2	S0	S2	S0	S2
Equities	24	21	-23	18	26	22	25	22
Bonds	21	17	34	21	21	17	15	7
STIR	3	3	18	11	3	4	6	10
FX	13	14	-5	1	13	13	10	3
Energy	7	8	24	13	7	7	3	-1
Metals	7	7	2	2	7	8	11	14
Agriculture	24	30	50	34	24	31	30	44
Panel C: Shares of the third moment of the 55-day return by asset class (%)								
	Forward		Reverse					
	S0	S2	S0	S2				
Equities	8	3	19	15				
Bonds	36	25	12	-6				
STIR	3	7	5	10				
FX	19	18	17	8				
Energy	26	41	6	-4				
Metals	1	1	10	24				
Agriculture	7	4	31	53				

Exhibit A20: **How Much Diversification Do the Filters Choose?** Filters of Exhibit 10, positions $x_t = CS_t$ fixed in the training sample, test-sample P&L. Panel A: Sharpe and convexity breadth of Corollary 1 computed from the eigenvalues of CP, with P the test-sample correlation matrix, and the test-sample Sharpe ratio and 55-day skewness. Panel B: share of each asset class in the mean absolute position and in the mean P&L. Panel C: share of each asset class in the third moment of the 55-day return, $\mathrm{Cov}(G_c, (G - \mathbb{E}G)^2)$ divided by the third central moment of G. Descriptive analysis fixed before the run (Section B).

E.8 Robustness details

Exhibit A21 reports the robustness tests of the first two results and the sensitivity of the third to the bootstrap block length. The interval of the Sharpe difference between S2 and S0 excludes zero for every block length, and the interval of the skewness difference includes it for every block length. The implementable comparison of Section 6 is borderline under the Ledoit–Wolf test: RP-PCA on the reference signal minus the reference system has a net Sharpe difference of 0.21 without a portfolio target (two-sided $p = 0.071$, against a 90% bootstrap interval of 0.01 to 0.41) and 0.03 with the ex-ante target ($p = 0.76$).

Panel A: Risk rules, classic CTA and LS(250,20), net, January 2001–July 2026

	Skewness	Difference to none	90% interval
Classic, no rule	1.35		
Classic, ex post (span 100)	0.79	-0.57	[-0.96, -0.15]
Classic, ex ante (span 30)	-0.04	-1.39	[-1.72, -0.85]
Classic, ex ante (span 60)	0.04	-1.31	[-1.63, -0.80]
Classic, ex ante (span 100)	0.10	-1.25	[-1.54, -0.76]
LS(250,20), no rule / ex ante (span 60)	0.41 / 0.07	-0.34	[-0.47, -0.12]

Panel B: Measures of skewness

	All / single market (H1)	Classic, no rule	Classic, ex ante	S2 − S0, forward	S2 − S0, reverse
Sample skewness, 55 days	0.49	1.35	0.04	0.16	-0.13
Kelly quantile skewness, 55 days	0.56	0.29	0.04	-0.04	0.07
Winsorized daily P&L (0.5%), 55 days	0.47	1.02	0.02	0.31	-0.11
Sample skewness, 21 days	0.59	1.44	-0.14	-0.86	-0.06
Sample skewness, 126 days	0.47	1.08	0.35	0.31	-0.02

Panel C: Bootstrap block length, S2 − S0 (90% intervals)

	Sharpe, mean block 21 / 63 / 252	Skewness, block 126 / 252 / 504
Forward split	[0.52, 1.13] / [0.54, 1.12] / [0.58, 1.10]	[-0.96, 0.75] / [-0.80, 0.75] / [-0.68, 0.80]
Reverse split	[0.25, 0.79] / [0.28, 0.79] / [0.30, 0.76]	[-0.47, 0.32] / [-0.42, 0.39] / [-0.44, 0.44]

Panel D: Leaving out one asset class

Class left out	K_2	K_3	Ratio realized / formula	S2 − S0 forward	S2 − S0 reverse
Equities	13.1	4.0	0.57 / 0.50	0.23	0.37
Bonds	13.7	3.3	0.57 / 0.55	0.71	0.47
STIR	14.2	3.8	0.50 / 0.52	0.82	0.50
FX	13.1	3.5	0.45 / 0.53	0.66	0.43
Energy	13.5	3.7	0.45 / 0.52	0.73	0.64
Metals	13.7	3.7	0.49 / 0.52	0.82	0.65
Agriculture	10.0	3.1	0.54 / 0.57	0.65	0.41

Exhibit A21: **Robustness Details.** Panel A: risk rules of Exhibit 9 for the classic CTA ($\alpha = 1$, monthly correlation, net of costs, 10% target) with the ex-ante span varied, and for the reference system; intervals: moving blocks of 252 days (500 draws). Panel B: H1, skewness of the all-market portfolio divided by the mean single-market skewness; H2, the classic CTA without a rule and with the ex-ante rule (span 60); H3, test-sample difference of S2 and S0 in the two splits of Exhibit 10. Kelly quantile skewness: $(Q_{0.9} + Q_{0.1} - 2Q_{0.5})/(Q_{0.9} - Q_{0.1})$, on a different scale from the moment skewness. Panel C: intervals of Exhibit 10 with other block lengths. Panel D: Sharpe and convexity breadth, the ratio of all-market to single-market skewness and its prediction $1/\sqrt{K_3}$, and the test-sample Sharpe difference of S2 and S0, with one asset class removed. Design fixed before the run (Section B).